Residues and Duality

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Preface

In the spring of 1963 I suggested to Grothendieck the possibility of my running a seminar at Harvard on his theory of duality for coherent sheaves — a theory which had been hinted at in his talk to the Séminaire Bourbaki in 1957 [**8**], and in his talk to the International Congress of Mathematicians in 1958 [**9**], but had never been developed systematically. He agreed, saying that he would provide an outline of the material, if I would fill in the details and write up lecture notes of the seminar. During the summer of 1963, he wrote a series of "pré-notes" [**10**] which were to be the basis for the seminar.

I quote from the preface of the pré-notes:

> Les présentes notes donnent une esquisse assez détaillée d'une théorie cohomologique de la dualité des Modules cohérents sur les préschémas. Les idées principales de la théorie m'étaient connues dès 1959, mais le manque de fondements adéquats d'Algèbre Homologique m'avait empêché d'aborder une rédaction d'ensemble. Cette lacune de fondements est sur le point d'être comblée par la thèse de Verdier, ce qui rend en principe possible un exposé satisfaisant. Il est d'ailleurs apparu depuis qu'il existe des théories cohomologiques de dualité formellement très analogues à celle développée ici dans toutes sortes d'autres contextes: faisceaux cohérents sur les espaces analytiques, faisceaux abéliens sur les espaces topologiques (Verdier), modules galoisiens (Verdier, Tate), faisceaux de torsion sur les schémas munis de leur topologie étale, corps de classe en tous genres . . . Cela me semble une raison assez sérieuse pour se familiariser avec le yoga général de la dualité dans un cas type, comme la théorie cohomologique des résidus.
>
> La théorie consiste pour l'essentiel dans des questions de variance: construction d'un foncteur $f^!$ et d'un homomorphisme-trace $\underline{\underline{R}}f_* f^! \to \mathrm{id}$. La construction donnée ici est compliquée et indirecte et n'est pas valable sous des conditions aussi générales qu'on est en droit de s'y attendre.

> Il faudra sans doute une idée nouvelle pour apporter des simplifications substantielles.

The seminar took place in the fall and winter of 1963–64, with the assistance of David Mumford, John Tate, Stephen Lichtenbaum, John Fogarty, and others, and gave rise to a series of six exposés which were circulated to a limited audience under the title "Séminaire Hartshorne". The present notes are a revised, expanded, and completed version of the previous notes.

I would like to take this opportunity to thank all those people who have helped in the course of this work, and in particular A. Grothendieck, who gave continual support and encouragement throughout the whole project.

R.H.
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Contents

Chapter 1

Introduction

The way up and the way down are one and the same.

–Heraclitus

The main purpose of these notes is to prove a duality theorem for cohomology of quasi-coherent sheaves, with respect to a proper morphism of locally noetherian preschemes. Various such theorems are already known. Typical is the duality theorem for a non-singular complete curve X over an algebraically closed field k, which says that

$$h^0(D) = h^1(K - D),$$

where D is a divisor, K is the canonical divisor, and

$$h^i(D) = \dim_k \mathrm{H}^i(X, \mathcal{O}_X(D))$$

for any i, and any divisor D. (See e.g. [**16**; **Ch. II**] for a proof.)

Various attempts were made to generalize this theorem to varieties of higher dimension, and as Zariski points out in his report [**20**], his generalization of a lemma of Enriques–Severi [**19**] is equivalent to the statement that for a normal projective variety X of dimension n over k, for any divisor D,

$$h^0(D) = h^n(K - D).$$

This is also equivalent to a theorem of Serre [**FAC**] §76, Thm. 4 on the vanishing of the cohomology group $\mathrm{H}^1(X, \mathcal{O}_X(-m))$ for m large and $\mathcal{L}$ locally free. Using a related theorem **[FAC]** §75, Thm. 3, Zariski shows how one can deduce on a non-singular projective variety the formula

$$h^i(D) = h^{n-i}(K - D)$$

for $0 \leq i \leq n$. In terms of sheaves, this result corresponds to the fact that the k-vector spaces

$$\mathrm{H}^i(X, \mathcal{F}) \quad \text{and} \quad \mathrm{H}^{n-i}\big(X, \mathcal{F}^\vee \otimes \omega\big)$$

are dual to each other, where $\mathcal{F}$ is a locally free sheaf, $\mathcal{F}^\vee$ is the dual sheaf $\underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{O}_X)$, and $\omega = \Omega^n_{X/k}$ is the sheaf of n-differentials on X. Serre **[15]** gives a proof of this same theorem by analytic methods for a compact complex analytic manifold X.

Grothendieck **[8]** gave some generalizations of these theorems for non-singular projective varieties and then in **[9]** announced the general theorem for schemes proper over a field, with arbitrary singularities, which is the subject of the present lecture notes.

To motivate the statement of our main theorem, let us consider the case of projective space $X = \mathbb{P}^n_k$ over an algebraically closed field k. Then there is a canonical isomorphism

$$\mathrm{H}^n(X, \omega) \cong k \tag{1.1}$$

where $\omega = \Omega^n_{X/k}$ is the sheaf of n-differentials. Combining this with the Yoneda pairing

$$\mathrm{H}^i(X, \mathcal{F}) \times \mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \to \mathrm{H}^n(X, \omega) \tag{1.2}$$

we obtain a pairing

$$\mathrm{H}^i(X, \mathcal{F}) \times \mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \to k \tag{1.3}$$

which one shows easily to be a perfect pairing **[SGA 62; expose 12]**. This generalizes the statements above, because for a locally free sheaf $\mathcal{F}$,

$$\mathrm{Ext}^{n-i}(\mathcal{F}, \omega) = \mathrm{Ext}^{n-i}\big(\mathcal{O}_X, \mathcal{F}^\vee \otimes \omega\big) = \mathrm{H}^{n-i}\big(X, \mathcal{F}^\vee \otimes \omega\big).$$

Another way of looking at our duality pairing is as an isomorphism

$$\mathrm{Ext}^{n-i}_X(\mathcal{F}, \omega) \cong \mathrm{Hom}_k\big(\mathrm{H}^i(X, \mathcal{F}), k\big). \tag{1.4}$$

Since everything is linear over k, we may introduce a k-vector space G, and have an isomorphism

$$\mathrm{Ext}^{n-i}_X\big(\mathcal{F}, G \otimes_k \omega\big) \to \mathrm{Hom}_k\big(\mathrm{H}^i(X, \mathcal{F}), G\big). \tag{1.5}$$

Before proceeding further, we must introduce the derived category. It will be discussed in detail in Chapter I, but for the moment it will be sufficient to know the following: For each abelian category $\mathcal{A}$, there is a category $\mathrm{D}(\mathcal{A})$, called the derived category of $\mathcal{A}$, whose objects are complexes of objects of $\mathcal{A}$. If $F\colon \mathcal{A} \to \mathcal{B}$ is an additive functor from one abelian category to another, then under reasonable conditions there is a right derived functor $\underline{\underline{\mathrm{R}}}F\colon \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ with the property that for any $X \in \mathrm{Ob}\,\mathcal{A}$, if X denotes also the complex which is X in degree zero, and zero elsewhere, then

$$\mathrm{H}^i(\underline{\underline{\mathrm{R}}}F(X)) = R^iF(X),$$

where R^iF is the ordinary i^{th} right derived functor of F. Finally, if $F\colon \mathcal{A} \to \mathcal{B}$ and $G\colon \mathcal{B} \to \mathcal{C}$ are two functors then

$$\underline{\underline{\mathrm{R}}}(G \circ F) = \underline{\underline{\mathrm{R}}}G \circ \underline{\underline{\mathrm{R}}}F.$$

This replaces the old-fashioned spectral sequence of a composite functor.

Now we can jazz up our duality for projective space as follows. We replace k by a prescheme Y, so that $X = \mathbb{P}^n_Y$. We consider the derived categories $\mathrm{D}(X)$ and $\mathrm{D}(Y)$ of the categories of $\mathcal{O}_X$-modules and $\mathcal{O}_Y$-modules, respectively. Then cohomology H^i becomes $\underline{\underline{\mathrm{R}}}f_*$, the derived functor of the direct image functor f_*, where $f\colon X \to Y$ is the projection. Ext becomes the derived functor $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}$ of Hom. We define

$$f^!(\mathcal{G}) = f^*(\mathcal{G}) \otimes \omega$$

for $\mathcal{G} \in \mathrm{D}(Y)$, and we replace $\mathcal{F}$ by a complex of sheaves $\mathcal{F} \in \mathrm{D}(X)$. Then the isomorphism (1.1) gives us an isomorphism

$$\underline{\underline{\mathrm{R}}}f_*\, f^!\mathcal{G} \xrightarrow{\sim} \mathcal{G} \tag{1.6}$$

which we call the trace map.

The Yoneda pairing reappears as a natural map

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_X\big(\mathcal{F}, f^!\mathcal{G}\big) \to \underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}}f_*\mathcal{F}, \underline{\underline{\mathrm{R}}}f_*\, f^!\mathcal{G}\big), \tag{1.7}$$

which, composed with the trace map (1.6) gives us the duality morphism

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_X\big(\mathcal{F}, f^!\mathcal{G}\big) \to \underline{\underline{\mathrm{R}}}\,\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}}f_*\mathcal{F}, \mathcal{G}\big) \tag{1.8}$$

which generalizes (1.5). This is easily proved to be an isomorphism ([III, 5.1] below) under suitable hypotheses on $Y, \mathcal{F}, \mathcal{G}$. In fact, the proof is nothing but "general nonsense" once one has the isomorphism (1.4).

Having examined the case of projective space, we can state the following ideal theorem, which is the primum mobile of these notes, although it may never appear explicitly in this form.

Ideal Theorem.

(a) For every morphism $f\colon X \to Y$ of finite type of preschemes, there is a functor $f^!$ such that

1) if $g\colon Y \to Z$ is a second morphism of finite type, then

$$(gf)^! = f^! \circ g^!;$$

2) if f is a smooth morphism, then

$$f^!(\mathcal{G}) = f^*(\mathcal{G}) \otimes \omega,$$

where $\omega = \Omega^n_{X/Y}$ is the sheaf of highest order differentials;

3) if f is a finite morphism, then

$$f^!(\mathcal{G}) = \underline{\mathrm{Hom}}_{\mathcal{O}_Y}\big(f_*\mathcal{O}_X, \mathcal{G}\big)^{\sim}.$$

(b) For every proper morphism $f\colon X \to Y$ of preschemes, there is a trace morphism

$$\mathrm{Tr}_f\colon \underline{\underline{\mathrm{R}}} f_* f^! \to \mathrm{id}$$

of functors from $\mathrm{D}(Y)$ to $\mathrm{D}(Y)$ such that

1) if $g\colon Y \to Z$ is a second proper morphism, then $\mathrm{Tr}_{g\circ f} = \mathrm{Tr}_g \circ \mathrm{Tr}_f$.

2) if $X = \mathbb{P}^n_Y$, then Tr_f is the map deduced from the canonical isomorphism $R^n f_*(\omega) \cong \mathcal{O}_Y$.

3) if f is a finite morphism, then Tr_f is obtained from the natural map "evaluation at one"

$$\underline{\mathrm{Hom}}_{\mathcal{O}_Y}\big(f_*\mathcal{O}_X, G\big) \to G.$$

(c) If $f\colon X \to Y$ is a proper morphism, then the duality morphism

$$\theta_f\colon \underline{\underline{\mathrm{R}}}\mathrm{Hom}_X\big(F, f^!G\big) \to \underline{\underline{\mathrm{R}}}\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}} f_*F, G\big)$$

obtained by composing the natural map (7) above with Tr_f, is an isomorphism for $F \in \mathrm{D}(X)$ and $G \in \mathrm{D}(Y)$.

It should be noted that we have deliberately left certain technical details out of the above statement, for ease of reading. Thus it seems reasonable to make these statements only for complexes of quasi-coherent sheaves, or rather for complexes of arbitrary sheaves whose cohomology sheaves are quasi-coherent. (We denote this category by $\mathrm{D}_{\mathrm{qc}}(Y)$ or $\mathrm{D}_{\mathrm{qc}}(X)$.) In fact, we give an example in the case of a finite morphism to show that the duality theorem (c) fails if G is not quasi-coherent (see example following [III, 6.7]). Secondly one must expect certain boundedness conditions on the complexes involved, namely F should be bounded above (we write $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$) and G should be bounded below ($G \in \mathrm{D}^+_{\mathrm{qc}}(Y)$) for the duality theorem. Finally, questions of variance will be very important in the proof, and so the equals signs in (a) and (b) must be taken with a grain of salt. We must preserve very carefully the distinction between "equals" and "is canonically isomorphic to". Hence we will have more precise statements below.

As to proving the ideal theorem, we succeed only partially. Certainly the conditions under which we can prove it will suffice for most applications, but it is unsatisfying to have restrictive hypotheses which are apparently not essential to the truth of the theorem. I mention four sets of hypotheses under which the theorem can be proved.

(i) For the category of noetherian preschemes of finite Krull dimension, and morphisms $f\colon X \to Y$ which can be factored through a suitable projective space $\mathbb{P}^N_Y$, we have (a1)–(a3) for $\mathrm{D}^+_{\mathrm{qc}}(Y)$ [I, 8.7], (b1)–(b3) for $\mathrm{D}^+_{\mathrm{qc}}(Y)$ [I, 10.5], and (c) for $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$ and $G \in \mathrm{D}^+_{\mathrm{qc}}(Y)$ [III, 11.1].

(ii) For the category of noetherian preschemes which admit dualizing complexes (see [V, §2]; this implies in particular that the preschemes have finite Krull dimension) and morphisms whose fibres are of bounded dimension, we have (a1)–(a3) and (b1)–(b3) for $\mathrm{D}^+_c(Y)$, and (c) for $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$, $G \in \mathrm{D}^+_c(Y)$ [VII, 3.4]. Here the subscript c denotes complexes with coherent cohomology sheaves.

(iii) For the category of noetherian preschemes of finite Krull dimension and smooth morphisms, we have (a1)–(a3) for $\mathrm{D}^+_{\mathrm{qc}}(Y)$, (b1)–(b3) for $G \in \mathrm{D}^b_{\mathrm{qc}}(Y)$, and (c) for $F \in \mathrm{D}^-_{\mathrm{qc}}(X)$ and $G \in \mathrm{D}^b_{\mathrm{qc}}(Y)$ [VII, 4.3]. Here the exponent "b" denotes complexes which are bounded in both directions, i.e., finite.

(iv) Recently P. Deligne has shown (unpublished) that for the category of noetherian preschemes, one can construct $f^!$ and Tr_f satisfying (a), (b), and (c), working with $F \in \mathrm{D}(\mathrm{Qco}(X))$ and $G \in \mathrm{D}^+(\mathrm{Qco}(Y))$, the derived categories of the categories of quasi-coherent sheaves on X and Y, respectively.

The principal difficulty has been the lack of a suitable construction of the functor $f^!$: to define it locally, and then glue. Our procedure is to define $f^!$ for a finite or a smooth morphism, where we have it given to us by (a2) and (a3) [III, §6,§11]. Thus by composition we can obtain it for any morphism which can be factored into a finite morphism followed by a smooth morphism [III, §8]. However, the derived category is not a local object, so we cannot glue these local determinations of $f^!$ to obtain a global one. We resort to a clumsy, round-about procedure of defining $f^!$ for a special class of complexes, called residual complexes [VI, §1], and then pulling ourselves up by our bootstraps to get it for arbitrary complexes [VII, §3]. But our result has the unpleasant hypotheses of (ii) above.

Deligne's construction of $f^!$ is entirely different, and proceeds essentially by representing the functor (for $F \in \mathrm{D}(X)$)

$$F \longmapsto \underline{\underline{\mathrm{R}}}\mathrm{Hom}_Y\big(\underline{\underline{\mathrm{R}}}f_*F, G\big).$$

This approach has the advantage of giving the duality theorem immediately. He also has a method for calculating $f^!G$ locally on X. However, it is not immediately clear from his construction that the properties (a2), (a3) and (b2), (b3) hold. In fact, their proof will probably require some knowledge of the duality theorem as we have proved it here. At least one can hope that some combination of the two approaches will provide substantial simplifications of the theory at a later date.

A second difficulty, which lurks on the fringes of these notes, is that the derived category seems to be a little too big when it comes to unbounded complexes. For example, one can have two unequal morphisms $f, g\colon X \to Y$ in $\mathrm{D}^+(\mathcal{A})$ (where $\mathcal{A}$ is an abelian category) such that their restrictions to each truncation of the complex X to a bounded complex are equal. This gives rise to some trouble with unbounded complexes (see the problem after [II, 5.7], the boundedness hypotheses in [IV, 3.1] and [VII, 4.3a], and the remark following [VI, 1.1c]). Perhaps one will have to replace D^+ by the categories $\operatorname{ind} D^b$ and $\operatorname{pro} D^b$, which however may not be triangulated categories.

Thirdly, some discussion of our noetherian hypotheses is in order. In the present state of the theory, the noetherian hypotheses are well entrenched. We have used them in [II, §7] for the structure of the injective objects in the category of sheaves on a locally noetherian prescheme, and its consequence that $\mathrm{D}^+(\mathrm{Qco}(X)) \to D^+_{\mathrm{qc}}(X)$ is an equivalence of categories. We have used them in the construction of the trace map for projective space [III, §4], in the construction of $f^!$ for a finite morphism [III, §6], in the whole theory of dualizing complexes [V, §2], in the definition of residual complexes [VI, §1], and so forth. Our finite Krull dimension hypotheses are often needed only to make possible the definition of $\mathbb{R}f_*$ and $f^!$ for unbounded complexes — a problem which will disappear when the second difficulty above is solved, and the relation between $\mathrm{D}(\mathbf{Qco}(X))$ and $\mathrm{D}_{\mathrm{qc}}(X)$ is better understood. It is certainly clear that our methods of proof rely heavily on noetherian hypotheses. I expect, however, that once a suitable statement is obtained, e.g., in case (iii) above, one could expect to prove the theorem without noetherian hypotheses, by reducing to the noetherian case. More satisfactory, of course, would be a treatment where noetherian hypotheses on the base were eliminated from the proofs as well as from the statements. When that is achieved, it will be reasonable to state the duality theorem for a proper morphism of ringed topos, whose fibres are noetherian schemes... At the present, however, this must remain a dream of things to come.

Now we will give the reader a brief description of the organization of these notes.

Chapter I gives the language of derived categories, which is used continually in the sequel. Sections 1–5, containing the definition of derived categories and derived functors, are essential, while sections 6 and 7 are refinements which are needed in proofs later on. This chapter is a self-contained treatment of the subject, and makes no reference to algebraic geometry, so can be used independently as an introduction to the notion of derived category. Sections 1–6 (except for the notion of localizing subcategory and the corresponding categories $\mathrm{K}_{A'}(\mathcal{A})$, $\mathrm{D}_{A'}(\mathcal{A})$, etc.) are taken almost without change from notes of Verdier, and should appear in his thesis [**18**].

Chapter II is a fairly systematic treatment of the applications of the language of derived categories to the category of sheaves on a prescheme. We consider the functors Γ, f_*, Hom, $\underline{\mathrm{Hom}}$, $\otimes$, f^*, their derived functors, and relations between these derived functors, such as associativity formulae. There is really no new mathematics involved here, since it is merely a translation into a new language of known results. However, some care is taken to see what hypotheses are needed. Only section 7 is notable new material. Here we give the structure of injective $\mathcal{O}_X$-modules on a locally noetherian prescheme X, showing that they are all direct sums of certain indecomposable injectives $J(x, x')$, for pairs of points x specializing to x' of X. This extends results of Matlis [**13**] and Gabriel [**5**] for the case of quasi-coherent sheaves.

Chapter III contains everything we can say about duality for projective morphisms, and if the reader cares only for them, he may stop at the end of this chapter. However, it should be noted that the following chapters, besides giving us the tools for the proof of the general duality theorem in Chapter VII, also give much new insight into the nature of duality for a projective morphism.

There are so many situations in which we have a functor $f^!$ like the one mentioned in the ideal theorem above, that we use different notations for them. Thus we have $f^\sharp$ for a smooth morphism in section 2, $f^\flat$ for a finite morphism in section 6, and

$f^!$ for an embeddable morphism in section 8. Later we will also have f^y, f^z [VI, §2] and f^Δ [VI, §3] for residual complexes.

In sections 3, 4, and 5 we recall the explicit calculations of cohomology for projective space, and give the old duality for projective space in the new language of derived categories. Sections 6 and 7 give the corresponding formalism for finite morphisms, and its relation to the case of smooth morphisms, so that in sections 8, 10, and 11 we can prove the ideal duality theorem for projective morphisms. Section 9 gives the formalism of a residue symbol which generalizes the classical residue of a differential on a curve. Even over the complex numbers, this important concept of residue for varieties of dimension greater than one was not known before.

In Chapter IV we study "local cohomology", or cohomology with restricted supports, of abelian sheaves on a locally noetherian topological space, generalizing results of [LC, §16]. In particular, we discuss various cohomological properties which a sheaf or complex of sheaves may have with respect to certain families of supports. This gives rise to the notions of depth, Cousin complex, Cohen–Macaulay complex, and Gorenstein complex. The results of this chapter are independent of all other chapters of these notes, and so may be of use elsewhere, although their only application for the moment is to the theory of duality on preschemes.

Chapter V discusses dualizing complexes (read section 0 to find out what one is) and gives a duality theorem for modules over a local ring, generalizing results of [LC, §16]. In particular, the dualizing functor

$$\underline{\mathrm{D}} = \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}(\,\bullet\,, R^\bullet)$$

treated here will be useful for our bootstrap operation (construction of $f^!$) later, because it interchanges $\otimes$ and $\underline{\mathrm{Hom}}$, $f^!$ and $\underline{\underline{\mathrm{L}}}f_*$, and commutes with $\underline{\underline{\mathrm{R}}}f_*$ (duality theorem).

In Chapter VI we prepare for the final duality theorem by giving the construction of $f^!$ and Tr_f for residual complexes. This is accomplished by a delicate glueing procedure which is the most difficult part of the theory, so we have given it in some detail. Perhaps some day this type of construction will be done more elegantly using the language of fibred categories and results of Giraud's thesis [**6**].

Chapter VII contains two main results. The first is the residue theorem, which generalizes the classical theorem that the sum of the residues of a differential on a curve is zero, and which is proved by reduction to that case. The second is the proof of the duality theorem for a proper morphism, which, now that all the functorial machinery has been set up, is little more than putting together the pieces of a jigsaw puzzle.

It remains to give the reader some perspective by listing some topics which have a logical place in these notes, but which are not here.

1. The cohomology class associated to a cycle. Let X be proper and smooth of dimension n over a field k. In [8, §4] it is shown how to associate to each non-singular subvariety Y of codimension p of X, a cohomology class

$$\mathrm{P}_X(Y) \in \mathrm{H}^p\big(X, \Omega^p_{X/k}\big).$$

Now this can be done for an arbitrary subvariety of X, using the remarks in [9]. One defines

$$\omega_Y = \mathrm{H}^{-n+p}\big(g^! k\big),$$

where $g\colon Y \to k$ is the projection. Then there is a canonical element

$$\eta \in \mathrm{Ext}^{-n+p}_{\mathcal{O}_Y}\big(\omega_Y, g^! k\big).$$

One defines a natural map

$$\Omega^{n-p}_{Y/k} \to \omega_Y,$$

which, together with η and the construction of [8, §4], gives the cohomology class $\mathrm{P}_X(Y)$. One proves the fundamental theorem that formation of the cohomology class associated to a cycle takes an intersection of cycles into the cup-product of their cohomology classes.

2. The theory of Poincaré duality and the Gysin homomorphism can be developed as in [8, §4].

3. A Lefschetz–Verdier fixed point formula for coherent sheaves on a scheme proper over a field, to generalize [21, Theorem 2]. In particular, the determination of the local contribution at a non-simple fixed point presents an interesting topic for future investigation.

4. The still lacking theory of duality for complexes with differential operators as boundary operator, its ties with ordinary singular homology theory and with vector bundles with integrable connections, as suggested in [22].

A remark on references: Theorem 5.1 of Chapter III is referred to as "Theorem 5.1" in Chapter III, and as [III, 5.1] elsewhere. References to the bibliography at the end are given by square brackets with an arabic numeral or some capital letters, e.g., [14, (31.I)] or [EGA III, 2.1.12].

Chapter 2

The Derived Category

2.1 Introduction

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, and let $F\colon \mathcal{A} \to \mathcal{B}$ be a functor. Our purpose is to define, functorially for each $X \in \operatorname{Ob} \mathcal{A}$, a complex $\underline{\underline{\mathrm{R}}}F(X)$ whose cohomology groups are the right derived functors of F acting on X, namely $\mathrm{R}^i F(X)$. In general, we will define $\underline{\underline{\mathrm{R}}}F(X^\bullet)$ for any complex $X^\bullet$ of objects of $\mathcal{A}$.

More precisely, $\underline{\underline{\mathrm{R}}}F$ will be a functor from the derived category $\mathrm{D}(\mathcal{A})$ of $\mathcal{A}$ to the derived category $\mathrm{D}(\mathcal{B})$ of $\mathcal{B}$.

The derived category $\mathrm{D}(\mathcal{A})$ is obtained as follows: one first considers the category $\mathrm{K}(\mathcal{A})$, whose objects are complexes of elements of $\mathcal{A}$, and whose morphisms are homotopy equivalence classes of morphisms of complexes. The category $\mathrm{D}(\mathcal{A})$ is obtained by "localizing" $\mathrm{K}(\mathcal{A})$ so that every morphism in $\mathrm{K}(\mathcal{A})$ which induces an isomorphism on cohomology becomes an isomorphism in $\mathrm{D}(\mathcal{A})$. This process of localization will be explained in general below.

Although $\mathcal{A}$ and $\mathcal{B}$ above are assumed to be abelian categories, the categories $\mathrm{K}(\mathcal{A})$, $\mathrm{D}(\mathcal{A})$, etc., are in general not abelian. However, they can be given a structure which carries enough information for our purposes, namely a structure of triangulated category. Thus we will be led to study triangulated categories.

2.2 Triangulated Categories

2.2.1 Definition

A triangulated category is an additive category $\mathcal{C}$, together with

a) an automorphism $T\colon \mathcal{C} \to \mathcal{C}$ called the *translation functor*, and

b) a collection of sextuples (X, Y, Z, u, v, w), called the *triangles* of $\mathcal{C}$,

where in each sextuple $X, Y, Z \in \operatorname{Ob}\mathcal{C}$ and

$$u\colon X \to Y, \quad v\colon Y \to Z, \quad w\colon Z \to T(X)$$

are morphisms. A triangle is usually written

$$\begin{tikzcd} X \arrow[r, "u"] & Y \arrow[d, "v"] \\ & Z \arrow[ul, "w"] \end{tikzcd}$$

A morphism of triangles

$$(X, Y, Z, u, v, w) \to (X', Y', Z', u', v', w')$$

is a commutative diagram

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[d, "f"'] & Y \arrow[r, "v"] \arrow[d, "g"'] & Z \arrow[r, "w"] \arrow[d, "h"'] & T(X) \arrow[d, "T(f)"'] \\ X' \arrow[r, "u'"] & Y' \arrow[r, "v'"] & Z' \arrow[r, "w'"] & T(X') \end{tikzcd}$$

This data is subject to the following axioms:

(TR1) (a) Every sextuple isomorphic to a triangle is a triangle.

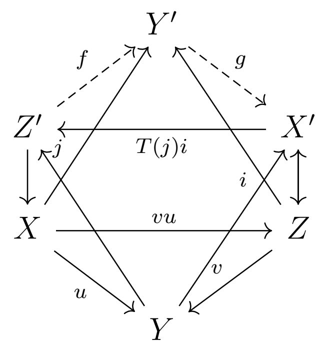
(b) Every morphism $u\colon X \to Y$ can be embedded in a triangle.

(c) The sextuple $(X, X, 0, \mathrm{id}_X, 0, 0)$ is a triangle.

(TR2) (X, Y, Z, u, v, w) is a triangle if and only if $(Y, Z, T(X), v, w, -T(u))$ is a triangle.

(TR3) Given two triangles (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w'), and morphisms $f\colon X \to X'$, $g\colon Y \to Y'$ such that $u'f = gu$, there exists a morphism $h\colon Z \to Z'$ (not necessarily unique) so that (f, g, h) is a morphism of triangles.

(TR4) (The octahedral axiom.)



Suppose given triangles

$$(X, Y, Z', u, j, \cdot),$$
$$(Y, Z, X', v, \cdot, i),$$
$$(X, Z, Y', vu, \cdot, \cdot)$$

Then there exist morphisms $f\colon Z' \to Y'$ and $g\colon Y' \to X'$ such that

$$(Z', Y', X', f, g, T(j)i)$$

is a triangle, and the two other faces of the octahedron with f, g as edges, are commutative diagrams.

Definition 1. *An additive functor $F\colon \mathcal{C} \to \mathcal{C}'$ from one triangulated category to another is called a* (covariant) ∂-functor *if it commutes with the translation functor and takes triangles into triangles. A contravariant ∂-functor takes triangles into triangles with the arrows reversed and sends the translation functor into its inverse.*

Definition 2. *An additive functor $H\colon \mathcal{C} \to \mathcal{A}$ from a triangulated category to an abelian category is called a* covariant cohomological functor *if whenever (X, Y, Z, u, v, w) is a triangle, the long sequence*

$$\cdots \to H(T^iX) \to H(T^iY) \to H(T^iZ) \to H(T^{i+1}X) \to \cdots$$

is exact (the morphisms being $H(T^iu)$, etc.). If H is a cohomological functor, we often write $H^i(X)$ for $H(T^iX)$, $i \in \mathbb{Z}$. One defines a contravariant cohomological functor by reversing the arrows.

Proposition 1. *a) The composition of any two consecutive morphisms in a triangle is zero.*

b) If $\mathcal{C}$ is a triangulated category and M an object of $\mathcal{C}$, then $\mathrm{Hom}_{\mathcal{C}}(M, \cdot)$ and $\mathrm{Hom}_{\mathcal{C}}(\cdot, M)$ are cohomological functors on $\mathcal{C}$.

c) If in the situation of (TR3) f and g are isomorphisms, then h is also an isomorphism.

Proof. a) Let (X, Y, Z, u, v, w) be a triangle. By (TR2) it is sufficient to show that $vu = 0$, Also by (TR2), $(Y, Z, T(X), v, w, -T(u))$ is a triangle. By (TR1), $(Z, Z, 0, \mathrm{id}_Z, 0, 0)$ is a triangle. We apply (TR3) to the maps $v\colon Y \to Z$ and $\mathrm{id}_Z\colon Z \to Z$, and conclude that there is a map $h : T(X) \to 0$ giving a morphism of triangles. It follows that $T(v)(-T(u)) = 0$, or, since T is an automorphism, $vu = 0$.

b) Let $M \in \mathrm{Ob}(\mathcal{C})$, and let (X, Y, Z, u, v, w) be a triangle. To show $\mathrm{Hom}_{\mathcal{C}}(M, \cdot)$ is a cohomological functor, it suffices by (TR2) to show the sequence

$$\mathrm{Hom}_{\mathcal{C}}(M, X) \to \mathrm{Hom}_{\mathcal{C}}(M, Y) \to \mathrm{Hom}_{\mathcal{C}}(M, Z)$$

is exact. By a) we know the composition is zero. Now suppose given $g \in \mathrm{Hom}_{\mathcal{C}}(M, Y)$ such that $vg \in \mathrm{Hom}_{\mathcal{C}}(M, Z)$ is zero. We apply (TR3) to the triangles $(M, M, 0, \mathrm{id}_M, 0, 0)$ and (X, Y, Z, u, v, w) and the map $g\colon M \to Y$ to conclude there exists $f\colon M \to X$ such that $uf = g$.

A similar proof shows that $\mathrm{Hom}_{\mathcal{C}}(\cdot, M)$ is a contravariant cohomological functor.

c) In the situation of (TR3) suppose that f and g are isomorphisms. We apply the cohomological functor $\mathrm{Hom}_{\mathcal{C}}(Z', \cdot)$ to the whole situation, and obtain an exact commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}(Z', X) \arrow[r] \arrow[d, "f_*"'] & \mathrm{Hom}(Z', Y) \arrow[r] \arrow[d, "g_*"'] & \mathrm{Hom}(Z', Z) \arrow[r] \arrow[d, "h_*"'] & \mathrm{Hom}(Z', T(X)) \arrow[r] \arrow[d, "T(f)_*"'] & \mathrm{Hom}(Z', T(Y)) \arrow[d, "T(g)_*"'] \\
\mathrm{Hom}(Z', X') \arrow[r] & \mathrm{Hom}(Z', Y') \arrow[r] & \mathrm{Hom}(Z', Z') \arrow[r] & \mathrm{Hom}(Z', T(X')) \arrow[r] & \mathrm{Hom}(Z', T(Y'))
\end{tikzcd}$$

where $f_* = \mathrm{Hom}(Z', f)$ etc. Now since f and g are isomorphisms in $\mathcal{C}$, it follows that $f_*, g_*, T(f)_*$, and $T(g)_*$ are isomorphisms of abelian groups. Hence by the five-lemma, h_* is an isomorphism. We conclude that there exists a $\varphi \in \mathrm{Hom}_{\mathcal{C}}(Z', Z)$ such that $h_*(\varphi) = h\varphi$ is $\mathrm{id}_{Z'} \in \mathrm{Hom}(Z', Z')$.

Similarly using the cohomological functor $\mathrm{Hom}_{\mathcal{C}}(\cdot, Z)$ we find there is a $\psi \in \mathrm{Hom}(Z', Z)$ such that $\psi h = \mathrm{id}_{Z'}$. It follows that $\varphi = \psi$ and h is an isomorphism. $\square$

2.3 K($\mathcal{A}$) is triangulated

Let $\mathcal{A}$ be an abelian category. A *complex* of objects of $\mathcal{A}$ is a collection $X^\bullet = (X^n)_{n\in\mathbb{Z}}$ of objects of $\mathcal{A}$ together with maps $d^n\colon X^n \to X^{n+1}$ such that $d^{n+1}d^n = 0$ for all $n \in \mathbb{Z}$.

A morphism f of complexes $X^\bullet \to Y^\bullet$ is a collection of maps $f^n\colon X^n \to Y^n$ which commute with differentials:

$$f^{n+1} d_X^n = d_Y^n f^n$$

for all n.

Two maps $f, g\colon X^\bullet \to Y^\bullet$ are *homotopic* if there exists a collection of maps $k = (k^n)$, $k^n\colon X^n \to X^{n-1}$ (not necessarily commuting with d) such that

$$f^n - g^n = d_Y^{n-1} k^n + k^{n+1} d_X^n$$

for all n. Homotopy is an equivalence relation, and compositions of homotopic maps are homotopic.

We define $\mathrm{K}(\mathcal{A})$ to be the category whose objects are complexes of objects of $\mathcal{A}$, and whose morphisms are homotopy equivalence classes of morphisms of complexes. A complex $X^\bullet$ is *bounded below* if $X^n = 0$ for $n \ll 0$. We denote by $\mathrm{K}^+(\mathcal{A})$ the full subcategory of $\mathrm{K}(\mathcal{A})$ of complexes bounded below. Similarly we define $\mathrm{K}^-(\mathcal{A})$ (bounded above) and $\mathrm{K}^b(\mathcal{A})$ (bounded on both sides).

We now define a triangulated structure on $\mathrm{K}(\mathcal{A})$ (resp. $\mathrm{K}^+(\mathcal{A})$, etc.). Let T be the left shift functor with sign change:

$$T(X^\bullet)^p = X^{p+1}, \qquad d_{T(X)} = -d_X.$$

We often write $X^\bullet[1]$ instead of $T(X^\bullet)$, and $X^\bullet[n]$ instead of $T^n(X^\bullet)$. If $u\colon X^\bullet \to Y^\bullet$ is any morphism, consider its *mapping cone* $Z^\bullet$ of u, defined as the complex

$$Z^\bullet = T(X^\bullet) \oplus Y^\bullet$$

with differential

$$d_Z = \begin{pmatrix} T(d_X) & T(u) \\ 0 & d_Y \end{pmatrix}.$$

There are natural morphisms

$$v\colon Y^\bullet \to Z^\bullet, \qquad w\colon Z^\bullet \to T(X^\bullet).$$

We define a triangle in $\mathrm{K}(\mathcal{A})$ to be any sextuple isomorphic to one obtained from a morphism u by this construction.

All axioms (TR1)–(TR4) are easily verified, once one notes that the mapping cone of the identity map $\mathrm{id}_{X^\bullet}$ is homotopic to 0. Indeed, the the mapping cone is $T(X^\bullet) \oplus X^\bullet$ as described above, matrix

$$k = \begin{pmatrix} 0 & 0 \\ \mathrm{id}_X & 0 \end{pmatrix}$$

is a homotopy operator.

Definition 3. *We define H to be the functor from $\mathrm{K}(\mathcal{A})$ to $\mathcal{A}$ which sends a complex $X^\bullet$ to its 0^{th} cohomology:*

$$H(X^\bullet) = \ker d^0 \big/ \operatorname{im} d^{-1}.$$

This is indeed a functor, since homotopic maps induce the same map on cohomology. We write $H^i = H \circ T^i$ for any $i \in \mathbb{Z}$.

Observe that H is a cohomological functor from $\mathrm{K}(\mathcal{A})$ to $\mathcal{A}$. Indeed, it suffices to check the long exact sequence for triangles coming from mapping cones, which is straightforward.

2.4 Localization of categories

Definition 4. *Let $\mathcal{C}$ be a category. A collection S of morphisms of $\mathcal{C}$ is called a* multiplicative system *if it satisfies the following axioms (FR1)–(FR3):*

(FR1) *If $f, g \in S$ and the composite fg exists, then $fg \in S$. For any $X \in \operatorname{Ob}\mathcal{C}$, $\mathrm{id}_X \in S$.*

(FR2) *Any diagram*

$$X \xrightarrow{u} Y \xleftarrow{s} Z, \quad s \in S$$

can be completed to a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{v} & Z \\ {\scriptstyle t}\downarrow & & \downarrow{\scriptstyle s} \\ X & \xrightarrow{u} & Y \end{array}$$

with $t \in S$. The analogous statement holds with all arrows reversed.

(FR3) *Let $f, g\colon X \to Y$ be morphisms in $\mathcal{C}$. The following are equivalent:*

(i) *There exists $s\colon Y \to Y'$ in S such that $sf = sg$;*

(ii) *There exists $t\colon X' \to X$ in S such that $ft = gt$.*

Definition 5. *Let $\mathcal{C}$ be a category and S a collection of morphisms in $\mathcal{C}$. The* localization of $\mathcal{C}$ with respect to S *is a category $\mathcal{C}_S$ together with a functor $Q\colon \mathcal{C} \to \mathcal{C}_S$ such that:*

(a) *$Q(s)$ is an isomorphism for every $s \in S$;*

(b) Any functor $F\colon \mathcal{C} \to \mathcal{D}$ sending each $s \in S$ to an isomorphism factors uniquely through Q.

Remark. One can show that such a localization exists without hypotheses on S, but we will not need this result.

Proposition 2. *Let $\mathcal{C}$ be a category and S a multiplicative system in $\mathcal{C}$. The localization $\mathcal{C}_S$ may be constructed as follows:* $\operatorname{Ob}\mathcal{C}_S = \operatorname{Ob}\mathcal{C}$, *and for all $X, Y \in \operatorname{Ob}\mathcal{C}$,*

$$\operatorname{Hom}_{\mathcal{C}_S}(X,Y) = \varinjlim_{I_X} \operatorname{Hom}_{\mathcal{C}}(X',Y),$$

where I_X is the category whose objects are morphisms $s\colon X' \to X$ in S, and morphisms are commutative diagrams:

$$\begin{tikzcd} X'' \arrow[r, "f"] \arrow[dr, "s2"'] & X' \arrow[d, "s1"] \\ & X \end{tikzcd}$$

Furthermore, if $\mathcal{C}$ is an additive category,, then $\mathcal{C}_S$ is also additive.

Proof. First observe, using (FR1), (FR2), and (FR3) that the category I_X satisfies the axioms L1, L2, L3 of [GT, Ch. I, §1], and hence behaves as well as an inductive system for taking limits. Thus a morphism of X to Y in C_S is represented by a diagram

$$\begin{tikzcd} & X' \arrow[dl, "s"'] \arrow[dr, "a"] & \\ X & & Y \end{tikzcd}$$

with $s \in S$. This diagram defines the same morphism as another one with $t \in S$

$$\begin{tikzcd} & X'' \arrow[dl, "t"'] \arrow[dr, "b"] & \\ X & & Y \end{tikzcd}$$

if and only if there is a morphism $u : X''' \longrightarrow X$ in S and morphisms $f : X''' \longrightarrow X'$ and $g : X''' \longrightarrow X''$ such that $sf = u = tg$ and $af = bg$.

To compose morphisms

$$
\begin{tikzcd}
 & X' \arrow[dl, "s"'] \arrow[dr, "a"] & \\
X & & Y
\end{tikzcd}
\quad \text{and} \quad
\begin{tikzcd}
 & Y' \arrow[dl, "t"'] \arrow[dr, "b"] & \\
Y & & Z
\end{tikzcd}
$$

we use (FR2) to find a commutative diagram

$$
\begin{tikzcd}
 & & X'' \arrow[dl, "t'"'] \arrow[dr, "c"] & & \\
 & X' \arrow[dl, "s"'] \arrow[dr, "a"] & & Y' \arrow[dl, "t"'] \arrow[dr, "b"] & \\
X & & Y & & Z
\end{tikzcd}
$$

with $t' \in S$, and then take X'', st', bc to be the composition. One verifies easily that the resulting morphism of X to Z is independent of the representatives of the morphisms of X to Y and Y to Z chosen, and is also independent of the commutative diagram chosen.

One can also verify easily that the functor $Q : C \longrightarrow C_S$ has the properties required and that C_S is additive if C is (again using L1, L2, L3 to show that the $\varinjlim$ is a group).

$\square$

Definition 6. *Let $\mathcal{C}$ be a triangulated category and S a multiplicative system in $\mathcal{C}$. We say S is* compatible with the triangulation *if:*

(FR4) $s \in S \implies T(s) \in S$, *where T is the translation functor;*

(FR5) Axiom (TR3) holds with the extra condition: if $f, g \in S$ are the given morphisms of triangles, then $h \in S$.

Proposition 3. *If $\mathcal{C}$ is triangulated and S a multiplicative system compatible with triangulation, then $\mathcal{C}_S$ has a unique structure of triangulated category such that Q is a ∂-functor, and Q has the universal property b) above for ∂-functors into triangulated categories.*

Proof. Left to the reader. It helps to observe that one can also calculate $\mathrm{Hom}_{C_S}(X, Y)$ as

$$\varinjlim_{J_Y} \mathrm{Hom}_C(X, Y')$$

where J_Y is the category whose objects are morphisms $s : Y \longrightarrow Y'$ in S, and whose maps are commutative diagrams

$$\begin{tikzcd} & Y \arrow[dl, "s_1"'] \arrow[dr, "s_2"] & \\ Y' \arrow[rr, "f"] & & Y'' \end{tikzcd}.$$

$\square$

Proposition 4. *Let $\mathcal{C}$ be a category, S a multiplicative system in $\mathcal{C}$, and $\mathcal{D} \subseteq \mathcal{C}$ a full subcategory (i.e. $X, Y \in \operatorname{Ob} D$ implies $\operatorname{Hom}_C(X,Y) = \operatorname{Hom}_D(X,Y)$) such that $S \cap \mathcal{D}$ is a multiplicative system in $\mathcal{D}$. Assume furthermore that one of the following two conditions holds:*

(i) For every $s\colon X' \to X$ in S with $X \in \operatorname{Ob}\mathcal{D}$, there exists $f\colon X'' \to X'$ with $X'' \in \operatorname{Ob}\mathcal{D}$ and $sf \in S$;

(ii) The same with all arrows reversed.

Then the natural functor

$$\mathcal{D}_{S\cap\mathcal{D}} \to \mathcal{C}_S$$

is fully faithful, i.e. $\mathcal{D}_{S\cap\mathcal{D}}$ can be identified with a full subcategory of $\mathcal{C}_S$.

Proposition 5. *Let $\mathcal{C}$ be a category, S a multiplicative system, $Q\colon \mathcal{C} \to \mathcal{C}_S$ the localization functor. Let $\mathcal{D}$ be any category, and $F, G\colon \mathcal{C}_S \to \mathcal{D}$ functors. Then the natural restriction map*

$$\alpha\colon \operatorname{Hom}(F,G) \to \operatorname{Hom}(FQ, GQ)$$

between functor morphism sets is bijective.

Proof. To give a morphism of functors $F \longrightarrow G$ means to give for each $X \in \operatorname{Ob} C_S$, a morphism $F(X) \longrightarrow G(X)$, such that if $X \longrightarrow Y$ is a morphism, then

$$\begin{tikzcd} F(X) \arrow[r] \arrow[d] & F(Y) \arrow[d] \\ G(X) \arrow[r] & G(Y) \end{tikzcd}$$

is a commutative diagram. Thus, since $\operatorname{Ob} C = \operatorname{Ob} C_S$, the map α is injective. To show α surjective, suppose given a morphism $FQ \longrightarrow GQ$. Then we have a morphism $F(X) \longrightarrow G(X)$ for each $X \in \mathrm{Ob}\, C = \mathrm{Ob}\, C_S$, and commutative diagrams for morphisms $X \longrightarrow Y$ in C. A morphism in C_S is represented by a diagram

$$\begin{tikzcd} & X' \arrow[dl, "s"'] \arrow[dr, "f"] & \\ X & & Y \end{tikzcd}$$

of morphisms in C, with $s \in S$. But for $s \in S$, $F(s)$ and $G(s)$ are isomorphisms, so we get the required commutative diagram.

□

2.5 Quasi-isomorphisms and the derived category

Let $\mathcal{A}$ be an abelian category, and let $\mathrm{K}(\mathcal{A})$ be the triangulated category of complexes up to homotopy introduced in §2. A morphism $f\colon X^\bullet \to Y^\bullet$ in $\mathrm{K}(\mathcal{A})$ is called a *quasi-isomorphism* if it induces an isomorphism on all cohomology groups. Denote by Qis the collection of all quasi-isomorphisms in $\mathrm{K}(\mathcal{A})$.

Proposition 6. Qis *is a multiplicative system in* $\mathrm{K}(\mathcal{A})$.

Proof. This is a consequence of the following more general proposition □

Proposition 7. *Let $\mathcal{C}$ be a triangulated category, $\mathcal{A}$ an abelian category, and $H\colon \mathcal{C} \to \mathcal{A}$ a cohomological functor. Let S be the class of morphisms $s \in \mathcal{C}$ such that $H(T^i(s))$ is an isomorphism for all $i \in \mathbb{Z}$. Then S is a multiplicative system in $\mathcal{C}$ and is compatible with the triangulation.*

Proof. We verify axioms (FR1)–(FR5). Axioms (FR1) and (FR4) are trivial. Axiom (FR5) follows from the long exact sequence of a cohomological functor and the five-lemma.

To prove (FR2), let a diagram is given:

$$X \xrightarrow{u} Y \xleftarrow{s} Z$$

with $s \in S$, Complete s to a triangle (Z, Y, N, s, f, g). Complete fu to a triangle (W, X, N, t, fu, h). Then (u, id_N) is a map of two sides of the second triangle into the first, so there is a map $v : W \longrightarrow Z$ giving a morphism of triangles.

$$\begin{tikzcd}
W \arrow[rr, "v"] \arrow[dd, "t"] & & Z \arrow[dd, "s"] & \\
& N \arrow[ul, "h"] \arrow[rr, "\mathrm{id}_N"] & & N \arrow[ul, "g"'] \\
X \arrow[ur, "fu"'] \arrow[rr, "u"] & & Y \arrow[ur, "f"] &
\end{tikzcd}$$

Now $sv = ut$, so it remains to prove $t \in S$. Indeed, since $s \in S$, we have $H(T^i(N)) = 0$ for all $i \in \mathbb{Z}$ by the long exact sequence of the first triangle. Applying this to the long exact sequence of the second triangle, we find $H(T^i(t))$ is an isomorphism for all $i \in \mathbb{Z}$.

The opposed statement of (FR2) is proved similarly.

To prove (FR3), we consider the morphism $f - g$, and reduce to showing the following two properties equivalent (where $f : X \longrightarrow Y$ is a morphism):

1. There exists an $s\colon Y \to Y' \in S$ such that $sf = 0$

2. There exists an $t\colon X' \to X \in S$ such that $ft = 0$

Suppose (i) holds.

$$\begin{tikzcd}
X' \arrow[d, "t"'] & Z \arrow[l, "w"'] \arrow[d, "v"] & \\
X \arrow[ur, "g"] \arrow[r, "f"'] & Y \arrow[r, "s"'] & Y' \arrow[ul, "u"']
\end{tikzcd}$$

By (TR1) and (TR2) we can find a triangle (Z, Y, Y', v, s, u) for suitable Z. Now $sf = 0$, so by Proposition 1.1 b), there is a map $g : X \longrightarrow Z$ such that $f = vg$. Again by (TR1) and (TR2) we can find a triangle (X', X, Z, t, g, w) for suitable X'. By the same Proposition applied to this second triangle, the existence of v implies that $ft = 0$. We need only show that $t \in S$. Since $s \in S$, $H(T^i(Z)) = 0$ for all $i \in \mathbb{Z}$, by the long exact sequence of cohomology. In turn, this implies that $t \in S$.

The implication (ii) $\Longrightarrow$ (i) is analogous. $\square$

Definition 7. *The* derived category *of $\mathcal{A}$ is*

$$\mathrm{D}(\mathcal{A}) := \mathrm{K}(\mathcal{A})_{\mathrm{Qis}}.$$

Similarly define

$$\mathrm{D}^+(\mathcal{A}) = \mathrm{K}^+(\mathcal{A})_{\mathrm{Qis}}, \quad \mathrm{D}^-(\mathcal{A}) = \mathrm{K}^-(\mathcal{A})_{\mathrm{Qis}}, \quad \mathrm{D}^b(\mathcal{A}) = \mathrm{K}^b(\mathcal{A})_{\mathrm{Qis}}.$$

By Proposition 1.3.3, these are full subcategories of $\mathrm{D}(\mathcal{A})$*, and*

$$\mathrm{D}^+(\mathcal{A}) \cap \mathrm{D}^-(\mathcal{A}) = \mathrm{D}^b(\mathcal{A}).$$

Definition 8. *Let* $\mathcal{A}$ *be abelian, and* $\mathcal{A}' \subset \mathcal{A}$ *a* thick *full abelian subcategory (closed under extensions in* $\mathcal{A}$*). Define* $\mathrm{K}_{\mathcal{A}'}(\mathcal{A})$ *to be the full subcategory of* $\mathrm{K}(\mathcal{A})$ *consisting of complexes* $X^\bullet$ *whose cohomology objects* $H^i(X^\bullet)$ *lie in* $\mathcal{A}'$*. Since* $\mathcal{A}'$ *is thick,* $\mathrm{K}_{\mathcal{A}'}(\mathcal{A})$ *is a triangulated subcategory of* $\mathrm{K}(\mathcal{A})$*.*

Define

$$\mathrm{D}_{\mathcal{A}'}(\mathcal{A}) := \mathrm{K}_{\mathcal{A}'}(\mathcal{A})_{\mathrm{Qis}}.$$

By Proposition 1.3.3, $\mathrm{D}_{\mathcal{A}'}(\mathcal{A})$ *is the full subcategory of* $\mathrm{D}(\mathcal{A})$ *of complexes with all cohomology in* $\mathcal{A}'$*. One defines similarly* $\mathrm{K}^+_{\mathcal{A}'}(\mathcal{A}), \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$*, etc., for bounded-below complexes.*

Example. To help understand the category $\mathrm{D}(\mathcal{A})$, let us ask: when does a morphism of complexes $f\colon X \to Y$ give the zero map in $\mathrm{D}(\mathcal{A})$? The condition is:

($*$) There exists an $s\colon Y \to Y'$ in Qis such that sf is homotopic to zero (or, equivalently, there exists a $t\colon X' \to X$ in Qis such that ft is homotopic to zero).

1. Of course, if $f \sim 0$ (f homotopic to zero), then f satisfies ($*$). The converse is false. For example, take X to be the complex

$$0 \to \mathbb{Z} \xrightarrow{2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0,$$

and f to be the identity $\mathrm{id}_X\colon X \to X$. Let $g\colon X \to 0$ be the zero map. Then $g \in$ Qis, and $gf = 0$, but f is not homotopic to zero, as one sees easily.

2. If f satisfies ($*$), then f induces the zero map on cohomology, but the converse is false. For example take $f\colon X \to Y$ as follows:

$$\begin{tikzcd} \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & \mathbb{Z} \arrow[r, "2"] \arrow[d, "1"] & \mathbb{Z} \arrow[r] \arrow[d, "2"] & 0 \arrow[r] \arrow[d] & \cdots \\ \cdots \arrow[r] & 0 \arrow[r] & \mathbb{Z} \arrow[r, "1"] & \mathbb{Z}/3\mathbb{Z} \arrow[r] & 0 \arrow[r] & \cdots \end{tikzcd}$$

Now f induces the zero map on cohomology, but there does not exist $t\colon X' \to X$ in Qis such that $ft \sim 0$. (The reader can supply the proof as follows: take a cycle $x \in X'$ such that $t(x)$ generates the single cohomology group $\mathbb{Z}/2\mathbb{Z}$ of X. If k is a homotopy operator for ft, show that $2k(x) = 1$, which is impossible.)

So for two maps $f, g\colon X \to Y$ of complexes, we see that the following implications are all strict:

$$\begin{aligned} f = g \implies & f \text{ homotopic to } g \\ \implies & f \text{ and } g \text{ give the same morphism in } \mathrm{D}(\mathcal{A}) \\ \implies & f \text{ and } g \text{ give the same map on cohomology.} \end{aligned}$$

Proposition 8. *The functor $\mathcal{A} \to \mathrm{D}(\mathcal{A})$ which sends each object X of $\mathcal{A}$ into the complex consisting of X in degree 0, and 0 elsewhere, gives an equivalence of the category $\mathcal{A}$ with the full subcategory of $\mathrm{D}(\mathcal{A})$ consisting of those complexes $X^\bullet$ such that $H^i(X^\bullet) = 0$ for $i \neq 0$.*

We now prove three lemmas, which give an alternative description of $\mathrm{D}^+(\mathcal{A})$ when $\mathcal{A}$ has enough injectives.

Lemma 1. *Let $\mathcal{A}$ be abelian, $f\colon Z^\bullet \to I^\bullet$ a morphism of complexes. Assume:*

(1) $Z^\bullet$ is acyclic;

(2) Each I^p is injective in $\mathcal{A}$;

(3) $I^\bullet$ is bounded below.

Then f is homotopic to zero.

Proof. Well known (and easy). □

Lemma 2. *Let $\mathcal{A}$ be abelian, $s\colon I^\bullet \to Y^\bullet$ a morphism of complexes. Assume:*

(1) s induces an isomorphism on cohomology;

(2) Each I^p is injective;

(3) $I^\bullet$ is bounded below.

Then s admits a homotopy inverse.

Proof. Suppose given $s\colon I^\bullet \to Y^\bullet$ as above. Let $Z^\bullet = T(I^\bullet) \oplus Y^\bullet$ be the mapping cone of s. Then $Z^\bullet$ is acyclic, and so the triangular morphism $v\colon Z^\bullet \to T(I^\bullet)$ satisfies the conditions of Lemma 1.4.4, and so is homotopic to zero. Let us call the homotopy operator

$$(k, t)\colon T(I^\bullet) \oplus Y^\bullet \longrightarrow I^\bullet.$$

Then we have the equation

$$v = (\mathrm{id}_{I^\bullet}, 0) = (k, t)d_Z + d_{I^\bullet}(k, t).$$

Separating the components, we find

$$\mathrm{id}_{T(I^\bullet)} = dk + kd + ts$$

and

$$dt - td = 0.$$

Thus $t\colon Y^\bullet \to I^\bullet$ is a morphism of complexes, and $\mathrm{id}_{I^\bullet}$ is homotopic to ts, so t is a homotopy inverse of s. □

Lemma 3. *Let $\mathcal{A}$ be an abelian category.*

1) Let $P \subseteq \operatorname{Ob}\mathcal{A}$ and assume:

(i) Every object of $\mathcal{A}$ admits an injection into an element of P.

Then every complex $X^\bullet \in \mathrm{K}^+(\mathcal{A})$ admits a quasi-isomorphism into a bounded-below complex $I^\bullet$ of objects of P.

2) Assume further that P satisfies:

(ii) If $0 \to X \to Y \to Z \to 0$ is exact and $X \in P$, then $Y \in P \iff Z \in P$;

(iii) There exists a positive integer n such that for any exact sequence

$$X^0 \to X^1 \to \cdots \to X^{n-1} \to X^n \to 0,$$

if $X^0, \ldots, X^{n-1} \in P$ then $X^n \in P$.

Then every $X^\bullet \in \mathrm{K}(\mathcal{A})$ admits a quasi-isomorphism into a complex $I^\bullet$ of objects of P.

3) Let $\mathcal{A}'$ be a thick subcategory of $\mathcal{A}$, and suppose $\mathcal{A}'$ has enough $\mathcal{A}$-injectives. Then every $X^\bullet \in \mathrm{K}^+_{\mathcal{A}'}(\mathcal{A})$ admits a quasi-isomorphism into a bounded-below complex $I^\bullet$ of $\mathcal{A}$-injective objects of $\mathcal{A}'$.

Proof of 1). We may assume $X^p = 0$ for $p < 0$. Embed $X^0 \hookrightarrow I^0$ with $I^0 \in P$. Having defined $I^0, I^1, \ldots, I^p$, choose $I^{p+1} \in P$ containing

$$I^p \big/ \operatorname{im} I^{p-1} \;\oplus\; X^{p+1},$$

and define differentials $I^p \to I^{p+1}$ and morphisms $X^{p+1} \to I^{p+1}$ canonically. One verifies $X^\bullet \to I^\bullet$ is a quasi-isomorphism, Note that in this construction all the maps $X^p \to I^p$ is injective. □

Proof of 2). Let $X^\bullet$ be any complex. For a fixed integer i_0, by 1), we can find a quasi- isomorphism of the truncated complex

$$0 \to 0 \to X^{i_0} \to X^{i_0+1} \to \cdots$$

into a complex $I^\bullet$ of objects of P, with each $X^i \to I^i$ injective. Define $X_0^\bullet$ by

$$\cdots \to X^{i_0-2} \to X^{i_0-1} \to I^{i_0} \to I^{i_0+1} \to \cdots .$$

Then $X^\bullet \to X_0^\bullet$ is a quasi-isomorphism, $X_0^i \in P$ for $i \geq i_0$, and each $X^i \to X_0^i$ is injective.

Suppose given a complex $X_1^\bullet$ with $X_1^i \in P$ for $i \geq i_1$, and let $i_2 < i_1$. Then we will construct a quasi-isomorphism $X_1^\bullet \to X_2^\bullet$ such that $X_2^i \in P$ for $i \geq i_2$, and $X_1^i = X_2^i$ for $i \geq i_1 + n$. (Here n is the integer of condition (iii) above.) Indeed, by the first step above, we can find a quasi-isomorphism $X_1^\bullet \to X'^\bullet$ such that $X'^i \in P$ for $i \geq i_2$, and each $X_1^i \to X'^i$ is injective. Let $Y^i = \operatorname{coker}(X_1^i \to X'^i)$. Then $Y^\bullet$ is an acyclic complex, and $Y^i \in P$ for $i \geq i_1$, by condition (ii) above. Hence $B^i(Y^\bullet) \in P$ for $i \geq i_1 + n$, by condition (iii). Now define $X_2^\bullet$ by

$$X_2^i = \begin{cases} X'^i & \text{for } i < i_1 + n, \\ B^i(X'^\bullet) \oplus_{X_1^{i-1}} X_1^i & \text{for } i = i_1 + n, \\ X_1^i & \text{for } i > i_1 + n. \end{cases}$$

One sees easily that $X_1^\bullet \to X_2^\bullet$ is a quasi-isomorphism. It follows from (ii) and the exact sequence

$$0 \longrightarrow X_1^i \longrightarrow B^i(X'^\bullet) \oplus_{X_1^{i-1}} X_1^i \longrightarrow B^i(Y^\bullet) \longrightarrow 0$$

that the middle term is in P, for $i \geq i_1 + n$, so $X_2^\bullet$ is as required.

Now, given a complex $X^\bullet \in \mathrm{K}(\mathcal{A})$, choose a sequence of integers $i_0 > i_1 > i_2 > \cdots$ tending to $-\infty$. Choose $X_0^\bullet$ for i_0 as in the first step, and choose $X_1^\bullet, X_2^\bullet, \ldots$ for $i_1, i_2, \ldots$ successively as in the second step. Then we have quasi-isomorphisms

$$X^\bullet \longrightarrow X_0^\bullet \longrightarrow X_1^\bullet \longrightarrow X_2^\bullet \longrightarrow \cdots ,$$

and for each i, the sequence $X^i \to X_0^i \to X_1^i \to \cdots$ is eventually constant, and eventually in P. Hence $I^\bullet = \varinjlim X_r^\bullet$ is the required complex of objects of P. □

Proof of 3). We may assume $X^i = 0$ for $i < 0$. Embed $H^0(X^\bullet)$ in I^0, an $\mathcal{A}$-injective of $\mathcal{A}'$, which is possible since $H^0(X^\bullet) \in \operatorname{Ob} \mathcal{A}'$. Extend this to a map $f^0 \colon X^0 \to I^0$, which is possible since I^0 is $\mathcal{A}$-injective.

Having defined $I^0, I^1, \ldots, I^p$ and $f^i\colon X^i \to I^i$ for $i = 0, \ldots, p$, choose I^{p+1} to be an $\mathcal{A}$-injective of $\mathcal{A}'$ containing

$$I^p \big/ \operatorname{im} I^{p-1} \oplus_{X^p} Z^{p+1}(X^\bullet). \tag{*}$$

We must check that this latter is in $\mathcal{A}'$. Indeed, $\mathcal{A}'$ is a thick subcategory, so it is sufficient to note that $I^p/\operatorname{im} I^{p-1} \in \mathcal{A}'$ (one shows by induction that $B^i(I^\bullet)$ and $Z^i(I^\bullet)$ are in $\mathcal{A}'$ for all i), and that the quotient of $(*)$ by $I^p/\operatorname{im} I^{p-1}$ is $H^{p+1}(X^\bullet)$, which is in $\mathcal{A}'$ by hypothesis. Extend the natural map $Z^{p+1}(X^\bullet) \to I^{p+1}$ to a map $f^{p+1}\colon X^{p+1} \to I^{p+1}$. One checks easily that the resulting map $f\colon X^\bullet \to I^\bullet$ is a quasi-isomorphism, as required. $\square$

Proposition 9. *Let $\mathcal{A}$ be abelian, and let $\mathcal{I}$ be the additive subcategory of injective objects of $\mathcal{A}$. The natural functor*

$$\alpha\colon\ \mathrm{K}^+(\mathcal{I}) \to \mathrm{D}^+(\mathcal{A})$$

is fully faithful. (Note that the results of section 3 carry over to additive subcategories of abelian categories.) Furthermore, if $\mathcal{A}$ has enough injectives (i.e., if every object of $\mathcal{A}$ admits an injection into an injective object), then α is an equivalence of categories.

Proof. We note that $\mathrm{Qis} \cap \mathrm{K}^+(\mathcal{I})$ is a multiplicative system in $\mathrm{K}^+(\mathcal{I})$, by Proposition 1.4.2, and we observe by Lemma 1.4.5 that condition (ii) of Proposition 1.3.3 is satisfied for $\mathrm{K}^+(\mathcal{I}) \subseteq \mathrm{K}^+(\mathcal{A})$ and Qis. Hence the natural functor

$$\mathrm{D}^+(\mathcal{I}) \longrightarrow \mathrm{D}^+(\mathcal{A})$$

is fully faithful. But on the other hand, Lemma 1.4.5 shows also that every quasi-isomorphism in $\mathrm{K}^+(\mathcal{I})$ is an isomorphism, hence $\mathrm{K}^+(\mathcal{I}) = \mathrm{D}^+(\mathcal{I})$.

Now if $\mathcal{A}$ has enough injectives, applying Lemma 1.4.6 in the case $\mathcal{A} = \mathcal{B}$ and $P =$ the injective objects, we see that every object of $\mathrm{D}^+(\mathcal{A})$ is isomorphic to an object in $\mathrm{K}^+(\mathcal{I})$, so α is an equivalence of categories. $\square$

Proposition 10. *Let $\mathcal{A}$ be abelian, $\mathcal{A}' \subseteq \mathcal{A}$ a thick abelian subcategory. Assume $\mathcal{A}'$ has enough $\mathcal{A}$-injectives, i.e every object of A' can be injected into an A-injective object of A'. Then the natural functor*

$$\mathrm{D}^+(\mathcal{A}') \to \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$$

is an equivalence of categories.

Proof. We apply Proposition 1.3.3 to the inclusion $\mathrm{K}^+(\mathcal{A}') \hookrightarrow \mathrm{K}^+(\mathcal{A})$. Clearly Qis is a multiplicative system in each. If $X^\bullet \to Y^\bullet$ is a quasi-isomorphism with $X^\bullet \in \mathrm{K}^+(\mathcal{A}')$, then $Y^\bullet$ has cohomology in $\mathcal{A}'$, and so by Lemma 1.4.6 admits a quasi-isomorphism $Y^\bullet \to I^\bullet$ with $I^\bullet \in \mathrm{K}^+(\mathcal{A}')$, each I^p injective in $\mathcal{A}$. Hence condition (ii) is satisfied, and so the functor

$$\mathrm{D}^+(\mathcal{A}') \longrightarrow \mathrm{D}^+(\mathcal{A})$$

is fully faithful. The same Lemma 1.4.6 also shows that the image is $\mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$. □

Exercise. We leave to the reader the analogous statements of the last five results in the case of projective objects of $\mathcal{A}$ and $\mathrm{D}^-(\mathcal{A})$.

2.6 Derived functors

We will treat only the question of *right derived covariant functors*, leaving the reader to make the obvious modifications for left derived covariant functors, and right and left derived contravariant functors.

Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and let $F\colon \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$ be a ∂-functor (cf. §1). Such is the case, for example, if we are given an additive functor $F\colon \mathcal{A} \to \mathcal{B}$. It extends to $\mathrm{K}(\mathcal{A})$.

In general, F will not take quasi-isomorphisms into quasi-isomorphisms – to say that it does is to say that it localizes and gives rise to a functor from $\mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$. That will be the case, for example, if F is an exact functor.

Thus we are led to ask if there is a functor from $\mathrm{D}(\mathcal{A})$ to $\mathrm{D}(\mathcal{B})$ which is at least close to F, and this gives rise to the notion of derived functor below. Before giving the definition we generalize slightly.

Definition 9. *Let $\mathcal{A}$ be abelian, and let $\mathrm{K}^*(\mathcal{A}) \subseteq \mathrm{K}(\mathcal{A})$ be a triangulated subcategory. By Proposition 1.4.2, $\mathrm{K}^*(\mathcal{A}) \cap \mathrm{Qis}$ is a multiplicative system in $\mathrm{K}^*(\mathcal{A})$.*

We say $\mathrm{K}^*(\mathcal{A})$ is a localizing subcategory *of $\mathrm{K}(\mathcal{A})$ if the natural functor*

$$\mathrm{K}^*(\mathcal{A})_{\mathrm{K}^*(\mathcal{A})\cap\mathrm{Qis}} \;\to\; \mathrm{K}(\mathcal{A})_{\mathrm{Qis}} = \mathrm{D}(\mathcal{A})$$

is fully faithful. In this case we write $\mathrm{D}^*(\mathcal{A})$ for the $\mathrm{K}^*(\mathcal{A})_{\mathrm{K}^*(\mathcal{A})\cap\mathrm{Qis}}$.

Examples.

1. The intersection of any family of localizing subcategories is localizing.

2. $\mathrm{K}^+(\mathcal{A}), \mathrm{K}^-(\mathcal{A}), \mathrm{K}^b(\mathcal{A})$ are localizing subcategories of $\mathrm{K}(\mathcal{A})$ (cf. §4).

3. If $\mathcal{A}' \subseteq \mathcal{A}$ is a thick subcategory, then $\mathrm{K}_{\mathcal{A}'}(\mathcal{A}), \mathrm{K}^+_{\mathcal{A}'}(\mathcal{A}), \mathrm{K}^-_{\mathcal{A}'}(\mathcal{A}), \mathrm{K}^b_{\mathcal{A}'}(\mathcal{A})$ are localizing subcategories of $\mathrm{K}(\mathcal{A})$ (cf. §4).

4. The complexes of finite injective dimension form a localizing subcategory $\mathrm{K}^+(\mathcal{A})_{\mathrm{fid}} \subseteq \mathrm{K}(\mathcal{A})$ (cf. Corollary 1.7.7).

Definition 10. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $\mathrm{K}^*(\mathcal{A}) \subseteq \mathrm{K}(\mathcal{A})$ a localizing subcategory, and*

$$F\colon\ \mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$$

a ∂-functor.

Denote by Q the localization functor $\mathrm{K}^*(\mathcal{A}) \to \mathrm{D}^*(\mathcal{A})$ (resp. $\mathrm{K}(\mathcal{B}) \to \mathrm{D}(\mathcal{B})$). The right derived functor *of F is a ∂-functor*

$$\underline{\underline{\mathrm{R}}}^*F\colon\ \mathrm{D}^*(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$$

together with a functor morphism

$$\xi\colon Q \circ F \ \to\ \underline{\underline{\mathrm{R}}}^*F \circ Q$$

of functors $\mathrm{K}^*(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$, satisfying the following universal property:

If $G\colon\ \mathrm{D}^*(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ is a ∂-functor and any functor morphism

$$\zeta\colon Q \circ F \ \to\ G \circ Q,$$

there exists a unique functor morphism

$$\eta\colon \underline{\underline{\mathrm{R}}}^*F \to G$$

such that

$$\zeta = (\eta \circ Q) \circ \xi.$$

Remarks.

1. If $\underline{\underline{\mathrm{R}}}^*F$ exists, it is unique up to unique isomorphism of functors.

2. For standard localizing subcategories $\mathrm{K}^+(\mathcal{A}), \mathrm{K}^-(\mathcal{A}), \mathrm{K}_{\mathcal{A}'}(\mathcal{A})$, we write

$$\underline{\underline{\mathrm{R}}}^+F, \quad \underline{\underline{\mathrm{R}}}^-F, \quad \underline{\underline{\mathrm{R}}}_{\mathcal{A}'}F$$

respectively; when no confusion arises, we write simply $\underline{\underline{\mathrm{R}}}F$.

3. We will write $\mathrm{R}^i F$ for $H^i(\underline{\underline{\mathrm{R}}}F)$, and it will follow from the results below that if F comes from a left-exact functor $F : A \longrightarrow B$, and if A has enough injectives, then these are the usual derived functors of F.

4. If $\varphi : F \longrightarrow G$ is a morphism of functors from $K^*(A)$ to $K(B)$, and if $\underline{\underline{\mathrm{R}}}F$ and $\underline{\underline{\mathrm{R}}}G$ both exist, then there is a unique morphism of functors

$$\underline{\underline{\mathrm{R}}}\varphi : \underline{\underline{\mathrm{R}}}F \longrightarrow \underline{\underline{\mathrm{R}}}G$$

compatible with the ξ's. This follows immediately from the definition.

5. Let $\mathrm{K}^{**}(\mathcal{A}) \subseteq \mathrm{K}^*(\mathcal{A})$ be localizing subcategories, if $F\colon \mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$ is a ∂-functor. and if $\underline{\underline{\mathrm{R}}}^*F$ and $\underline{\underline{\mathrm{R}}}^{**}(F|_{\mathrm{K}^{**}(\mathcal{A})})$ exist, there is a natural functor morphism

$$\underline{\underline{\mathrm{R}}}^{**}\big(F|_{\mathrm{K}^{**}(\mathcal{A})}\big) \;\to\; \underline{\underline{\mathrm{R}}}^*F\big|_{\mathrm{D}^{**}(\mathcal{A})}.$$

We do not know if this is an isomorphism in general, but it holds in all our intended applications (cf. Corollary 1.5.3).

Theorem 1 (Existence of Derived Functors)**.** *Let $\mathcal{A}, \mathcal{B}, \mathrm{K}^*(\mathcal{A}), F$ be as above. Suppose there exists a triangulated subcategory $\mathcal{L} \subseteq \mathrm{K}^*(\mathcal{A})$ such that:*

(1) Every object of $\mathrm{K}^*(\mathcal{A})$ admits a quasi-isomorphism into some object of $\mathcal{L}$;

(2) For every acyclic complex $I^\bullet \in \mathrm{Ob}\,\mathcal{L}$ (i.e. $H^i(I^\bullet) = 0$ for all i), the complex $F(I^\bullet)$ is also acyclic.

Then the right derived functor $(\underline{\underline{\mathrm{R}}}^*F, \xi)$ exists. Furthermore, for any $I^\bullet \in \mathrm{Ob}\,\mathcal{L}$, the map

$$\xi(I^\bullet)\colon Q \circ F(I^\bullet) \to \underline{\underline{\mathrm{R}}}^*F \circ Q(I^\bullet)$$

is an isomorphism in $\mathrm{D}(\mathcal{B})$.

Proof. First, the restriction $F|_{\mathcal{L}}$ sends quasi-isomorphisms to quasi-isomorphisms. Indeed, let $s\colon I_1^\bullet \to I_2^\bullet$ be a quasi-isomorphism in $\mathcal{L}$; complete s to a triangle in $\mathrm{K}^*(\mathcal{A})$, and let $J^\bullet$ be the third side of that triangle. Then $J^\bullet$ is acyclic, so $F(J^\bullet)$ is acyclic. Hence $F(s)$ is a quasi-isomorphism.

Thus F passes to the quotient to give a functor

$$\overline{F}\colon \mathcal{L}_{\mathrm{Qis}} \to \mathrm{D}(\mathcal{B})$$

satisfying $\overline{F} \circ Q = Q \circ F$ on $\mathcal{L}$ (we denote as usual by Q the morphism from a category to its localization).

Second, by the condition of Proposition 1.3.3, (ii) are satisfied for $\mathcal{L}, \mathrm{K}^*(\mathcal{A}), \mathrm{Qis}$, and so the natural functor

$$T\colon \mathcal{L}_{\mathrm{Qis}} \to \mathrm{D}^*(\mathcal{A})$$

is an equivalence of categories, using 1) above. Let U be a quasi-inverse of T, i.e. a functor $U\colon\ \mathrm{D}^*(\mathcal{A}) \to \mathcal{L}_{\mathrm{Qis}}$ with functorial isomorphisms

$$\alpha\colon\ \mathrm{id}_{\mathcal{L}_{\mathrm{Qis}}} \to U \circ T, \qquad \beta\colon\ \mathrm{id}_{\mathrm{D}^*(\mathcal{A})} \to T \circ U.$$

Define

$$\underline{\underline{\mathrm{R}}}^* F := \overline{F} \circ U.$$

We now define the functor morphism

$$\xi\colon Q \circ F \to \underline{\underline{\mathrm{R}}}^* F \circ Q = \overline{F} \circ U \circ Q$$

as follows. Given $X^\bullet \in \mathrm{Ob}\,\mathrm{K}^*(\mathcal{A})$, choose $I^\bullet \in \mathrm{Ob}\,\mathcal{L}$ such that $Q(I^\bullet) \cong U \circ Q(X^\bullet)$ in $\mathrm{D}^*(\mathcal{A})$.

We have an isomorphism in $\mathrm{D}^*(\mathcal{A})$,

$$\beta\big(Q(X^\bullet)\big)\colon Q(X^\bullet) \xrightarrow{\sim} T \circ U\big(Q(X^\bullet)\big) = T\big(Q(I^\bullet)\big).$$

This isomorphism may be represented by a diagram in $\mathrm{K}^*(\mathcal{A})$

$$X^\bullet \xrightarrow{t} Y^\bullet \xleftarrow{s} I^\bullet \qquad\qquad (*)$$

with $s, t \in \mathrm{Qis}$ and $Y^\bullet \in \mathrm{Ob}\,\mathrm{K}^*(\mathcal{A})$. Furthermore by hypothesis (1), we may assume $Y^\bullet \in \mathrm{Ob}\,\mathcal{L}$. Applying the functor F, we obtain a diagram in $\mathrm{K}(\mathcal{B})$:

$$F(X^\bullet) \xrightarrow{F(t)} F(Y^\bullet) \xleftarrow{F(s)} F(I^\bullet).$$

As observed previously, $F(s)$ is also a quasi-isomorphisms.

This induces a morphism in $\mathrm{D}(\mathcal{B})$:

$$\xi(X^\bullet)\colon Q \circ F(X^\bullet) \to Q \circ F(I^\bullet) = \overline{F} \circ Q(I^\bullet) = \overline{F} \circ U \circ Q(X^\bullet) = \underline{\underline{\mathrm{R}}}^* F \circ Q(X^\bullet).$$

One can now check without difficulty that $\xi(X^\bullet)$ does not depend on the choice of the diagram (*) above, that ξ defines a functor morphism $Q \circ F \to \underline{\underline{\mathrm{R}}}^* F \circ Q$, and that the pair $(\underline{\underline{\mathrm{R}}}^* F, \xi)$ satisfies the universal property of the right derived functor.

If $X^\bullet \in \mathrm{Ob}\,\mathcal{L}$, then $F(t)$ is also a quasi-isomorphism in the construction above, so $\xi(X^\bullet)$ is an isomorphism in $\mathrm{D}(\mathcal{B})$, as required. $\square$

Proposition 11. *Let $\mathcal{A}, \mathcal{B}, \mathrm{K}^*(\mathcal{A}), F$ be as above, and let $\mathrm{K}^{**}(\mathcal{A}) \subseteq \mathrm{K}^*(\mathcal{A})$ be another localizing subcategory of $\mathrm{K}(\mathcal{A})$. Suppose there exists a triangulated subcategory $\mathcal{L} \subseteq \mathrm{K}^*(\mathcal{A})$ satisfying hypotheses (1),(2) of Theorem 1.5.1, and such that $\mathcal{L} \cap \mathrm{K}^{**}(\mathcal{A})$ satisfies hypothesis (1) for $\mathrm{K}^{**}(\mathcal{A})$. Then $\underline{\underline{\mathrm{R}}}^* F$ and $\underline{\underline{\mathrm{R}}}^{**}\big(F|_{\mathrm{K}^{**}(\mathcal{A})}\big)$ both exist, and the natural morphism*

$$\underline{\underline{\mathrm{R}}}^{**}\big(F|_{\mathrm{K}^{**}(\mathcal{A})}\big) \;\to\; \underline{\underline{\mathrm{R}}}^* F\big|_{\mathrm{D}^{**}(\mathcal{A})}$$

is an isomorphism.

Proof. Existence follows immediately from Theorem 1.5.1. For the isomorphism, any object of $\mathrm{D}^{**}(\mathcal{A})$ is isomorphic to one coming from an object of $\mathcal{L}$, we may assume that $X'' = Q(I^\bullet)$ for some $I^\bullet \in \operatorname{Ob}\mathcal{L} \cap \mathrm{K}^{**}(\mathcal{A})$. The claim then follows from the final assertion of Theorem 1.5.1. $\square$

Corollary 1. *α) Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F\colon \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}(\mathcal{B})$ a ∂-functor (e.g., induced by an additive functor $F_0\colon \mathcal{A} \to \mathcal{B}$). If $\mathcal{A}$ has enough injectives, then $\underline{\underline{\mathrm{R}}}^+ F$ exists.*

β) Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F\colon \mathcal{A} \to \mathcal{B}$ additive. Suppose there exists an subset $P \subseteq \operatorname{Ob}\mathcal{A}$ satisfying (i),(ii) of Lemma 1.4.6, and:

(iv) F sends short exact sequences in $\mathcal{A}$ to short exact sequences in $\mathcal{B}$.

Then $\underline{\underline{\mathrm{R}}}^+ F$ exists (we also denote by F its extension to a ∂-functor $\mathrm{K}^+(\mathcal{A}) \to \mathrm{K}^+(\mathcal{B})$).

γ) Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $F\colon \mathcal{A} \to \mathcal{B}$ additive. Assume:

(a) The hypotheses of β hold;

(b) F has finite cohomological dimension: there exists $n > 0$ such that $\mathrm{R}^i F(Y) = 0$ for all $Y \in \operatorname{Ob}\mathcal{A}$ and all $i > n$.

Then $\underline{\underline{\mathrm{R}}} F$ exists on all of $\mathrm{D}(\mathcal{A})$, and its restriction to $\mathrm{D}^+(\mathcal{A})$ coincides with $\underline{\underline{\mathrm{R}}}^+ F$.

Remark. Case α is a special case of β: if $\mathcal{A}$ has enough injectives, the class of injective objects satisfies (i),(ii),(iv) for any additive F.

Proof. α) Let $\mathcal{L} \subseteq \mathrm{K}^+(\mathcal{A})$ be the triangulated subcategory of bounded-below injective complexes. By Lemma 1.4.6(1), every object of $\mathrm{K}^+(\mathcal{A})$ admits a quasi-isomorphism into an object of $\mathcal{L}$. By Lemma 1.4.5, every quasi-isomorphism in $\mathcal{L}$ is an isomorphism, so acyclic objects of $\mathcal{L}$ are zero in $\mathrm{K}(\mathcal{A})$. Thus condition (2) of Theorem 1.5.1 holds for any ∂-functor F, so $\underline{\underline{\mathrm{R}}}^+ F$ exists.

β) Take $\mathcal{L} \subseteq \mathrm{K}^+(\mathcal{A})$ to be the triangulated subcategory of complexes of objects of P (Note $\mathcal{L}$ is a triangulated subcategory because P is stable under direct sums by

(ii).). Condition (1) holds of the theorem is satisfied as above. For (2), let $Z^\bullet \in \mathcal{L}$ be acyclic; using (ii) repeatedly, all kernels $\ker d_Z^p$ lie in P. By (iv), F preserves exactness, so $F(Z^\bullet)$ is acyclic.

γ) Let P' be the collection of F-acyclic objects of $\mathcal{A}$ ($\mathrm{R}^i\, F(X) = 0, \forall i > 0$). Then P' satisfies (i),(ii),(iii) of Lemma 1.4.6. Let $\mathcal{L} \subseteq \mathrm{K}(\mathcal{A})$ be complexes of objects of P'. By Lemma 1.4.6 and the argument for β, the hypotheses of Theorem 1.5.1 are satisfied, so $\underline{\underline{\mathrm{R}}}F$ exists.

By Proposition 1.5.2, the restriction of $\underline{\underline{\mathrm{R}}}F$ to $\mathrm{D}^+(\mathcal{A})$ is isomorphic to $\underline{\underline{\mathrm{R}}}^+F$. □

Proposition 12. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories, $\mathrm{K}^*(\mathcal{A}) \subseteq \mathrm{K}(\mathcal{A})$, $\mathrm{K}^\dagger(\mathcal{B}) \subseteq \mathrm{K}(\mathcal{B})$ localizing subcategories, and let*

$$F\colon\ \mathrm{K}^*(\mathcal{A}) \to \mathrm{K}(\mathcal{B}), \qquad G\colon\ \mathrm{K}^\dagger(\mathcal{B}) \to \mathrm{K}(\mathcal{C})$$

be ∂-functors.

a) *Assume $F(\mathrm{K}^*(\mathcal{A})) \subseteq \mathrm{K}^\dagger(\mathcal{B})$, that $\underline{\underline{\mathrm{R}}}^*F, \underline{\underline{\mathrm{R}}}^\dagger G, \underline{\underline{\mathrm{R}}}^*(G \circ F)$ exist, and $\underline{\underline{\mathrm{R}}}^*F(\mathrm{D}^*(\mathcal{A})) \subseteq \mathrm{D}^\dagger(\mathcal{B})$. Then there exists a unique functor morphism*

$$\zeta\colon\ \underline{\underline{\mathrm{R}}}^*(G \circ F) \to \underline{\underline{\mathrm{R}}}^\dagger G \circ \underline{\underline{\mathrm{R}}}^* F$$

so the the diagram is commutative.

$$\begin{tikzcd}
Q \cdot G \cdot F \arrow[r, "\xi_G"] \arrow[d, "\xi_{G\cdot F}"'] & \underline{\underline{\mathrm{R}}}^+ G \cdot Q \cdot F \arrow[d, "\xi_F"] \\
\underline{\underline{\mathrm{R}}}^*(G \cdot F) \cdot Q \arrow[r, "\zeta\cdot Q"] & \underline{\underline{\mathrm{R}}}^+ G \cdot \underline{\underline{\mathrm{R}}}^* F \cdot Q
\end{tikzcd}$$

b) *Suppose there exist triangulated subcategories $\mathcal{L} \subseteq \mathrm{K}^*(\mathcal{A})$, $\mathrm{M} \subseteq \mathrm{K}^\dagger(\mathcal{B})$ satisfying Theorem 1.5.1's hypotheses for F, G respectively, and $F(\mathcal{L}) \subseteq \mathrm{M}$. Then the hypotheses of (a) hold, and the canonical morphism ζ is an isomorphism.*

Proof. Straightforward □

Remarks.

1. If F, G, H are three consecutive functors, then there is a commutative diagram of ζs (provided they all exist):

$$\begin{tikzcd}
\underline{\underline{\mathrm{R}}}(H \cdot G \cdot F) \arrow[r, "\xi_{G\cdot F,H}"] \arrow[d, "\zeta_{F,H\cdot G}"'] & \underline{\underline{\mathrm{R}}}H \cdot \underline{\underline{\mathrm{R}}}(G \cdot F) \arrow[d, "\zeta_{F,G}"] \\
\underline{\underline{\mathrm{R}}}(H \cdot G) \cdot \underline{\underline{\mathrm{R}}}F \arrow[r] & \underline{\underline{\mathrm{R}}}H \cdot \underline{\underline{\mathrm{R}}}G \cdot \underline{\underline{\mathrm{R}}}F
\end{tikzcd}$$

2. This proposition shows the convenience of derived functors in the context of derived categories. What used to be a spectral sequence becomes now simply a composition of functors. (And of course one can recover the old spectral sequence from this proposition by taking cohomology and using the spectral sequence of a double complex.)

Corollary 2. *Left to the reader: Illustrate the Proposition in the style of Corollary 5.3.*

Proposition 13. *Let $\mathcal{A}$ be an abelian category, let $\mathcal{A}'$ be a thick abelian subcategory, let $\mathcal{B}$ be another abelian category, and let $F\colon \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}^+(\mathcal{B})$ be a ∂-functor. Suppose that $\underline{\underline{\mathrm{R}}}^+F$ and $\underline{\underline{\mathrm{R}}}^+\big(F|_{\mathcal{A}'}\big)$ both exist. Then there is a natural morphism*

$$\zeta\colon \underline{\underline{\mathrm{R}}}^+\big(F|_{\mathcal{A}'}\big) \longrightarrow \varphi \circ \underline{\underline{\mathrm{R}}}^+F$$

of functors from $\mathrm{D}^+(\mathcal{A}')$ to $\mathrm{D}(\mathcal{B})$, where

$$\varphi\colon\ \mathrm{D}^+(\mathcal{A}') \longrightarrow \mathrm{D}^+(\mathcal{A})$$

is the natural functor. If furthermore $\mathcal{A}'$ has enough $\mathcal{A}$-injectives, and $\mathcal{A}$ has enough injectives, then ζ is an isomorphism.

Proof. The existence of ζ follows from the definition of the derived functor. If $\mathcal{A}'$ has enough $\mathcal{A}$-injectives, and $\mathcal{A}$ has enough injectives, then we can use $\mathcal{A}$-injectives to calculate both functors above, by Corollary 1.5.3 α, and so ζ is an isomorphism. (Recall by Proposition 1.4.8 that φ is an equivalence of categories in that case.) □

2.7 Examples. Ext and $\underline{\underline{\mathrm{R}}}$ Hom

Definition 11. *Let $\mathcal{A}$ be an abelian category, and let $X^\bullet, Y^\bullet \in \mathrm{D}(\mathcal{A})$. Define the* ith hyperext groups *by*

$$\mathrm{Ext}^i\left(X^\bullet, Y^\bullet\right) = \mathrm{Hom}_{\mathrm{D}(\mathcal{A})}\left(X^\bullet, T^i(Y^\bullet)\right).$$

Remarks.

1. If $X^\bullet, Y^\bullet \in \mathrm{D}^+(\mathcal{A})$, then Ext^i computed in $\mathrm{D}^+(\mathcal{A})$ agrees with the above, since $\mathrm{D}^+(\mathcal{A})$ is a full subcategory of $\mathrm{D}(\mathcal{A})$.

2. This gives in particular a definition of $\mathrm{Ext}^i(X, Y)$ for any $X, Y \in \mathcal{A}$. We shall see that if $\mathcal{A}$ has enough injectives, this coincides with the classical definition of Ext groups.

Proposition 14. *Let*

$$0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$$

be a short exact sequence of complexes over $\mathcal{A}$, and let $V^\bullet$ be another complex. Then we have long exact sequences

$$\cdots \to \mathrm{Ext}^i(V^\bullet, X^\bullet) \to \mathrm{Ext}^i(V^\bullet, Y^\bullet) \to \mathrm{Ext}^i(V^\bullet, Z^\bullet) \to \mathrm{Ext}^{i+1}(V^\bullet, X^\bullet) \to \cdots$$

and

$$\cdots \to \mathrm{Ext}^i(Z^\bullet, V^\bullet) \to \mathrm{Ext}^i(Y^\bullet, V^\bullet) \to \mathrm{Ext}^i(X^\bullet, V^\bullet) \to \mathrm{Ext}^{i+1}(Z^\bullet, V^\bullet) \to \cdots$$

Proof. Let $W^\bullet$ be the third side of a triangle on $X^\bullet \to Y^\bullet$.

$$\begin{tikzcd}
 & & W^\bullet \arrow[dl] \arrow[dr, dashed, "g"] & & \\
0 \arrow[r] & X^\bullet \arrow[r] & Y^\bullet \arrow[r] \arrow[u] & Z^\bullet \arrow[r] & 0
\end{tikzcd}$$

Then by Proposition 1.1(b), there is a morphism of complexes $g \colon W^\bullet \to Z^\bullet$. Using the long exact sequence of cohomology of the short exact sequence, and of the triangle, and using the five-lemma, we see that g is a quasi-isomorphism, i.e., an isomorphism in $\mathrm{D}(\mathcal{A})$. Hence we may replace $Z^\bullet$ by $W^\bullet$ in the conclusion, which then follows from the same proposition. □

Remark. It follows from the proof that any short exact sequence of complexes

$$0 \to X^\bullet \to Y^\bullet \to Z^\bullet \to 0$$

determines a morphism $Z^\bullet \to T(X^\bullet)$ in $\mathrm{D}(\mathcal{A})$, making $X^\bullet, Y^\bullet, Z^\bullet$ into a triangle.

Now we will define a functor whose cohomology gives the Ext groups.

Definition 12. *For complexes $X^\bullet, Y^\bullet$ over $\mathcal{A}$, define the complex* $\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)$ *by*

$$\mathrm{Hom}^n\left(X^\bullet, Y^\bullet\right) = \prod_{p\in\mathbb{Z}} \mathrm{Hom}_{\mathcal{A}}\left(X^p, Y^{p+n}\right),$$

with differential

$$d^n = \prod_{p\in\mathbb{Z}} \left(d_X^{p-1} + (-1)^{n+1} d_Y^{p+n}\right).$$

Notice under this definition the n-cycles of $\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)$ *correspond bijectively to morphisms $X^\bullet \to T^n(Y^\bullet)$ of complexes. The n-boundaries correspond to morphisms homotopic to zero. In other words*

$$H^n\big(\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)\big) \cong \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}\left(X^\bullet, T^n(Y^\bullet)\right).$$

Clearly $\mathrm{Hom}^\bullet$ *is a bi-∂-functor*

$$\mathrm{Hom}^\bullet\colon\ \mathrm{K}(\mathcal{A})^\circ \times \mathrm{K}(\mathcal{A}) \to \mathrm{K}(\mathbf{Ab}).$$

If $\mathcal{A}$ has enough injectives, we can calculate its derived functors.

Lemma 4. *Let $X^\bullet \in \mathrm{K}(\mathcal{A})$ be a complex, $Y^\bullet \in \mathrm{K}^+(\mathcal{A})$ be a complex of injectives. Assume either*

(a) $Y^\bullet$ is acyclic, or

(b) $X^\bullet$ is acyclic.

Then $\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)$ *is acyclic.*

Proof. By the remark above, one has only to check that any morphism of $X^\bullet$ to $T^n(Y^\bullet)$ is homotopic to zero, or, since $T^n(Y^\bullet)$ also satisfies the hypotheses of the lemma, it is enough to show that any morphism of $X^\bullet$ to $Y^\bullet$ is homotopic to zero. In case (a), $Y^\bullet$ is split exact, and it is easy to construct the homotopy (left to reader). In case (b) the result follows from Lemma 1. □

Suppose $\mathcal{A}$ has enough injectives, and let $\mathcal{L} \subseteq \mathrm{K}^+(\mathcal{A})$ be the triangulated subcategory of bounded-below injective complexes. By Lemma 1.6.2(a), for fixed $X^\bullet \in \mathrm{K}(\mathcal{A})$, the functor

$$\mathrm{Hom}^\bullet(X^\bullet, \cdot\,)\colon\ \mathrm{K}^+(\mathcal{A}) \to \mathrm{K}(\mathbf{Ab})$$

satisfies the hypotheses of Theorem 1.5.1, hence admits a right derived functor. This is functorial in $X^\bullet$, giving a bi-∂-functor

$$\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\,\mathrm{Hom}^\bullet\colon\ \mathrm{K}(\mathcal{A})^\circ \times \mathrm{D}^+(\mathcal{A}) \to \mathrm{D}(\mathbf{Ab}).$$

Now using the lemma, part b), we see that this functor is "exa in the first variable, i.e., takes acyclic complexes into acyc complexes, and hence passes to the quotient, giving a trivial derived functor

$$\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\operatorname{Hom}^{\bullet}: \quad D(A)^{\circ} \times D^{+}(A) \longrightarrow D(\mathbf{Ab}).$$

(We will denote this functor by $\underline{\underline{\mathrm{R}}}\operatorname{Hom}^{\bullet}$ when no confusion can

Suppose on the other hand that A has enough projectives by the usual process of "reversing the arrows" we see that the also a functor

$$\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\operatorname{Hom}^{\bullet}: \quad D^{-}(A)^{\circ} \times D(A) \longrightarrow D(A).$$

Now if A has enough injectives and enough projectives, then functors $\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\operatorname{Hom}^{\bullet}$ and $\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\operatorname{Hom}^{\bullet}$ are defined on $D^{-}(A)^{\circ} \times D^{+}$

and are canonically isomorphic, as we see by the lemma below. Thus we are justified in using the ambiguous notation $\underline{\underline{\mathrm{R}}}\operatorname{Hom}^{\bullet}$.

Lemma 5. *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories,*

$$T\colon\ \mathrm{K}^{*}(\mathcal{A}) \times \mathrm{K}^{\dagger}(\mathcal{B}) \to \mathrm{K}(\mathcal{C})$$

a bi-∂-functor. Suppose that the derived functors $\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\underline{\mathrm{R}}}_{\mathrm{II}}T$ and $\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\underline{\underline{\mathrm{R}}}_{\mathrm{I}}T$ both exists, (where the subscripts I, II *denote the derived functor in the first or second variable, respectively). Then there is a unique isomorphism between them compatible with the morphisms $\xi_1 : T \to \underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\underline{\mathrm{R}}}_{\mathrm{II}}T$ and $\xi_2 : T \to \underline{\underline{\mathrm{R}}}_{\mathrm{II}}\underline{\underline{\mathrm{R}}}_{\mathrm{I}}T$*

Proof. Immediate from the universal property of derived functors. □

Theorem 2 (Yoneda)**.** *Let $\mathcal{A}$ have enough injectives. For all $X^{\bullet} \in \mathrm{D}(\mathcal{A})$, $Y^{\bullet} \in \mathrm{D}^{+}(\mathcal{A})$,*

$$H^{i}\big(\underline{\underline{\mathrm{R}}}\operatorname{Hom}^{\bullet}(X^{\bullet}, Y^{\bullet})\big) = \operatorname{Ext}^{i}\big(X^{\bullet}, Y^{\bullet}\big).$$

Corollary 3. *If $\mathcal{A}$ has enough injectives, the derived-category definition of* $\operatorname{Ext}^{i}(X, Y)$ *coincides with the classical* Ext *groups for all $X, Y \in \mathcal{A}$.*

Proof of Corollary 1.6.5. Take an injective resolution $I^{\bullet}$ of Y. Then

$$\operatorname{Ext}^{i}(X, Y) = H^{i}\big(\operatorname{Hom}^{\bullet}(X, I^{\bullet})\big),$$

which is the standard definition. □

Proof of Theorem 1.6.4. Choose a quasi-isomorphism $s\colon Y^\bullet \to I^\bullet$ with $I^\bullet$ a bounded-below injective complex. Then

$$\mathrm{Ext}^i(X^\bullet, Y^\bullet) = \mathrm{Ext}^i(X^\bullet, I^\bullet) = \mathrm{Hom}_{\mathrm{D}(\mathcal{A})}\big(X^\bullet, T^i(I^\bullet)\big).$$

By Lemma 1.4.5, one sees that every morphism in $\mathrm{D}(\mathcal{A})$ of a complex X" to a complex of injectives bounded below, say $T^i(I^\bullet)$, is represented by an actual morphism of complexes. Hence the above is equal to

$$\begin{aligned}\mathrm{Hom}_{\mathrm{D}(\mathcal{A})}\big(X^\bullet, T^i(I^\bullet)\big) &= \mathrm{Hom}_{\mathrm{K}(\mathcal{A})}\big(X^\bullet, T^i(I^\bullet)\big)\\ &= H^i\big(\mathrm{Hom}^\bullet(X^\bullet, I^\bullet)\big)\\ &= H^i\big(\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}^\bullet(X^\bullet, Y^\bullet)\big).\end{aligned}$$

□

2.8 Way-out functors and Isomorphisms

Definition 13. *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, and let $F\colon \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$ be a (covariant) ∂-functor. We say F is* way-out (right) *if for every $n_1 \in \mathbb{Z}$, there exists $n_2 \in \mathbb{Z}$ such that for any complex $X^\bullet \in \mathrm{D}(\mathcal{A})$ satisfying $H^i(X^\bullet) = 0$ for all $i < n_2$, we have $H^i(F(X^\bullet)) = 0$ for all $i < n_1$.*

One defines way-out left *and* way-out in both directions *similarly. For a contravariant ∂-functor $F\colon \mathrm{D}(\mathcal{A}) \to \mathrm{D}(\mathcal{B})$, the definition of way-out right is the same, except reversing the inequality to $i > n_2$.*

Examples.

1. If $F_0\colon \mathcal{A} \to \mathcal{B}$ is additive and satisfies the hypotheses of Corollary 1.5.3(α) or (β), then $\underline{\underline{\mathrm{R}}}^+F$ is way-out right. If F_0 satisfies (γ), then $\underline{\underline{\mathrm{R}}}F$ is way-out in both directions.

2. For unbounded $X^\bullet \in \mathrm{D}(\mathcal{A})$, the functor $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}(X^\bullet, Y^\bullet)$ for $Y^\bullet \in \mathrm{Ob}\,\mathrm{D}^+(\mathcal{A})$ is generally not way-out in $Y^\bullet$.

Proposition 15 (Lemma on Way-out Functors)**.** *Let $\mathcal{A}, \mathcal{B}$ be abelian categories, $\mathcal{A}' \subseteq \mathcal{A}$ a thick subcategory. Let F, G be ∂-functors from $\mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$ or $\mathrm{D}_{\mathcal{A}'}(\mathcal{A})$ to $\mathrm{D}(\mathcal{B})$, and $\eta\colon F \to G$ a morphism of functors.*

(i) If $\eta(X)$ is an isomorphism for all $X \in \mathrm{Ob}\,\mathcal{A}'$, then $\eta(X^\bullet)$ is an isomorphism for all $X^\bullet \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$.

(ii) *If $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$, and F, G are way-out right, then $\eta(X^\bullet)$ is an isomorphism for all $X^\bullet \in \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$.*

(iii) *If $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$, and F, G are way-out in both directions, then $\eta(X^\bullet)$ is an isomorphism for all $X^\bullet \in \mathrm{D}_{\mathcal{A}'}(\mathcal{A})$.*

(iv) *Let $P \subseteq \operatorname{Ob}\mathcal{A}'$ be such that every object of $\mathcal{A}'$ embeds into an object of P. If $\eta(X)$ is an isomorphism for all $X \in P$, and F, G are way-out right, then $\eta(X)$ is an isomorphism for all $X \in \operatorname{Ob}\mathcal{A}'$.*

Definition 14. *Let $X^\bullet$ be a complex over abelian category $\mathcal{A}$, $n \in \mathbb{Z}$. Define truncation functors:*

$$\begin{aligned}
\tau_{>n}(X^\bullet)&\colon\ \cdots \to 0 \to X^{n+1} \to X^{n+2} \to \cdots\,,\\
\tau_{\leq n}(X^\bullet)&\colon\ \cdots \to X^{n-1} \to X^{n} \to 0 \to \cdots\,,\\
\sigma_{>n}(X^\bullet)&\colon\ \cdots \to 0 \to \operatorname{im} d^n \to X^{n+1} \to X^{n+2} \to \cdots\,,\\
\sigma_{\leq n}(X^\bullet)&\colon\ \cdots \to X^{n-1} \to \ker d^n \to 0 \to \cdots\,.
\end{aligned}$$

There are natural morphisms inducing short exact sequences of complexes:

$$0 \to \tau_{>n}(X^\bullet) \to X^\bullet \to \tau_{\leq n}(X^\bullet) \to 0, \tag{2.1}$$

$$0 \to \sigma_{\leq n}(X^\bullet) \to X^\bullet \to \sigma_{>n}(X^\bullet) \to 0. \tag{2.2}$$

The σ-truncations satisfy:

$$H^i\big(\sigma_{>n}(X^\bullet)\big) = \begin{cases} H^i(X^\bullet) & i > n, \\ 0 & i \leq n; \end{cases} \qquad H^i\big(\sigma_{\leq n}(X^\bullet)\big) = \begin{cases} H^i(X^\bullet) & i \leq n, \\ 0 & i > n. \end{cases}$$

Lemma 6. *For any complex $X^\bullet$ over $\mathcal{A}$, there exist triangles in $\mathrm{D}(\mathcal{A})$:*

(3)

$$\begin{tikzcd}
& \tau_{>n}(X^\bullet) \arrow[dl] & \\
\tau_{\geq n}(X^\bullet) \arrow[rr] & & X^n \arrow[ul]
\end{tikzcd}$$

(4)

$$\begin{tikzcd}
& \sigma_{>n}(X^\bullet) \arrow[dl] & \\
H^n(X^\bullet) \arrow[rr] & & \sigma_{\geq n}(X^\bullet) \arrow[ul]
\end{tikzcd}$$

Proof. Triangle (3) comes from the short exact sequence

$$0 \to \tau_{>n}(X^\bullet) \to \tau_{\geq n}(X^\bullet) \to X^n \to 0.$$

For the second, define $\sigma'_{\geq n}(X^\bullet)$ by

$$\cdots \to 0 \to X^n/\operatorname{im} d^{n-1} \to X^{n+1} \to \cdots .$$

There is a quasi-isomorphism $\sigma_{\geq n}(X^\bullet) \to \sigma'_{\geq n}(X^\bullet)$, and an exact sequence

$$0 \to H^n(X^\bullet) \to \sigma'_{\geq n}(X^\bullet) \to \sigma_{>n}(X^\bullet) \to 0.$$

This induces the required triangle in $\mathrm{D}(\mathcal{A})$. □

Proof of Proposition 1.7.1. (i) Let $X^\bullet \in \operatorname{Ob} \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$. We prove, by descending induction on n, that

$$\eta\big(\sigma_{>n}(X^\bullet)\big) : F(\sigma_{>n}(X^\bullet)) \to G(\sigma_{>n}(X^\bullet))$$

is an isomorphism for all n. If n is large enough, then $\sigma_{>n}(X^\bullet)$ has zero cohomology, since $X^\bullet$ has bounded cohomology. Hence it is the zero object of $\mathrm{D}(\mathcal{A})$, and η of it is an isomorphism. The induction step follows from the hypotheses and Proposition 1.1(c), applied to the triangle (4) above. Now for n small enough, the natural map $X^\bullet \to \sigma_{>n}(X^\bullet)$ is a quasi-isomorphism, hence an isomorphism in $\mathrm{D}(\mathcal{A})$, and we are done.

(ii) Let $X^\bullet \in \operatorname{Ob} \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$. To show $\eta(X^\bullet)$ is an isomorphism, it is sufficient to show that

$$H^j(\eta(X^\bullet)) : H^j(F(X^\bullet)) \to H^j(G(X^\bullet))$$

is an isomorphism for all $j \in \mathbb{Z}$. Given j, let $n_1 \geq j+2$, and choose n_2 as in the definition of way-out functors, to work for F and G both. From the exact sequence (2) above, we have a triangle in $\mathrm{D}(\mathcal{A})$.

$$\begin{tikzcd}
 & \tau_{>n_2}(X^\bullet) \arrow[dl] & \\
\tau_{\geq n_2}(X^\bullet) \arrow[rr] & & X^\bullet \arrow[ul]
\end{tikzcd}$$

Since

$$H^i(\sigma_{>n_2}(X^\bullet)) = 0 \quad \text{for } i \leq n_2,$$

we have

$$H^i\big(F(\sigma_{>n_2}(X^\bullet))\big) = 0 = H^i\big(G(\sigma_{>n_2}(X^\bullet))\big)$$

for $i < n_1$, in particular for $i = j, j+1$. Therefore from the long exact sequence of cohomology we get isomorphisms

$$H^j\big(F(\sigma_{\leq n_2}(X^\bullet))\big) \xrightarrow{\sim} H^j\big(F(X^\bullet)\big),$$
$$H^j\big(G(\sigma_{\leq n_2}(X^\bullet))\big) \xrightarrow{\sim} H^j\big(G(X^\bullet)\big).$$

But $\sigma_{\leq n_2}(X^\bullet) \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$, so η is an isomorphism on it by what we have just proved. Hence $H^j(\eta(X^\bullet))$ is an isomorphism, as required.

(iii) Given $X^\bullet \in \mathrm{D}_{\mathcal{A}'}(\mathcal{A})$, we treat first $\sigma_{\leq 0}(X^\bullet)$ and $\sigma_{>0}(X^\bullet)$ by case (ii), then glue the results by a triangle deduced from the exact sequence (2).

(iv) Given $X \in \operatorname{Ob}\mathcal{A}'$, we can find by Lemma 1.4.6 a resolution $I^\bullet \in \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$ of X, with each $I^p \in P$. Thus it is sufficient to show that $\eta(I^\bullet)$ is an isomorphism. This follows by applying the technique of (i) and (ii), but using the functors τ instead of σ. □

Proposition 16. *Let $\mathcal{A}$, $\mathcal{B}$ be abelian categories, and let $\mathcal{A}' \subseteq \mathcal{A}$, $\mathcal{B}' \subseteq \mathcal{B}$ be thick abelian subcategories, F be a ∂-functor from $\mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$ or $\mathrm{D}_{\mathcal{A}'}(\mathcal{A})$ to $\mathrm{D}(\mathcal{B})$,.*

- *(i) If $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in \operatorname{Ob}\mathcal{A}'$, then $F(X^\bullet) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X^\bullet \in \mathrm{D}^b_{\mathcal{A}'}(\mathcal{A})$.*
- *(ii) If $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in \operatorname{Ob}\mathcal{A}'$ and F is way-out right, then $F(X^\bullet) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X^\bullet \in \mathrm{D}^+_{\mathcal{A}'}(\mathcal{A})$.*
- *(iii) As in (ii) if F is way-out in both directions, then $F(X^\bullet) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X^\bullet \in \mathrm{D}_{\mathcal{A}'}(\mathcal{A})$.*
- *(iv) Let $P \subseteq \operatorname{Ob}\mathcal{A}'$ such that every object of $\mathcal{A}'$ embeds into P. If $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in P$ and F is way-out right, then $F(X) \in \mathrm{D}_{\mathcal{B}'}(\mathcal{B})$ for all $X \in \operatorname{Ob}\mathcal{A}'$.*

Proof. The proof is analogous to that of the previous proposition, and will be left to the reader. □

Proposition 17. *Let $\mathcal{A}$ have enough injectives, $F\colon \mathcal{A} \to \mathcal{B}$ additive of cohomological dimension $\leq n$. Let $P \subseteq \operatorname{Ob}\mathcal{A}$ be all X with $\mathrm{R}^i F(X) = 0$ for $i \neq n$. Assume every object of $\mathcal{A}$ is a quotient of an object in P. Set $G = \mathrm{R}^n F$. Then $\underline{\underline{\mathrm{R}}}F, \underline{\underline{\mathrm{L}}}G$ exist, and there is a functorial isomorphism*

$$\psi\colon \underline{\underline{\mathrm{R}}}F \xrightarrow{\sim} \underline{\underline{\mathrm{L}}}G[-n],$$

where $[-n]$ denotes shifting right by n degrees.

Proof. $\underline{\underline{\mathrm{R}}}F$ exists by Corollary 1. To see that $\underline{\underline{\mathrm{L}}}G$ exists, we apply Theorem 1 with the arrows reversed. By hypothesis P satisfies condition (i)* of Lemma 3 (where $*$ denotes reversal of arrows), and one sees easily from the definition of P that it satisfies also (ii)*, (iii)*, and (iv)* of Corollary 1. Therefore, if we let $\mathcal{L} \subseteq \mathrm{K}(\mathcal{A})$ be the triangulated subcategory consisting of complexes of elements of P, we see as in the proof of Corollary 1(γ) that the hypotheses 1)* and 2)$^* = 2$) of the theorem are satisfied, and so $\underline{\underline{\mathrm{L}}}G$ exists.

To define the morphism ψ, we observe, as in the proof of Theorem 1, that $\mathcal{L}_{\mathrm{Qis}} \to \mathrm{D}(\mathcal{A})$ is an equivalence of categories, so it will be sufficient to define ψ on $\mathcal{L}_{\mathrm{Qis}}$. Or, using Proposition 5, it will be sufficient to define ψ on $\mathcal{L}$ itself. In other words, for every complex $X^\bullet$ of objects of P we must give a morphism in $\mathrm{D}(\mathcal{B})$,

$$\psi(X^\bullet) \colon \underline{\underline{\mathrm{R}}}F(X^\bullet) \to \underline{\underline{\mathrm{L}}}G(X^\bullet)[-n],$$

such that whenever $f \colon X^\bullet \to Y^\bullet$ is a morphism of such complexes, there is the usual commutative diagram $\underline{\underline{\mathrm{L}}}G(f) \cdot \psi(X^\bullet) = \psi(Y^\bullet) \cdot \underline{\underline{\mathrm{R}}}F(f)$.

Definition 15. *Let $\mathcal{A}$ be an abelian category, $X^\bullet$ a complex of objects of $\mathcal{A}$. We refer to [M, Ch. IV, §4] for the definition of double complexes, maps and homotopies of double complexes, and injective resolutions of complexes. We recall the following facts. A* (right) Cartan–Eilenberg resolution *of $X^\bullet$ is a double complex $C^{\cdot\cdot}$ of injective objects of $\mathcal{A}$, with $C^{p,q} = 0$ if $p < 0$, together with an augmentation morphism $\epsilon \colon X^\bullet \to C^{\bullet,0}$ such that for every $p \in \mathbb{Z}$, the induced maps*

$$B^p(\epsilon) \colon B^p(X^\bullet) \to B^p_{\mathrm{I}}(C^{\cdot\cdot}), \qquad H^p(\epsilon) \colon H^p(X^\bullet) \to H^p_{\mathrm{I}}(C^{\cdot\cdot})$$

are injective resolutions in the usual sense (cf. [M, Ch. XVII, §1] where this is called an injective resolution of $X^\bullet$).

Lemma 7.

a) If $\mathcal{A}$ has enough injectives, every complex $X^\bullet$ of objects of $\mathcal{A}$ has a Cartan–Eilenberg resolution.

b) If $f \colon X^\bullet \to Y^\bullet$ is a morphism of complexes, and $C^{\cdot\cdot}, D^{\cdot\cdot}$ are Cartan–Eilenberg resolutions of $X^\bullet, Y^\bullet$, then there exists a map $F \colon C^{\cdot\cdot} \to D^{\cdot\cdot}$ of double complexes lying over f.

c) If $f, g \colon X^\bullet \to Y^\bullet$ are homotopic maps, and $F, G \colon C^{\cdot\cdot} \to D^{\cdot\cdot}$ are maps of Cartan-Eilenberg resolutions lying over f, g, then F is homotopic to G.

d) If $F, G\colon C^{\cdot\cdot} \to D^{\cdot\cdot}$ are homotopic maps of double complexes, then $s(F), s(G)\colon s(C^{\cdot\cdot}) \to s(D^{\cdot\cdot})$ are homotopic maps of the associated simple complexes.

e) If $F, G\colon C^{\cdot\cdot} \to D^{\cdot\cdot}$ are homotopic maps of double complexes, and $\sigma^{\mathrm{II}}_{\leq n}$ denotes truncation with respect to the second differential d_2, then the restrictions of F and G to maps $\sigma^{\mathrm{II}}_{\leq n}(C^{\cdot\cdot}) \to \sigma^{\mathrm{II}}_{\leq n}(D^{\cdot\cdot})$ are homotopic.

f) If $\epsilon\colon X^\bullet \to C^{\cdot\cdot}$ augments $X^\bullet$ to a double complex $C^{\cdot\cdot}$ with $C^{p,q} = 0$ for $q < 0$ and $q > n_0$ (for suitable n_0), and such that for each p the map $X^p \to C^{p,\bullet}$ is a resolution, then the natural map $X^\bullet \to s(C^{\cdot\cdot})$ into the associated simple complex is a quasi-isomorphism.

Proof of Lemma 1.7.5. Parts a), b), and c) are in [M, Ch. XVII, Prop. 1.2]. Part d) is in [M, Ch. IV, §4]. Part e) is easy. Part f) follows either from the spectral sequence of a double complex [EGA 0_{III} 11.3.3(ii)] or by a direct calculation. □

Proof of Proposition 1.7.4, continued. Given $X^\bullet \in \operatorname{Ob}\mathcal{L}$, choose a Cartan–Eilenberg resolution $C^{\cdot\cdot}$ of $X^\bullet$ (exists by Lemma 1.7.5 a)). Define $C'^{\cdot\cdot} = \sigma^{\mathrm{II}}_{\leq n}(C^{\cdot\cdot})$; explicitly

$$C'^{p,q} = \begin{cases} C^{p,q} & q < n, \\ \ker d_2^{p,n} & q = n, \\ 0 & q > n. \end{cases}$$

For each $p \in \mathbb{Z}$, we have an exact sequence

$$0 \to X^p \to C'^{p,0} \to C'^{p,1} \to \cdots \to C'^{p,n-1} \to C'^{p,n} \to 0.$$

The terms $C'^{p,0}, \ldots, C'^{p,n-1}$ are injective and $X^p \in P$ implies $C'^{p,n}$ is F-acyclic. Hence this resolution may be used to calculate $\underline{\underline{\mathrm{R}}}F(X^p)$ (see end of Theorem 1.5.1), giving an isomorphism

$$\varphi(t^p)\colon G(X^p) = \mathrm{R}^n F(X^p) \xrightarrow{\sim} F(C'^{p,n})\big/ \operatorname{im} F(C'^{p,n-1}),$$

where t^p is the quasi-isomorphism $X^p \to C^{p,\bullet}$. This may be used to construct a map

$$\alpha^p\colon F(C^{p,\bullet}) \to G(X^p),$$

whence a map of double complexes

$$\alpha\colon F(C'^{\cdot\cdot}) \to G(X^\bullet),$$

where $G(X^\bullet)$ is concentrated in the n-th row.

Passing to the associated simple complexes we obtain

$$s(\alpha)\colon F\big(s(C'^{\cdot\cdot})\big) \to G(X^\bullet)[-n].$$

Now $u\colon X^\bullet \to s(C'^{\cdot\cdot})$ is a quasi-isomorphism into a complex of F-acyclic objects, by Lemma 1.7.5(f), so there is an isomorphism

$$\varphi(u)\colon \underline{\underline{\mathrm{R}}}F(X^\bullet) \xrightarrow{\sim} F\big(s(C'^{\cdot\cdot})\big).$$

Since $X^\bullet$ consists of G-acyclic objects, there is an isomorphism

$$\varphi(\mathrm{id}_{X^\bullet})\colon G(X^\bullet) \xrightarrow{\sim} \underline{\underline{\mathrm{L}}}G(X^\bullet).$$

Composing, we define

$$\psi(X^\bullet)\colon \underline{\underline{\mathrm{R}}}F(X^\bullet) \longrightarrow \underline{\underline{\mathrm{L}}}G(X^\bullet)[-n].$$

I do not care whether $\psi(X^\bullet)$ is dependent of the choice of the Cartan–Eilenberg resolution $C^{\cdot\cdot}$. It will be sufficient to verify that whenever $f\colon X^\bullet \to Y^\bullet$ is a morphism of complexes in $\mathcal{L}$, then $\psi(X^\bullet)$ and $\psi(Y^\bullet)$ defined as above, fit into a commutative diagram. This is not hard to show using the results of the lemma above, and can safely be left to the reader.

Thus we have a morphism of functors $\psi\colon \underline{\underline{\mathrm{R}}}F \to \underline{\underline{\mathrm{L}}}G[-n]$.

To show that ψ is an isomorphism, we note that $\underline{\underline{\mathrm{R}}}F$ and $\underline{\underline{\mathrm{L}}}G$ are way-out in both directions, and so, by the lemma on way-out functors (Proposition 15), we reduce to showing $\psi(X)$ is an isomorphism for any $X \in P$. But that is clear from the construction. □

Proposition 18. *Let $\mathcal{A}$ have enough injectives, $X^\bullet \in \mathrm{K}^+(\mathcal{A})$. The following are equivalent:*

(i) $X^\bullet$ admits a quasi-isomorphism $X^\bullet \to I^\bullet$ into a bounded complex of injective objects of $\mathcal{A}$.

(ii) The functor $\underline{\underline{\mathrm{R}}}\operatorname{Hom}(\,\cdot\,, X^\bullet)\colon \mathrm{D}(\mathcal{A})^\circ \to \mathrm{D}(\mathbf{Ab})$ is way-out left (hence way-out both ways).

(iii) There exists $n_0 \in \mathbb{Z}$ such that $\operatorname{Ext}^i(Y, X^\bullet) = 0$ for all $Y \in \operatorname{Ob}\mathcal{A}$ and all $i > n_0$.

Proof. $(i) \Rightarrow (ii)$: We may assume that $X^\bullet$ is a bounded complex of injectives, and calculate $F(Y^\bullet)$ as $\operatorname{Hom}(Y^\bullet, X^\bullet)$. Then it is clear that F is way-out in both directions.

(ii) $\Rightarrow$ *(iii)*: Since F is way-out left, choose n_0 such that whenever $Y^\bullet \in \mathrm{D}(\mathcal{A})$ and $H^i(Y^\bullet) = 0$ for $i < n_0$, then $H^i(F(Y^\bullet)) = 0$ for $i > 0$. Then this n_0 will do. Indeed, let $Y \in \operatorname{Ob}\mathcal{A}$, and let $Y'^\bullet$ be the complex $T^{-n_0}(Y)$. Then $H^i(Y'^\bullet) = 0$ for $i < n_0$, so $H^i(F(Y'^\bullet)) = 0$ for $i > 0$. But $H^i(F(Y'^\bullet)) = H^i(F(T^{-n_0}Y)) = H^{i+n_0}(F(Y))$. This says $\operatorname{Ext}^j(Y, X^\bullet) = 0$ for $j > n_0$, as required.

(iii) $\Rightarrow$ *(i)*: Let $X^\bullet$ satisfy condition (iii) for n_0. Since $\mathcal{A}$ has enough injectives, we can find an injective resolution of $X^\bullet$ (Lemma 3), i.e., a quasi-isomorphism $s\colon X^\bullet \to X'^\bullet$ where $X'^\bullet$ is a bounded-below complex of injective objects of $\mathcal{A}$.

I claim that $H^i(X'^\bullet) = 0$ for all $i > n_0$. Suppose to the contrary that $H^m(X'^\bullet) \neq 0$ for some $m > n_0$. Then from the exact sequence

$$0 \to B^m(X'^\bullet) \to Z^m(X'^\bullet) \to H^m(X'^\bullet) \to 0$$

we see that the first inclusion is strict. Hence there exists a $Y \in \operatorname{Ob}\mathcal{A}$ (e.g., $Y = Z^m(X'^\bullet)$) such that the inclusion

$$\operatorname{Hom}(Y, B^m(X'^\bullet)) \to \operatorname{Hom}(Y, Z^m(X'^\bullet))$$

is strict. However, there is a commutative diagram

$$\begin{tikzcd}
\operatorname{Hom}(Y, B^m(X'^\bullet)) \arrow[r] \arrow[d] & \operatorname{Hom}(Y, Z^m(X'^\bullet)) \arrow[d, "\cong"] \\
B^m\big(\operatorname{Hom}^\bullet(Y, X'^\bullet)\big) \arrow[r] & Z^m\big(\operatorname{Hom}^\bullet(Y, X'^\bullet)\big)
\end{tikzcd}$$

with an isomorphism on the right since Hom is left exact. But by hypothesis $\operatorname{Ext}^m(Y, X^\bullet) = H^m(\operatorname{Hom}^\bullet(Y, X'^\bullet)) = 0$, so the arrow on top is an isomorphism, which gives a contradiction. Hence $H^i(X'^\bullet) = 0$ for $i > n_0$, as claimed.

It follows that the map

$$i\colon \sigma_{\leq n}(X'^\bullet) \to X'^\bullet$$

is a quasi-isomorphism for any $n \geq n_0$ (using the notation introduced earlier in this section). Now I claim that $Z^{n+1}(X'^\bullet) = B^{n+1}(X'^\bullet)$ is injective for any $n \geq n_0$. This will show that $\sigma_{\leq n+1}(X'^\bullet)$ is a bounded complex of injectives.

Consider the exact sequence of complexes

$$0 \to \sigma_{\leq n}(X'^\bullet) \to \tau_{\leq n}(X'^\bullet) \to B^{n+1}(X'^\bullet)[-n] \to 0$$

for $n \geq n_0$. We will show that $B^{n+1} = B^{n+1}(X'^\bullet)$ is injective by showing that $\operatorname{Ext}^1(Y, B^{n+1}) = 0$ for all $Y \in \operatorname{Ob}\mathcal{A}$. For any Y, we have

$$\operatorname{Ext}^1(Y, B^{n+1}) = \operatorname{Ext}^{n+1}\big(Y, B^{n+1}[-n]\big)$$

and there is an exact sequence

$$\cdots \to \mathrm{Ext}^{n+1}\left(Y, \tau_{\le n}(X'^{\bullet})\right) \to \mathrm{Ext}^{n+1}\left(Y, B^{n+1}[-n]\right) \to \mathrm{Ext}^{n+2}\left(Y, \sigma_{\le n}(X'^{\bullet})\right) \to \cdots$$

The group on the left is zero since $\tau_{\le n}(X'^{\bullet})$ is an injective complex which stops in degree n. The group on the right is zero by our hypothesis and the fact that $\sigma_{\le n}(X'^{\bullet})$ is isomorphic to $X'^{\bullet}$ in $\mathrm{D}^+(\mathcal{A})$. Hence the middle group is zero.

Now for $n \ge n_0$ we have seen that $\sigma_{\le n}(X'^{\bullet})$ is a bounded complex of injectives, and that the map

$$i\colon \sigma_{\le n}(X'^{\bullet}) \to X'^{\bullet}$$

is a quasi-isomorphism. By Lemma 2, i has a homotopy inverse, say f. Therefore the composition

$$fs\colon X^{\bullet} \to \sigma_{\le n}(X'^{\bullet})$$

gives the required quasi-isomorphism of $X^{\bullet}$ into a bounded complex of injectives. □

Definition 16. *If $X^{\bullet} \in \mathrm{K}^+(\mathcal{A})$ satisfies the equivalent conditions of Proposition 1.7.6, we say $X^{\bullet}$ has* finite injective dimension*.*

Corollary 4. *The complexes of finite injective dimension form a triangulated localizing subcategory of* $\mathrm{K}(\mathcal{A})$*.*

Proof. The fact that they form a triangulated subcategory follows e.g. from condition (iii) of Proposition 18 and the long exact sequence of Ext. We see that the subcategory is localizing by applying Proposition 4(i), since anything in $\mathrm{K}^+(\mathcal{A})$ quasi-isomorphic to an element of the subcategory is already in the subcategory, and $\mathrm{K}^+(\mathcal{A})$ is a localizing subcategory of $\mathrm{K}(\mathcal{A})$. □

Definition 17. *Denote by $\mathrm{K}^+(\mathcal{A})_{\mathrm{fid}}$ the full subcategory of $\mathrm{K}^+(\mathcal{A})$ consisting of complexes of finite injective dimension. The corresponding derived subcategory is* $\mathrm{D}^+(\mathcal{A})_{\mathrm{fid}}$.

Remark. These subcategories are essentially different from the subcategories of the form $\mathrm{K}_{\mathcal{A}'}(\mathcal{A})$ and $\mathrm{D}_{\mathcal{A}'}(\mathcal{A})$ we have considered before, because they cannot be characterized by putting conditions on the cohomology objects $H^i(X^{\bullet})$.

Chapter 3

Application to Preschemes

In this chapter we will discuss the various categories of sheaves we plan to use in the sequel, their derived categories, various functors between them, and relations among these functors. In §7 we give the structure of injective sheaves of $\mathcal{O}_X$-modules on a locally noetherian prescheme X.

3.1 Categories of sheaves

Let $(X, \mathcal{O}_X)$ be a prescheme. We will denote by $\mathrm{Mod}(X)$ the category of sheaves of $\mathcal{O}_X$-modules, and we will denote its derived category by $\mathrm{D}(X) = \mathrm{D}(\mathrm{Mod}(X))$.

We will consider the thick subcategories of $\mathrm{Mod}(X)$ consisting of the quasi-coherent sheaves, denoted by $\mathrm{Qco}(X)$, and the coherent sheaves, denoted by $\mathrm{Coh}(X)$. (Cf. [EGA O, §5] for definitions and the fact that these are thick subcategories.) We will write $\mathrm{D}_{\mathrm{qc}}(X)$ for $\mathrm{D}_{\mathrm{Qco}(X)}(\mathrm{Mod}(X))$ and $\mathrm{D}_{\mathrm{c}}(X)$ for $\mathrm{D}_{\mathrm{Coh}(X)}(\mathrm{Mod}(X))$, using the notation of [I.4]. Similarly $\mathrm{D}^+_{\mathrm{qc}}(X)$ etc.

Proposition 19. *There are enough injectives in the categories* $\mathrm{Mod}(X)$ *and* $\mathrm{Qco}(X)$.

Proof. This is well known, e.g., [T, 1.10.1], or, for $\mathrm{Mod}(X)$, [G, II 7.1.1]. □

Proposition 20. *Every sheaf of $\mathcal{O}_X$-modules is a quotient of a flat $\mathcal{O}_X$-module.*

Proof. If U is an open subset of X, we denote by $\mathcal{O}_{X,U}$ the sheaf which is $\mathcal{O}_X$ on U and 0 outside [G, II Thm. 2.9.1]. Then $\mathcal{O}_{X,U}$ is a flat $\mathcal{O}_X$-module, and any direct sum of these is also flat. If $\mathcal{F}$ is any sheaf of $\mathcal{O}_X$-modules, let $s_i \in \Gamma(U_i, \mathcal{F})$ be a set of generators of $\mathcal{F}$ as an $\mathcal{O}_X$-module. Then $\mathcal{F}$ is a quotient of $\bigoplus_i \mathcal{O}_{X,U_i}$, by sending $1 \in \Gamma(U_i, \mathcal{O}_{X,U_i})$ to s_i. □

3.2 The derived functors of f_* and Γ

Let $f\colon X \to Y$ be a morphism of preschemes. Then the functor f_* sends $\mathrm{Mod}(X)$ to $\mathrm{Mod}(Y)$. Since the first of these categories has enough injectives, we have a derived functor

$$\underline{\underline{\mathrm{R}}}^+ f_*\colon\ \mathrm{D}^+(X) \longrightarrow \mathrm{D}^+(Y).$$

There are two cases in which we can define $\underline{\underline{\mathrm{R}}}f_*$ for unbounded complexes. If X is noetherian of finite Krull dimension n, then $H^i(X, \mathcal{F}) = 0$ for $i > n$ and every sheaf $\mathcal{F}$ of abelian groups on X [T 3.6.5]. Note that if $\mathcal{F}$ is an $\mathcal{O}_X$-module, then its cohomology as an $\mathcal{O}_X$-module is the same as its cohomology as an abelian sheaf [EGA 0_{III} 12.1.1]. Therefore we have $\mathrm{R}^i f_*(\mathcal{F}) = 0$ for $i > n$, since this sheaf is the sheaf associated to the presheaf $V \mapsto H^i(f^{-1}(V), \mathcal{F})$. Thus f_* has finite cohomological dimension on $\mathrm{Mod}(X)$, and using [I.5.3(γ)] we can define

$$\underline{\underline{\mathrm{R}}} f_*\colon\ \mathrm{D}(X) \longrightarrow \mathrm{D}(Y).$$

On the other hand, if f_* has finite cohomological dimension on $\mathrm{Qco}(X)$, we can define

$$\underline{\underline{\mathrm{R}}} f_*\colon\ \mathrm{D}(\mathrm{Qco}(X)) \longrightarrow \mathrm{D}(\mathrm{Qco}(Y)).$$

This will be the case if

a) f is separated, quasi-compact, and Y is quasi-compact, by [EGA III 1.4.12], or

b) if X, Y are locally noetherian, f is of finite type, and the fibres of f are of bounded dimension (use [EGA III 4.1.5] and [T 3.6.5]).

We will often make statements in only one of these two contexts, and will let the reader make the necessary modifications for the other. We will also write $\underline{\underline{\mathrm{R}}}f_*$ instead of $\underline{\underline{\mathrm{R}}}^+f_*$, when no confusion can arise.

Proposition 21. *If $f\colon X \to Y$ is separated and quasi-compact, then $\underline{\underline{\mathrm{R}}}f_*$ takes $\mathrm{D}^+_{\mathrm{qc}}(X)$ into $\mathrm{D}^+_{\mathrm{qc}}(Y)$. If, furthermore, X is noetherian of finite Krull dimension, $\underline{\underline{\mathrm{R}}}f_*$ takes $\mathrm{D}_{\mathrm{qc}}(X)$ into $\mathrm{D}_{\mathrm{qc}}(Y)$.*

Proof. By [EGA III 1.4.10], if $\mathcal{F}$ is quasi-coherent on X, then all the $\mathrm{R}^i f_*(\mathcal{F})$ are quasi-coherent sheaves on Y. In other words, $\mathrm{R}^i f_*(\mathcal{F}) \in \mathrm{D}^+_{\mathrm{qc}}(Y)$. But $\underline{\underline{\mathrm{R}}}f_*$ is a way-out functor, so the result follows from [I.7.3]. □

Proposition 22. *If $f\colon X \to Y$ is a proper morphism, and Y is locally noetherian, then $\underline{\underline{\mathrm{R}}}f_*$ sends $\mathrm{D}^+_{\mathrm{c}}(X)$ into $\mathrm{D}^+_{\mathrm{c}}(Y)$. If, furthermore, X is noetherian of finite Krull dimension, $\underline{\underline{\mathrm{R}}}f_*$ takes $\mathrm{D}_{\mathrm{c}}(X)$ into $\mathrm{D}_{\mathrm{c}}(Y)$.*

Proof. Same as the previous proposition, using [EGA III 3.2.1]. □

If X is a prescheme, the global section functor Γ has a right derived functor

$$\underline{\underline{\mathrm{R}}}^+\Gamma\colon\ \mathrm{D}^+(X) \longrightarrow \mathrm{D}(\mathrm{Ab}).$$

As above, we can extend the domain of definition of $\underline{\underline{\mathrm{R}}}\Gamma$ to $\mathrm{D}(X)$ if X is noetherian of finite Krull dimension.

3.3 The derived functor of $\underline{\mathrm{Hom}}^\bullet$

Let X be a prescheme, and let $F^\bullet, G^\bullet \in \mathrm{K}(\mathrm{Mod}(X))$. Then we define $\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet) \in \mathrm{K}(\mathrm{Mod}(X))$ as in [I.6] by

$$\underline{\mathrm{Hom}}^n(F^\bullet, G^\bullet) = \prod_{p\in\mathbb{Z}} \underline{\mathrm{Hom}}_{\mathrm{Mod}(X)}(F^p, G^{p+n})$$

and

$$d^n = d_F + (-1)^{n+1} d_G.$$

Then $\underline{\mathrm{Hom}}^\bullet$ is a bi-∂-functor

$$\underline{\mathrm{Hom}}^\bullet\colon\ \mathrm{K}(\mathrm{Mod}(X))^\bullet \times \mathrm{K}(\mathrm{Mod}(X)) \longrightarrow \mathrm{K}(\mathrm{Mod}(X)).$$

Lemma 8. *Let $F^\bullet$ be a complex of $\mathcal{O}_X$-modules, and let $G^\bullet$ be a complex of injective $\mathcal{O}_X$-modules, bounded below. Assume that either $F^\bullet$ or $G^\bullet$ is acyclic. Then $\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet)$ is acyclic.*

Proof. It is enough to show for each open subset U of X that $\Gamma(U, \underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet)) = \mathrm{Hom}^\bullet(F^\bullet|_U, G^\bullet|_U)$ is acyclic. But this follows from [I.6.2] applied to the abelian category of $\mathcal{O}_U$-modules, since if G is an injective $\mathcal{O}_X$-module, then $G|_U$ is an injective $\mathcal{O}_X$-module. □

Using this Lemma, we can follow the method of [I.6] and obtain a derived functor

$$\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet\colon\ \mathrm{D}(X)^\bullet \times \mathrm{D}^+(X) \longrightarrow \mathrm{D}(X)$$

(more precisely $\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\underline{\mathrm{Hom}}^\bullet$).

Exercise. Suppose that X is locally noetherian, and that every coherent $\mathcal{O}_X$-module is a quotient of a locally free $\mathcal{O}_X$-module of finite rank (e.g. X quasi-projective over a field). Show then that

$$\underline{\underline{\mathrm{R}}}_{\mathrm{II}}\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\mathrm{Hom}}^\bullet\colon\ \mathrm{D}_{\mathrm{c}}^-(X)^\bullet \times \mathrm{D}(X) \longrightarrow \mathrm{D}(X)$$

exists.

Problem. Without the hypotheses of the Exercise, study sheaves on X which are acyclic for $\underline{\mathrm{Hom}}$ in the first variable. Find out whether there are enough of them, and hence whether $\underline{\underline{\mathrm{R}}}_{\mathrm{I}}\underline{\mathrm{Hom}}^\bullet$ exists.

Definition 18. *If $F^\bullet \in \mathrm{D}(X)$ and $G^\bullet \in \mathrm{D}^+(X)$, we define the* local hyperext

$$\underline{\mathrm{Ext}}^i(F^\bullet, G^\bullet) = H^i\big(\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet)\big).$$

Lemma 9. *Let X be a locally noetherian prescheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module, and let $\mathcal{G}$ be a coherent (resp. quasi-coherent) $\mathcal{O}_X$-module. Then for all $i \geq 0$, $\underline{\mathrm{Ext}}^i(\mathcal{F}, \mathcal{G})$ is coherent (resp. quasi-coherent).*

Proof. The coherent case is [EGA 0_{III} 12.3.3], and the quasi-coherent case is proved similarly. □

Proposition 23. *Let X be a locally noetherian prescheme. Let $F^\bullet \in \mathrm{D}_{\mathrm{c}}(X)$ and $G^\bullet \in \mathrm{D}^+_{\mathrm{c}}(X)$ (resp. $\mathrm{D}^+_{\mathrm{qc}}(X)$). Assume either*

a) $F^\bullet \in \mathrm{D}_{\mathrm{c}}(X)$, or

b) $G^\bullet \in \mathrm{D}^+_{\mathrm{c}}(X)_{\mathrm{fid}}$ (cf. [I.7]).

Then $\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet) \in \mathrm{D}_{\mathrm{c}}(X)$ (resp. $\mathrm{D}_{\mathrm{qc}}(X)$).

Proof. This follows from the lemma and [I.7.3] applied twice. The details are left to the reader. □

3.4 The derived functors of $\otimes$ and f^*

Let X be a prescheme, and let $F^\bullet, G^\bullet \in \mathrm{K}(\mathrm{Mod}(X))$. Then we define the tensor product $F^\bullet \otimes G^\bullet$ to be the simple complex associated to the double complex $(F^p \otimes G^q)$, i.e.,

$$(F^\bullet \otimes G^\bullet)^n = \bigoplus_{p+q=n} F^p \otimes G^q$$

and

$$d = d_F + (-1)^n d_G.$$

Homotopies carry over to the tensor product, so we have a functor

$$\otimes\colon\ \mathrm{K}(\mathrm{Mod}(X)) \times \mathrm{K}(\mathrm{Mod}(X)) \longrightarrow \mathrm{K}(\mathrm{Mod}(X)).$$

Lemma 10. *Let $F^\bullet$ be a complex of $\mathcal{O}_X$-modules, and let $G^\bullet$ be a complex of flat $\mathcal{O}_X$-modules, bounded above. Assume that either*

a) $G^\bullet$ is acyclic, or

b) $F^\bullet$ is acyclic,

and assume also that either

1) $F^\bullet$ is bounded above, or

2) $G^\bullet$ is bounded in both directions.

Then $F^\bullet \otimes G^\bullet$ is acyclic.

Proof. Let $K^{\bullet\bullet}$ be the double complex $K^{pq} = F^p \otimes G^q$. Then there are spectral sequences [EGA 0_{III} 11.3.2]

$$''E_2^{pq} = H_{\mathrm{I}}^p H_{\mathrm{II}}^q (K^{\bullet\bullet}) \Longrightarrow E^n = H^n(F^\bullet \otimes G^\bullet)$$

and

$$''E_2^{pq} = H_{\mathrm{II}}^p H_{\mathrm{I}}^q (K^{\bullet\bullet}) \Longrightarrow E^n = H^n(F^\bullet \otimes G^\bullet).$$

Our hypotheses 1) or 2) imply that these spectral sequences are biregular. In case a), one shows that $B^q(G^\bullet) = Z^q(G^\bullet)$ is flat for each q, and so $F^p \otimes G^\bullet$ is acyclic for each p. This implies that $'E_2^{pq} = 0$ for all p, q, and hence $E^n = 0$, and $F^\bullet \otimes G^\bullet$ is acyclic. In case b), $F^\bullet \otimes G^q$ is acyclic for each q, since G^q is flat, and so $''E_2^{pq} = 0$ for each p, q. Hence again $E^n = 0$, and $F^\bullet \otimes G^\bullet$ is acyclic. $\square$

Now let $F^\bullet \in \mathrm{K}^-(\mathrm{Mod}(X))$, and let $\mathcal{L} \subseteq \mathrm{K}^-(\mathrm{Mod}(X))$ be the triangulated subcategory of complexes of flat $\mathcal{O}_X$-modules. Then by Proposition 2.1.2 and the lemma, part a1), $\mathcal{L}$ satisfies the hypotheses of [I.5.1] for the functor

$$F^\bullet \otimes \cdot \colon\ \mathrm{K}^-(\mathrm{Mod}(X)) \longrightarrow \mathrm{K}(\mathrm{Mod}(X)),$$

and hence

$$\underline{\underline{\mathrm{L}}}_{\mathrm{II}}\otimes\colon\ \mathrm{K}^-(\mathrm{Mod}(X)) \times \mathrm{D}^-(X) \longrightarrow \mathrm{D}(X)$$

exists. (It is clearly functorial in $F^\bullet$.)

By the lemma, part b1), this functor is exact in the first variable, hence passes to the quotient to give

$$\underline{\underline{\mathrm{L}}}_{\mathrm{I}}\underline{\underline{\mathrm{L}}}_{\mathrm{II}}\otimes\colon\ \mathrm{D}(X) \times \mathrm{D}^-(X) \longrightarrow \mathrm{D}(X).$$

Of course we can derive $\otimes$ in the first variable first, then in the second variable, and by [I.6.3] we get the same result. We will therefore use the ambiguous notation $F^\bullet \underline{\underline{\otimes}} G^\bullet$ for $\underline{\underline{\mathrm{L}}}_{\mathrm{I}}\underline{\underline{\mathrm{L}}}_{\mathrm{II}} \otimes (F^\bullet, G^\bullet)$.

Definition 19. *If $F^\bullet, G^\bullet \in \mathrm{D}^-(X)$, we define the* local hyperTor

$$\underline{\mathrm{Tor}}_i(F^\bullet, G^\bullet) = H^{-i}(F^\bullet \underline{\underline{\otimes}} G^\bullet).$$

Proposition 24. *Let X be a prescheme, and let $F^\bullet \in \mathrm{Ob}\,\mathrm{K}^b(\mathrm{Mod}(X))$. Then the following conditions are equivalent:*

(i) There is a quasi-isomorphism $s\colon G^\bullet \to F^\bullet$ where $G^\bullet$ is a bounded complex of flat $\mathcal{O}_X$-modules.

(ii) The functor $F^\bullet \underline{\underline{\otimes}} \cdot$ from $\mathrm{D}^-(X)$ to $\mathrm{D}(X)$ is way-out right (and hence way-out in both directions).

(iii) There is an n_0 such that $\underline{\mathrm{Tor}}_i(F^\bullet, \mathcal{G}) = 0$ for all $\mathcal{O}_X$-modules $\mathcal{G}$, and all $i > n_0$.

Proof. The proof is entirely analogous to the proof of [I.7.6] and will be left to the reader. Important points are 1) an $\mathcal{O}_X$-module F is flat if and only if $\underline{\mathrm{Tor}}_1(F, \mathcal{G}) = 0$ for all $\mathcal{O}_X$-modules $\mathcal{G}$, 2) if $Z \to B \to 0$ is a surjection of $\mathcal{O}_X$-modules with $Z \neq B$, then there exists an $\mathcal{O}_X$-module $\mathcal{G}$ (for example $\mathcal{G} = \mathcal{O}_X$) such that $Z \otimes \mathcal{G} \neq B \otimes \mathcal{G}$, and 3) the analogue of [I.4.5] for flat modules fails. Instead, we use the commutative diagram

$$\begin{tikzcd} F'^\bullet \arrow[r] \arrow[d] & F^\bullet \arrow[d] \\ \sigma_{\geq n}(F'^\bullet) \arrow[r] & \sigma_{\geq n}(F^\bullet) \end{tikzcd}$$

where $F'^\bullet$ is a (not necessarily bounded) flat resolution of $F^\bullet$. Since $F^\bullet$ itself is a bounded complex, for n small enough, $\sigma_{\geq n}(F^\bullet)$ is equal to $F^\bullet$. Hence $\sigma_{\geq n}(F'^\bullet) \to F^\bullet$ is the required quasi-isomorphism. $\square$

Remark. We see from the proof that it is sufficient in (iii) to consider only quasi-coherent sheaves $\mathcal{G}$. Indeed, it is sufficient to consider only the sheaves $\mathcal{O}_X$, and $k(x)$ for every point $x \in X$. Similarly, in (ii) it is sufficient to consider the restriction of the functor $F^\bullet \underline{\underline{\otimes}} \cdot$ to $\mathrm{D}^-_{\mathrm{qc}}(X)$.

Definition 20. *If $F^\bullet \in \mathrm{Ob}\,\mathrm{K}^b(\mathrm{Mod}(X))$ satisfies the equivalent conditions of the Proposition, we say that $F^\bullet$ has* finite Tor-dimension. *These complexes form a localizing subcategory of* $\mathrm{K}(\mathrm{Mod}(X))$, *which we denote by* $\mathrm{K}^b(\mathrm{Mod}(X))_{\mathrm{fTd}}$. *The corresponding subcategory of* $\mathrm{D}(X)$ *is denoted by* $\mathrm{D}^b(X)_{\mathrm{fTd}}$.

Proof of the localizing property. Cf. [I.7.7]. □

Now let $F^\bullet \in \mathrm{Ob}\,\mathrm{K}(\mathrm{Mod}(X))$, and consider the functor $F^\bullet \otimes \cdot \colon \mathrm{K}^b(\mathrm{Mod}(X))_{\mathrm{fTd}} \to \mathrm{K}(\mathrm{Mod}(X))$.

Let $\mathcal{L} \subseteq \mathrm{K}^b(\mathrm{Mod}(X))_{\mathrm{fTd}}$ be the subcategory of complexes of flat modules. Then by Proposition 24 and Lemma 10, part a2), $\mathcal{L}$ satisfies the hypotheses of [I.5.1] for this functor. Hence we can take the left derived functor in the second variable, and obtain

$$\underline{\underline{\mathrm{L}}}_{\mathrm{II}}\otimes\colon \mathrm{K}(\mathrm{Mod}(X)) \times \mathrm{D}^b(X)_{\mathrm{fTd}} \longrightarrow \mathrm{D}(X).$$

By Lemma 10, part b2), this gives rise to a functor

$$\underline{\underline{\mathrm{L}}}_{\mathrm{I}}\underline{\underline{\mathrm{L}}}_{\mathrm{II}}\otimes\colon \mathrm{D}(X) \times \mathrm{D}^b(X)_{\mathrm{fTd}} \longrightarrow \mathrm{D}(X)$$

which we will also denote by $\underline{\underline{\otimes}}$.

Problem. Does the functor

$$\otimes\colon \mathrm{K}(\mathrm{Mod}(X)) \times \mathrm{K}^b(\mathrm{Mod}(X))_{\mathrm{fTd}} \longrightarrow \mathrm{K}(\mathrm{Mod}(X))$$

admit a left derived functor in the first variable? Given $G^\bullet \in \mathrm{Ob}\,\mathrm{K}^b(\mathrm{Mod}(X))_{\mathrm{fTd}}$, does there exist a subcategory $\mathcal{L} \subseteq \mathrm{K}(\mathrm{Mod}(X))$ satisfying the hypotheses of [I.5.1] for $\cdot \otimes G^\bullet$?

Proposition 25. *Let X be a prescheme (resp. a locally noetherian prescheme), and let $F^\bullet \in \mathrm{D}_{\mathrm{qc}}(X)$ (resp. $\mathrm{D}_{\mathrm{c}}(X)$) and let $G^\bullet \in \mathrm{D}_{\mathrm{qc}}(X)$ (resp. $\mathrm{D}_{\mathrm{c}}(X)$). Assume either*

a) $F^\bullet \in \mathrm{D}^-(X)$, or

b) $G^\bullet \in \mathrm{D}^b(X)_{\mathrm{fTd}}$.

Then $F^\bullet \underline{\underline{\otimes}} G^\bullet \in \mathrm{D}_{\mathrm{qc}}(X)$ (resp. $\mathrm{D}_{\mathrm{c}}(X)$).

Proof. Using [I.7.3] as before, it is enough to show that if $\mathcal{F}, \mathcal{G}$ are quasi-coherent (resp. coherent) $\mathcal{O}_X$-modules, then $\underline{\mathrm{Tor}}_i(\mathcal{F}, \mathcal{G})$ is quasi-coherent (resp. coherent) for all $i \geq 0$. The question is local on X, so we may assume X affine. Then $\mathcal{F}$ has a resolution by quasi-coherent (resp. coherent) $\mathcal{O}_X$-modules, namely direct sums (resp. finite direct sums) of copies of $\mathcal{O}_X$. We can use this resolution to calculate the $\underline{\mathrm{Tor}}_i(\mathcal{F}, \mathcal{G})$. □

Suppose now that $f\colon X \to Y$ is a morphism of preschemes. Then we have

$$f^*\colon \operatorname{Mod}(Y) \longrightarrow \operatorname{Mod}(X)$$

and we can take its left-derived functor

$$\underline{\underline{\mathrm{L}}} f^*\colon \mathrm{D}^-(Y) \longrightarrow \mathrm{D}^-(X)$$

since there are enough flat $\mathcal{O}_Y$-modules, and they are f^*-acyclic. If f^* has finite cohomological dimension on $\operatorname{Mod}(Y)$, then we say f has finite Tor-dimension, and we can extend the domain of definition of $\underline{\underline{\mathrm{L}}} f^*$ to

$$\underline{\underline{\mathrm{L}}} f^*\colon \mathrm{D}(Y) \longrightarrow \mathrm{D}(X).$$

Proposition 26. *Let $f\colon X \to Y$ be a morphism of preschemes (resp. locally noetherian preschemes). Then $\underline{\underline{\mathrm{L}}} f^*$ takes $\mathrm{D}^-_{\mathrm{qc}}(Y)$ (resp. $\mathrm{D}^-_{\mathrm{c}}(Y)$) into $\mathrm{D}^-_{\mathrm{qc}}(X)$ (resp. $\mathrm{D}^-_{\mathrm{c}}(X)$). If f is of finite Tor-dimension, the same is true for unbounded complexes.*

Proof. Left to reader. □

3.5 Relations among the derived functors

In this section we will make a list of various natural homomorphisms and isomorphisms among the derived functors discussed in the previous sections.

Proposition 27. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be two morphisms of preschemes. Then there is a natural isomorphism*

$$\zeta^+\colon \underline{\underline{\mathrm{R}}}^+(g_* \cdot f_*) \xrightarrow{\sim} \underline{\underline{\mathrm{R}}}^+ g_* \cdot \underline{\underline{\mathrm{R}}}^+ f_*$$

of functors from $\mathrm{D}^+(X)$ to $\mathrm{D}^+(Z)$.

Suppose furthermore that X and Y are noetherian of finite Krull dimension. Then there is a natural isomorphism

$$\zeta\colon \underline{\underline{\mathrm{R}}}(g_* \cdot f_*) \xrightarrow{\sim} \underline{\underline{\mathrm{R}}} g_* \cdot \underline{\underline{\mathrm{R}}} f_*$$

of functors from $\mathrm{D}(X)$ to $\mathrm{D}(Z)$.

Proof. We use [I.5.4b]. For the first statement, let $\mathcal{L} \subseteq \mathrm{K}^+(\operatorname{Mod}(X))$ be the complexes of injective sheaves, and let $\mathcal{M} \subseteq \mathrm{K}^+(\operatorname{Mod}(Y))$ be the complexes of g_*-acyclic sheaves. Then $f_*(\mathcal{L}) \subseteq \mathcal{M}$ (indeed, f_* of an injective sheaf is flasque [EGA 0_{III} 12.2.4] and a flasque sheaf is g_*-acyclic [EGA 0_{III} 12.2.1]). Then the hypotheses are satisfied, so we have ζ^+.

For the second statement, let $\mathcal{L} \subseteq \mathrm{K}(\mathrm{Mod}(X))$ be the complexes of sheaves which are f_*-acyclic and g_*f_*-acyclic. Let $\mathcal{M} \subseteq \mathrm{K}(\mathrm{Mod}(Y))$ be the complexes of g_*-acyclic sheaves. Then by the first statement of the proposition we see that $f_*(\mathcal{L}) \subseteq \mathcal{M}$. Indeed, if F is both f_*-acyclic and g_*f_*-acyclic, then $\underline{\underline{\mathrm{R}}} f_*(F) \cong f_*(F)$, so for $i > 0$, $\mathrm{R}^i\, g_*(f_*F) \cong \mathrm{R}^i(g_*f_*)(F) = 0$, and f_*F is g_*-acyclic. On the other hand, one sees easily that the collection P of sheaves on X which are f_*-acyclic and g_*f_*-acyclic satisfies (i) and (ii) of [I.4.6], so $\mathcal{L}$ and $\mathcal{M}$ satisfy the hypotheses of [I.5.1], and we have the isomorphism ζ by [I.5.4b]. □

Proposition 28. *Let $f\colon X \to Y$ be a morphism of preschemes. Then there is a natural isomorphism*

$$\zeta^+ \colon \underline{\underline{\mathrm{R}}}^+\Gamma(X,\cdot) \xrightarrow{\sim} \underline{\underline{\mathrm{R}}}^+\Gamma(Y, \underline{\underline{\mathrm{R}}} f_*(\cdot))$$

of functors from $\mathrm{D}^+(X)$ to $\mathrm{D}^+(\mathrm{Ab})$.

If furthermore X and Y are noetherian of finite Krull dimension, then there is a natural isomorphism

$$\zeta \colon \underline{\underline{\mathrm{R}}}\Gamma(X,\cdot) \xrightarrow{\sim} \underline{\underline{\mathrm{R}}}\Gamma(Y, \underline{\underline{\mathrm{R}}} f_*(\cdot))$$

of functors from $\mathrm{D}(X)$ to $\mathrm{D}(\mathrm{Ab})$.

Proof. Similar to proof of the previous proposition. □

Proposition 29. *Let X be a prescheme. Then there is a natural isomorphism*

$$\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}^\bullet(F^\bullet, G^\bullet) \xrightarrow{\sim} \underline{\underline{\mathrm{R}}}\Gamma(X, \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet))$$

of bi-functors from $\mathrm{D}^-(X)^\bullet \times \mathrm{D}^+(X)$ to $\mathrm{D}(\mathrm{Ab})$.

Proof. Use [I.5.4]. Note that for any two sheaves F and G, $\mathrm{Hom}(F,G) = \Gamma(\underline{\mathrm{Hom}}(F,G))$. Also note that if G is injective, then $\underline{\mathrm{Hom}}(F,G)$ is flasque [G, II 7.3.2], and hence Γ-acyclic. □

Proposition 30. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be morphisms of preschemes. Then there is a natural isomorphism*

$$\zeta^- \colon \underline{\underline{\mathrm{L}}}^-(f^* \cdot g^*) \xrightarrow{\sim} \underline{\underline{\mathrm{L}}}^- f^* \cdot \underline{\underline{\mathrm{L}}}^- g^*$$

of functors from $\mathrm{D}^-(Z)$ to $\mathrm{D}^-(X)$.

If furthermore f and g have finite Tor-dimension (see §4) then so does $g \circ f$, and there is a natural isomorphism

$$\zeta \colon \underline{\underline{\mathrm{L}}}(g \circ f)^* \xrightarrow{\sim} \underline{\underline{\mathrm{L}}} f^* \cdot \underline{\underline{\mathrm{L}}} g^*$$

of functors from $\mathrm{D}(Z)$ to $\mathrm{D}(X)$.

Proof. Left to reader. (Note that g^* takes flat $\mathcal{O}_Z$-modules into flat $\mathcal{O}_Y$-modules.) $\square$

Proposition 31. *Let $f\colon X \to Y$ be a morphism of preschemes, with X noetherian of finite Krull dimension. Then there is a natural functorial homomorphism*

$$\underline{\underline{\mathrm{R}}}f_*\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_X(F^\bullet, G^\bullet) \longrightarrow \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_Y(\underline{\underline{\mathrm{R}}}f_*F^\bullet, \underline{\underline{\mathrm{R}}}f_*G^\bullet)$$

for $F^\bullet \in \mathrm{D}^-(X)$ and $G^\bullet \in \mathrm{D}^+(X)$.

Proof. Note first that our hypotheses on f, Y, $F^\bullet$, and $G^\bullet$ ensure that both objects above are defined (cf. §2 and §3 above). We wish to define a morphism between two functors from $\mathrm{D}^-(X)^\bullet \times \mathrm{D}^+(X)$ to $\mathrm{D}(Y)$. Let $\mathcal{L} \subseteq \mathrm{K}^-(\mathrm{Mod}(X))$ be the complexes of f_*-acyclic objects, and let $\mathcal{M} \subseteq \mathrm{K}^+(\mathrm{Mod}(X))$ be the complexes of injective objects. Then, as we have seen before ([I, §5] and §2 above) the natural functors

$$\mathcal{L}_{\mathrm{qis}} \longrightarrow \mathrm{D}^-(X)$$

and

$$\mathcal{M}_{\mathrm{qis}} \longrightarrow \mathrm{D}^+(X)$$

are equivalences of categories. Hence it will be sufficient to define a morphism between the extensions of our functors to functors from ${\mathcal{L}_{\mathrm{qis}}}^\bullet \times {\mathcal{M}_{\mathrm{qis}}}^\bullet$ to $\mathrm{D}(Y)$. But now, using [I.3.4] applied to the triangulated categories $\mathcal{L}$ and $\mathcal{M}$, we see that it is enough to define a morphism of functors

$$\underline{\underline{\mathrm{R}}}f_*\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_X(QF^\bullet, QG^\bullet) \longrightarrow \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_Y(\underline{\underline{\mathrm{R}}}f_*QF^\bullet, \underline{\underline{\mathrm{R}}}f_*QG^\bullet)$$

(where Q denotes the localization functor from $\mathcal{L}$ to $\mathcal{L}_{\mathrm{qis}}$ or $\mathcal{M}$ to $\mathcal{M}_{\mathrm{qis}}$) for $F^\bullet \in \mathcal{L}$ and $G^\bullet \in \mathcal{M}$.

We now make explicit the morphisms ξ between a functor and its derived functor (cf. definition of derived functor, [I.5]) and obtain the following diagram:

$$\begin{tikzcd}
Qf_*\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet) \arrow[r, "(1)", "\xi_{f_*}"'] \arrow[d, "\varphi"'] & \underline{\underline{\mathrm{R}}}f_*\, Q\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet) \arrow[d, "\xi_{\underline{\mathrm{Hom}}^\bullet}"', "(2)"] \\
Q\underline{\mathrm{Hom}}^\bullet(f_*F^\bullet, f_*G^\bullet) \arrow[d, "\xi_{\underline{\mathrm{Hom}}^\bullet}"', "(3)"] & \underline{\underline{\mathrm{R}}}f_*\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(QF^\bullet, QG^\bullet) \arrow[dd, dotted, "\psi"'] \\
\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(Qf_*F^\bullet, Qf_*G^\bullet) \arrow[d, "\xi_{f_*}"', "(4)"] & \\
\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(Qf_*F^\bullet, \underline{\underline{\mathrm{R}}}f_*QG^\bullet) \arrow[r, "(5)", "\xi_{f_*}"'] & \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(\underline{\underline{\mathrm{R}}}f_*QF^\bullet, \underline{\underline{\mathrm{R}}}f_*QG^\bullet)
\end{tikzcd}$$

Here φ is deduced from the well-known natural map

$$f_*\underline{\mathrm{Hom}}_X(F,G) \longrightarrow \underline{\mathrm{Hom}}_Y(f_*F, f_*G)$$

for any two sheaves of $\mathcal{O}_X$-modules F, G. Now we use the last statement of [I.5.1] to deduce that certain of the ξ's are isomorphisms.

(1) Since $G^\bullet$ is made of injective sheaves, $\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet)$ is made of flasque sheaves [G, II 7.3.2] (note only finite products are involved since $F^\bullet \in \mathrm{K}^-$ and $G^\bullet \in \mathrm{K}^+$). But flasque sheaves are f_*-acyclic, so they can be used to calculate $\underline{\underline{\mathrm{R}}}f_*$, and hence the map ξ_{f_*} denoted by (1) above is an isomorphism.

(2) $G^\bullet$ is injective (or more precisely "$\mathcal{M}$ satisfies the hypotheses of [I.5.1] for $\underline{\mathrm{Hom}}^\bullet(F^\bullet, \cdot)$") and so $\xi_{\underline{\mathrm{Hom}}^\bullet}$ is an isomorphism.

(3) We can say nothing here.

(4) $G^\bullet$ is injective, so ξ_{f_*} is an isomorphism.

(5) $F^\bullet$ is made of f_*-acyclic sheaves, so ξ_{f_*} here is an isomorphism.

Therefore, since (1), (2), (4), and (5) are isomorphisms, there is a unique

$$\psi\colon \underline{\underline{\mathrm{R}}}f_*\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(QF^\bullet, QG^\bullet) \longrightarrow \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(\underline{\underline{\mathrm{R}}}f_*QF^\bullet, \underline{\underline{\mathrm{R}}}f_*QG^\bullet)$$

making the diagram commutative. This is the desired morphism of functors. $\square$

Remark. We have given the above proof in some detail to show the method. Faced with a similar situation in the sequel, we will say simply "we may assume $F^\bullet$ is made of f_*-acyclic objects, and $G^\bullet$ of injective objects", and then we will drop all Q's, and write "=" for any ξ which is an isomorphism. In other words, we use the convention that we may erase the $\underline{\underline{\mathrm{R}}}$ before a functor applied to an argument for which ξ is an isomorphism. So if $F^\bullet$ is a complex of f_*-acyclic objects, we will write $\underline{\underline{\mathrm{R}}}f_*(F^\bullet) = f_*F^\bullet$.

Proposition 32 (Projection formula)**.** *Let $f\colon X \to Y$ be a quasi-compact morphism of noetherian preschemes, of finite Krull dimension. Then there is a natural functorial isomorphism*

$$\underline{\underline{\mathrm{R}}}f_*(F^\bullet)\underline{\underline{\otimes}}_Y G^\bullet \xrightarrow{\sim} \underline{\underline{\mathrm{R}}}f_*(F^\bullet \underline{\underline{\otimes}}_X \underline{\underline{\mathrm{L}}}f^*G^\bullet)$$

for $F^\bullet \in \mathrm{D}^-(X)$ and $G^\bullet \in \mathrm{D}^-_{\mathrm{qc}}(Y)$.

Proof. Note first that both sides are defined. To define the morphism, we may assume that $F^\bullet$ is a complex of f_*-acyclic sheaves, and that $G^\bullet$ is a complex of $\mathcal{O}_Y$-flat sheaves. Then we get the morphism by composing the usual projection formula for sheaves with suitable ξ's:

$$f_*(F^\bullet) \otimes_Y G^\bullet \xrightarrow{\sim} f_*(F^\bullet \otimes_X f^* G^\bullet) \xrightarrow{\xi} \underline{\underline{\mathrm{R}}} f_*(F^\bullet \otimes_X f^* G^\bullet)$$

using the conventions of the remark above.

To show it is an isomorphism, we must show for each i that the map of sheaves

$$H^i\big(\underline{\underline{\mathrm{R}}} f_*(F^\bullet) \underline{\underline{\otimes}}_Y G^\bullet\big) \longrightarrow H^i\big(\underline{\underline{\mathrm{R}}} f_*(F^\bullet \underline{\underline{\otimes}}_X \underline{\underline{\mathrm{L}}} f^* G^\bullet)\big)$$

is an isomorphism. But this question is local on Y. Furthermore, the functors involved are all compatible with localization on Y, so we may assume Y affine. Now, going back to the original morphism, note that both sides are way-out left functors in $G^\bullet$. Note also that every quasi-coherent $\mathcal{O}_Y$-module is a quotient of a free $\mathcal{O}_Y$-module, since Y is affine. Hence, using the Lemma on Way-Out Functors [I.7.1 (ii) and (iv), dual statement], we reduce to the case where $G^\bullet$ is reduced to a single sheaf G, which is a free $\mathcal{O}_Y$-module. But now X, Y, and f are quasi-compact, so everything commutes with infinite direct sums, and we reduce to the case $G = \mathcal{O}_Y$. (The noetherian hypothesis ensures that $\mathrm{R}^i f_*$ commutes with direct sums, cf. [T 3.6.2] and [G, II §4.12].) Then we have $\underline{\underline{\mathrm{R}}} f_*(F^\bullet)$ on each side and we are done. □

Remark. Again we have given a proof in some detail to show the method. In the sequel, we may leave many of these details to the reader.

Corollary 5. *Let f, X, Y be as in the Proposition, and assume that f has finite Tor-dimension. Let $F^\bullet \in \mathrm{D}^b(X)$ have finite Tor-dimension (see Definition 4.3 above). Then $\underline{\underline{\mathrm{R}}} f_* F^\bullet \in \mathrm{D}^b(Y)$ also has finite Tor-dimension.*

Proof. We use condition (ii) of Proposition 4.2. Since f has finite Tor-dimension, the functor on $\mathrm{D}^-_{\mathrm{qc}}(Y)$

$$G^\bullet \longmapsto \underline{\underline{\mathrm{L}}} f^* G^\bullet$$

is way-out right. Since $F^\bullet$ also has finite Tor-dimension,

$$G^\bullet \longmapsto F^\bullet \underline{\underline{\otimes}}_X \underline{\underline{\mathrm{L}}} f^* G^\bullet$$

is way-out right, and so

$$G^\bullet \longmapsto \underline{\underline{\mathrm{R}}} f_*(F^\bullet \underline{\underline{\otimes}}_X \underline{\underline{\mathrm{L}}} f^* G^\bullet)$$

is way-out right. By the Proposition, this implies that the functor

$$G^\bullet \longmapsto \underline{\underline{\mathrm{R}}} f_*(F^\bullet) \underline{\underline{\otimes}}_Y G^\bullet$$

is way-out right (for $G^\bullet \in \mathrm{D}^-_{\mathrm{qc}}(Y)$). But this implies that $\underline{\underline{\mathrm{R}}} f_*(F^\bullet)$ has finite Tor-dimension (using the Remark following Proposition 4.2). □

Problem. Let f, X, Y be as in the Proposition, and assume that f has finite Tor-dimension. Then the functors $\underline{\underline{\mathrm{R}}} f_*(F^\bullet) \underline{\underline{\otimes}}_Y G^\bullet$ and $\underline{\underline{\mathrm{R}}} f_*(F^\bullet \underline{\underline{\otimes}}_X \underline{\underline{\mathrm{L}}} f^* G^\bullet)$ are both defined for $F^\bullet \in \mathrm{D}^b(X)_{\mathrm{fTd}}$, and $G^\bullet \in \mathrm{D}_{\mathrm{qc}}(Y)$, using the result of the Corollary. Are they isomorphic? The problem is to define a morphism between them, since once we have a morphism, it is easy to prove that it is an isomorphism, using the lemma on way-out functors.

Proposition 33. *Let $f\colon X \to Y$ be a flat morphism of preschemes. Then there is a natural functorial homomorphism*

$$f^* \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_Y(F^\bullet, G^\bullet) \longrightarrow \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_X(f^*F^\bullet, f^*G^\bullet)$$

for $F^\bullet \in \mathrm{D}(Y)$ and $G^\bullet \in \mathrm{D}^+(Y)$. If furthermore Y is locally noetherian, and $F^\bullet \in \mathrm{D}^-_{\mathrm{c}}(Y)$, it is an isomorphism. (We write f^* instead of $\underline{\underline{\mathrm{L}}} f^*$ since it is an exact functor.)

Proof. To define the map, we may assume $G^\bullet$ is a complex of injective sheaves, and then use the natural map of sheaves

$$f^* \underline{\mathrm{Hom}}_Y(F, G) \longrightarrow \underline{\mathrm{Hom}}_X(f^*F, f^*G).$$

To show the isomorphism, we may assume that Y is affine, and then we reduce to the case $F = \mathcal{O}_Y$ by the lemma on way-out functors. □

Proposition 34. *Let $f\colon X \to Y$ be a morphism of preschemes. Then there is a natural functorial isomorphism*

$$\underline{\underline{\mathrm{L}}} f^*(F^\bullet) \underline{\underline{\otimes}}_X \underline{\underline{\mathrm{L}}} f^*(G^\bullet) \xrightarrow{\sim} \underline{\underline{\mathrm{L}}} f^*(F^\bullet \underline{\underline{\otimes}}_Y G^\bullet)$$

for $F^\bullet, G^\bullet \in \mathrm{D}^-(Y)$.

Proof. Left to reader. □

Proposition 35. *Let $f\colon X \to Y$ be a morphism of noetherian preschemes of finite Krull dimension. Then there is a natural functorial homomorphism*

$$\rho\colon F^\bullet \longrightarrow \underline{\underline{\mathrm{R}}} f_* \underline{\underline{\mathrm{L}}} f^* F^\bullet$$

for $F^\bullet \in \mathrm{D}^-(Y)$, which gives rise by Proposition 5.5 to a natural functorial homomorphism

$$\tau\colon \underline{\underline{\mathrm{R}}} f_* \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_X(\underline{\underline{\mathrm{L}}} f^* F^\bullet, G^\bullet) \longrightarrow \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet_Y(F^\bullet, \underline{\underline{\mathrm{R}}} f_* G^\bullet)$$

for $F^\bullet \in \mathrm{D}^-(Y)$ and $G^\bullet \in \mathrm{D}^+(X)$.

If furthermore $F^\bullet \in \mathrm{D}^-_{\mathrm{c}}(Y)$, then τ is an isomorphism.

Proof. To define ρ, we may assume $F^\bullet$ is a complex of flat $\mathcal{O}_Y$-modules, compose the natural map $F^\bullet \to f_* f^* F^\bullet$ with ξ_{f_*}. To check that τ is an isomorphism, we may assume Y affine, and then reduce to the case $F^\bullet = \mathcal{O}_Y$. Then $\underline{\underline{\mathrm{L}}} f^* F^\bullet = \mathcal{O}_X$, and we have simply $\underline{\underline{\mathrm{R}}} f_* G^\bullet$ on each side. □

Corollary 6. *Under the hypotheses of the Proposition, we have*

$$\mathrm{Hom}_{\mathrm{D}(X)}(\underline{\underline{\mathrm{L}}} f^* F^\bullet, G^\bullet) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{D}(Y)}(F^\bullet, \underline{\underline{\mathrm{R}}} f_* G^\bullet)$$

in other words, $\underline{\underline{\mathrm{L}}} f^*$ and $\underline{\underline{\mathrm{R}}} f_*$ are adjoint functors from $\mathrm{D}^-_{\mathrm{c}}(Y)$ to $\mathrm{D}^-_{\mathrm{c}}(X)$ and $\mathrm{D}^+(X)$ to $\mathrm{D}^+(Y)$, respectively.

Proof. Apply $H^0 \underline{\underline{\mathrm{R}}}\Gamma$ to both sides of the isomorphism τ of the Proposition, and use Propositions 5.2, 5.3, and [I.6.4]. □

Proposition 36. *Let $f\colon X \to Y$ be a morphism of finite type of noetherian preschemes of finite Krull dimension. Let $u\colon Y' \to Y$ be a flat morphism, let $X' = X \times_Y Y'$, and let $v\colon X' \to X$ and $g\colon X' \to Y'$ be the projections, as shown:*

$$\begin{tikzcd} X' \arrow[r, "v"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "u"] & Y \end{tikzcd}$$

Then there is a natural functorial isomorphism

$$u^* \underline{\underline{\mathrm{R}}} f_* F^\bullet \xrightarrow{\sim} \underline{\underline{\mathrm{R}}} g_* v^* F^\bullet$$

for $F^\bullet \in \mathrm{D}_{\mathrm{qc}}(X)$.

Proof. To define the morphism, we may assume $F^\bullet$ is a complex of f_*-acyclic sheaves, and use the natural map $u^* f_* F^\bullet \to g_* v^* F^\bullet$ followed by ξ_{g_*}. Both sides are way-out in both directions, so to prove the isomorphism we reduce by the lemma on way-out functors to the case of a single quasi-coherent sheaf F on Y. Then we must show that for each i,

$$H^i(u^* \underline{\underline{\mathrm{R}}} f_* F) \xrightarrow{\sim} H^i(\underline{\underline{\mathrm{R}}} g_* v^* F).$$

This is [EGA III 1.4.15]. (Recall that the proof uses Čech cohomology, hence the quasi-coherence hypothesis.) □

Proposition 37. *Let X be a prescheme. Then there are natural functorial isomorphisms*

$$F^\bullet \underline{\underline{\otimes}} G^\bullet \xrightarrow{\sim} G^\bullet \underline{\underline{\otimes}} F^\bullet$$

and

$$F^\bullet \underline{\underline{\otimes}} (G^\bullet \underline{\underline{\otimes}} H^\bullet) \xrightarrow{\sim} (F^\bullet \underline{\underline{\otimes}} G^\bullet) \underline{\underline{\otimes}} H^\bullet$$

for all $F^\bullet, G^\bullet, H^\bullet \in \mathrm{D}^-(X)$.

Proof. Left to reader. □

Proposition 38. *Let X be a prescheme. Then there is a natural functorial homomorphism*

$$\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet) \underline{\underline{\otimes}} H^\bullet \longrightarrow \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet \underline{\underline{\otimes}} H^\bullet)$$

for $F^\bullet \in \mathrm{D}(X)$, $G^\bullet \in \mathrm{D}^+(X)$, and $H^\bullet \in \mathrm{D}^b(X)_{\mathrm{fTd}}$. If furthermore X is locally noetherian, and $F^\bullet \in \mathrm{D}^-_{\mathrm{c}}(X)$, then it is an isomorphism.

Proof. Left to reader. □

Proposition 39. *Let X be a locally noetherian prescheme, and assume that every coherent sheaf on X is a quotient of a locally free sheaf of finite rank (lffr). Then there is a natural functorial isomorphism*

$$\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet, \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(G^\bullet, H^\bullet)) \xrightarrow{\sim} \underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(F^\bullet \underline{\underline{\otimes}} G^\bullet, H^\bullet)$$

for $F^\bullet, G^\bullet \in \mathrm{D}^-_{\mathrm{c}}(X)$, and $H^\bullet \in \mathrm{D}^+(X)$.

Proof. To define the morphism, we use the result of the exercise in §3 above. We take resolutions of $F^\bullet$ and $G^\bullet$ by lffr's, and use them to calculate $\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet$. Note that any lffr is flat, and so if $F^\bullet, G^\bullet$ are complexes of lffr's, then $F^\bullet \underline{\underline{\otimes}} G^\bullet = F^\bullet \otimes G^\bullet$, which is also a complex of lffr's. For the isomorphism, we use the lemma on way-out functors and reduce to the case $F^\bullet = \mathcal{O}_X$. □

Proposition 40. *Let X be a prescheme, and let $L^\bullet$ be a bounded complex of locally free sheaves of finite rank. Let $L^{\bullet\vee} = \underline{\mathrm{Hom}}^\bullet(L^\bullet, \mathcal{O}_X)$. Then there are natural functorial isomorphisms*

$$\underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet)\underline{\underline{\otimes}}L^\bullet \xrightarrow{\sim} \underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet(F^\bullet, G^\bullet\underline{\underline{\otimes}}L^\bullet) \cong \underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet(F^\bullet\underline{\underline{\otimes}}L^{\bullet\vee}, G^\bullet)$$

for all $F^\bullet \in \mathrm{D}^-(X)$ and $G^\bullet \in \mathrm{D}^+(X)$.

Proof. Easy once one notes that the corresponding formulae hold for sheaves, and if G is an injective sheaf, and L a lffr, then $G \otimes L$ is injective. □

3.6 Compatibilities among the relations of §5

In situations involving three or more derived functors, there may be different ways of composing the homomorphisms and isomorphisms of §5 to obtain a homomorphism or isomorphism of functors. One would like to know that the result is independent of any choices. We give three examples.

1. Let $f\colon X \to Y$, $g\colon Y \to Z$, and $h\colon Z \to W$ be three morphisms of preschemes. Then by Proposition 5.1 there are isomorphisms

$$\begin{tikzcd} \underline{\underline{\mathrm{R}}}^+(h_*g_*f_*) \arrow[r, "\sim"] \arrow[d, "\sim"'] & \underline{\underline{\mathrm{R}}}^+h_*\underline{\underline{\mathrm{R}}}^+(g_*f_*) \arrow[d, "\sim"] \\ \underline{\underline{\mathrm{R}}}^+(h_*g_*)\underline{\underline{\mathrm{R}}}^+(f_*) \arrow[r, "\sim"] & \underline{\underline{\mathrm{R}}}^+h_*\underline{\underline{\mathrm{R}}}^+g_*\underline{\underline{\mathrm{R}}}^+f_* \end{tikzcd}$$

   We would like to know that this diagram is commutative.

2. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be morphisms of noetherian preschemes of finite Krull dimension. Then there are functorial homomorphisms

$$\begin{tikzcd} \underline{\underline{\mathrm{R}}}(g_*f_*)\underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet_X(F^\bullet, G^\bullet) \arrow[r, "\sim"] \arrow[dd] & \underline{\underline{\mathrm{R}}}g_*\underline{\underline{\mathrm{R}}}f_*\underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet_X(F^\bullet, G^\bullet) \arrow[d] \\ & \underline{\underline{\mathrm{R}}}g_*\underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet_Y(\underline{\underline{\mathrm{R}}}f_*F^\bullet, \underline{\underline{\mathrm{R}}}f_*G^\bullet) \arrow[d] \\ \underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet_Z(\underline{\underline{\mathrm{R}}}(g_*f_*)F^\bullet, \underline{\underline{\mathrm{R}}}(g_*f_*)G^\bullet) \arrow[r, "\sim"] & \underline{\underline{\mathrm{R}}\mathrm{Hom}}^\bullet_Z(\underline{\underline{\mathrm{R}}}g_*\underline{\underline{\mathrm{R}}}f_*F^\bullet, \underline{\underline{\mathrm{R}}}g_*\underline{\underline{\mathrm{R}}}f_*G^\bullet) \end{tikzcd}$$

for $F^\bullet \in \mathrm{D}^-(X)$ and $G^\bullet \in \mathrm{D}^+(X)$. Here the horizontal arrows are deduced from Proposition 5.1, and the vertical arrows from Proposition 5.5, and we have tacitly included in the left-hand vertical arrow a double use of the natural isomorphism $g_* f_* \cong (gf)_*$. We would like to know that this diagram is commutative.

3. Let X be a prescheme. Then there are functorial isomorphisms

$$\begin{tikzcd}
F^\bullet \underline{\underline{\otimes}} (G^\bullet \underline{\underline{\otimes}} (H^\bullet \underline{\underline{\otimes}} I^\bullet)) \arrow[r, "\sim"] \arrow[dd] & F^\bullet \underline{\underline{\otimes}} ((G^\bullet \underline{\underline{\otimes}} H^\bullet) \underline{\underline{\otimes}} I^\bullet) \arrow[d, "\wr"] \\
 & (F^\bullet \underline{\underline{\otimes}} (G^\bullet \underline{\underline{\otimes}} H^\bullet)) \underline{\underline{\otimes}} I^\bullet \arrow[d, "\wr"] \\
(F^\bullet \underline{\underline{\otimes}} G^\bullet) \underline{\underline{\otimes}} (H^\bullet \underline{\underline{\otimes}} I^\bullet) \arrow[r, "\sim"] & ((F^\bullet \underline{\underline{\otimes}} G^\bullet) \underline{\underline{\otimes}} H^\bullet) \underline{\underline{\otimes}} I^\bullet
\end{tikzcd}$$

for $F^\bullet, G^\bullet, H^\bullet, I^\bullet \in \mathrm{D}^-(X)$, by Proposition 5.13. We would like to know that this diagram is commutative.

In the first example, the commutativity follows from the Remark after [I.5.4]. In the second and third examples, we note that the homomorphisms and isomorphisms referred to are uniquely determined by the corresponding natural maps for sheaves, and by the condition that they commute with the relevant morphisms ξ which occur in the definition of the derived functor. The required commutativities then follow from the analogous results for sheaves, which we may assume known.

Now these examples are only three of many more similar compatibilities which will come immediately to the reader's mind. I could make a big list, and in principle could prove each one on the list. However, I would be sure to need some more later, and already the list of ones I can think of off-hand is too cumbersome to write down. And since the chore of inventing these diagrams and checking their commutativity is almost mechanical, the reader would not want to read them, nor I write them. It would be comforting to know that such a list existed, or to have a meta-theorem saying that any such diagram one would dream up is commutative. However, both of these possibilities seem of an order of complexity too great to treat in these notes.

Unfortunately, I will have to use many of these compatibilities in an essential way in what is to follow. Perhaps for each theorem in the sequel one could make a list of exactly which compatibilities are needed, and verify them, but even that is too clumsy at this stage. So I must ask the reader's indulgence. I believe in the truth of the theorems stated, and I hope to convince him of their truth also. But I have not verified every commutative diagram which is necessary for a rigorous proof, and I do not suppose that any reader will have the patience to do so either.

In particular, I believe that all reasonable compatibilities one could imagine between the "natural" homomorphisms and isomorphisms of §5 are true, and so in the sequel I will write simply "=" instead of naming the isomorphism (provided, of course, the relevant hypotheses are satisfied). However, in later chapters we will deal with more homomorphisms and isomorphisms, where the compatibilities one can imagine are not always obvious, and may even be false (e.g., where there is a question of sign). Hence I will name those isomorphisms and list (in principle) the compatibilities we will need, and keep track of them.

For a completely satisfactory treatment of this question of "natural" isomorphisms and their compatibilities, we must await future developments. Mac Lane [12 pp. 14,15] refers to the problem in the context of the associativity of the tensor product, where he says that he does not know even a finite number of diagrams such as the one in example 3 above, which together imply that all such diagrams involving associativities are commutative. Perhaps the language of fibred categories [SGA 60–61, exposé VI] or the techniques of [Giraud, thesis] will supply what is needed.

3.7 Injective sheaves on a locally noetherian prescheme

In this section we give the structure of the injective objects in the category $\mathrm{Mod}(X)$ of all $\mathcal{O}_X$-modules on a locally noetherian prescheme X. We show in particular that every quasi-coherent $\mathcal{O}_X$-module can be embedded in a quasi-coherent injective $\mathcal{O}_X$-module, and hence that the natural functor

$$\mathrm{D}^+(\mathrm{Qco}(X)) \longrightarrow \mathrm{D}^+_{\mathrm{qc}}(X)$$

is an equivalence of categories.

We use results of Gabriel [5], which in turn were inspired from results of Matlis [13] in the case of noetherian rings.

Definition 21. *Let A be an abelian category, and let M be an object of A. An* injective hull *of M is an injective object I of A, together with a monomorphism $M \to I$, such that if N is a non-zero subobject of I, then $M \cap N \neq 0$.*

Theorem 3. *[5, Ch. II, Thm. 2]. If A is an abelian category with generators and exact direct limits, then every object has an injective hull.*

Examples. The category of modules over a commutative ring has generators and exact direct limits, hence has injective hulls.

The category $\mathrm{Mod}(X)$ of $\mathcal{O}_X$-modules on an arbitrary prescheme X has generators and exact direct limits, hence has injective hulls.

Definition 22 (5, Ch. II, §4)**.** . *An object of an abelian category A is* noetherian *if every ascending chain of subobjects is stationary. An abelian category A is* locally noetherian *if it has exact direct limits, and has a family of generators consisting of noetherian objects of A.*

Examples. The category of modules over a noetherian ring is locally noetherian — the ring itself is a noetherian generator for the category.

The category $\mathrm{Qco}(X)$ of quasi-coherent sheaves on a noetherian prescheme X is locally noetherian, with the coherent sheaves forming a family of noetherian generators. This example was studied by Gabriel [5, Ch. VI].

Proposition 41. *[5, Ch. IV, Prop. 6]. Any direct sum of injective objects in a locally noetherian category is injective.*

Theorem 4. *[5, Ch. IV, Thm. 2]. Let A be a locally noetherian category. Then every injective object I of A is isomorphic to a direct sum $\bigoplus_{j\in J} I_j$ of indecomposable injectives I_j. Furthermore, if $\bigoplus_{k\in K} I_k$ is a second such decomposition, then there is a bijection $h\colon J \to K$ such that $I_j \cong I_{h(j)}$ for each j.*

Proposition 42. *[13]. Let A be a noetherian ring. Then the indecomposable injective A-modules are precisely the injective hulls $I(\mathfrak{p})$ of $k(\mathfrak{p})$ over A, where $\mathfrak{p}$ ranges over the prime ideals of A.*

We also have information about the structure of one of these injective hulls [Matlis, ibid.]:

Proposition 43. *Let A be a noetherian ring, let $\mathfrak{p}$ be a prime ideal of A, and let I be an injective hull of the residue field $k(\mathfrak{p})$ of $\mathfrak{p}$. Then one can write I as a direct limit of the submodules*

$$0 \subseteq E^1 \subseteq E^2 \subseteq \cdots$$

where $E^i = \mathrm{Hom}_A(A/\mathfrak{p}^i A, I)$ is an Artin module over the local ring $A_\mathfrak{p}$, and where, for each i,

$$\dim_{k(\mathfrak{p})}(E^{i+1}/E^i) = \dim_{k(\mathfrak{p})}((\mathfrak{p}^i/\mathfrak{p}^{i+1}) \otimes_A k(\mathfrak{p})).$$

Now we apply these results to locally noetherian preschemes, giving first a special case.

Definition 23. *Let X be a prescheme, x a point of X, and M an $\mathcal{O}_{X,x}$-module. Then we define $i_x(M)$ to be the sheaf $i_x(\widetilde{M})$ on X, where $i\colon \operatorname{Spec}\mathcal{O}_{X,x} \to X$ is the natural inclusion, and $\widetilde{M}$ is the sheaf on $\operatorname{Spec}\mathcal{O}_{X,x}$ associated to M.*

Remark. We will be particularly interested in the case where $\operatorname{Supp} M$ is just the closed point x of $\operatorname{Spec}\mathcal{O}_{x,X}$. In that case $\Gamma(U, i_x(M)) = M$ if $x \in U$, and 0 otherwise, i.e., it is a simple sheaf on the closed subset $\overline{\{x\}}$ of X. Moreover, for any $\mathcal{O}_X$-module F, we have

$$\operatorname{Hom}_X(F, i_x(M)) = \operatorname{Hom}_{\operatorname{Spec}\mathcal{O}_{X,x}}(i^*(F), \widetilde{M}) = \operatorname{Hom}_{\mathcal{O}_{X,x}}(F_x, M)$$

where F_x is the stalk of F at x.

Proposition 44. *Let X be a locally noetherian prescheme, let x be a point of X, and let I be an injective hull of $k(x)$ over the local ring $\mathcal{O}_{x,X}$ of x. Then $i_x(I)$ is a (quasi-coherent) injective $\mathcal{O}_X$-module.*

Proof. By Proposition 7.5 above, I has support at the closed point of $\operatorname{Spec}\mathcal{O}_{X,x}$. Therefore, if $F \subseteq G$ are two $\mathcal{O}_X$-modules, and $\phi\colon F \to i_x(I)$ is a homomorphism of sheaves, ϕ gives a map $F_x \to I$, which extends to a map $G_x \to I$ since I is an injective $\mathcal{O}_{X,x}$-module. But by the Remark above, this gives a map of G to $i_x(I)$ extending ϕ, so $i_x(I)$ is an injective $\mathcal{O}_X$-module. □

Lemma 11. *Let X be a noetherian prescheme, and let $\mathcal{G} \subseteq \mathcal{O}_X$ be a sheaf of ideals (not necessarily quasi-coherent). Then there are a finite number of open subsets $U_i \subseteq X$, and a finite number of sections $s_{ij} \in \Gamma(U_i, \mathcal{G})$ for each i, such that at each point $x \in X$, the sections s_{ij}, for those i such that $x \in U_i$, generate the stalk $\mathcal{G}_x$ as an $\mathcal{O}_{X,x}$-module.*

Proof. Since X can be covered by a finite number of open affines, we may as well assume X is affine, equal to $\operatorname{Spec} A$ for a suitable noetherian ring A.

Let $x \in X$. For each open affine neighborhood U of x of the form $X_f = \operatorname{Spec} A_f$ for $f \in A$, consider the ideal $\mathfrak{a}(U) = \rho^{-1}\Gamma(U, \mathcal{G})$ of A, where

$$\rho\colon A = \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(U, \mathcal{O}_X)$$

is the natural restriction. Clearly a smaller neighborhood gives a larger ideal. Hence by the a.c.c. in A, there is such a neighborhood U of x giving a maximal $\mathfrak{a}(U_x) = \mathfrak{a}_x$. Hence, since the rings $\Gamma(U, \mathcal{O}_X)$ are localizations of A, we have for every such open neighborhood $U \subseteq U_x$ of x that

$$\Gamma(U, \mathcal{G}) = \mathfrak{a}_x \otimes_A \Gamma(U, \mathcal{O}_X) = \Gamma(U_x, \mathcal{G}) \otimes_{\Gamma(U_x, \mathcal{O}_X)} \Gamma(U, \mathcal{O}_X). \tag{1}$$

Now if x' is a specialization of x in U_x, i.e., $x' \in \overline{\{x\}} \cap U_x$, then every neighborhood of x' contains x. So by (1) we see that the stalk $\mathcal{G}_x$ can be generated by sections $s_i \in \Gamma(U_x, \mathcal{G})$, which, being an ideal in a noetherian ring, has a finite number of generators. (Recall that open sets of the form X_f form a base for the topology of X.) Thus we have established the following result:

($*$) For each point $x \in X$, one can find an open affine neighborhood U_x of x, and a finite number of sections $s_i \in \Gamma(U_x, \mathcal{G})$, such that for every $x' \in \overline{\{x\}} \cap U_x$, the s_i generate the stalk $\mathcal{G}_{x'}$.

Now we prove the statement of the Lemma. By quasi-compacity, we can find an open subset U of X (possibly empty) which is maximal with the property that the lemma is true for $\mathcal{G}|_U$. Suppose that $U \neq X$. Let x be the generic point of an irreducible component of $X - U$, and choose a neighborhood U_x with the property ($*$) above, and also so small that it does not meet any other irreducible component of $X - U$. Then every point of $U \cup U_x$ lies either in U or in $\overline{\{x\}} \cap U_x$, so that the lemma is also true for $\mathcal{G}|_{U \cup U_x}$. We conclude that $U = X$, which proves the Lemma. □

Theorem 5. *Let X be a locally noetherian prescheme. Then the category* $\mathrm{Mod}(X)$ *of all $\mathcal{O}_X$-modules is a locally noetherian category.*

Proof. $\mathrm{Mod}(X)$ clearly has exact direct limits. We must provide it with a family of noetherian generators. I claim the sheaves $\mathcal{O}_U$, where U is a noetherian open affine of X, and $\mathcal{O}_U$ is the sheaf $\mathcal{O}_X$ on U and zero outside, will do. They clearly form a family of generators. To see that $\mathcal{O}_U$ is noetherian, let $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \cdots$ be an increasing sequence of subsheaves. We may assume $X = U$ is noetherian. Let $\mathcal{G} = \bigcup \mathcal{G}_k$. Then by the Lemma, there are a finite number of open sets U_i and sections $s_{ij} \in \Gamma(U_i, \mathcal{G})$ which generate the stalk of $\mathcal{G}$ at each point. For each s_{ij}, we can cover U_i with a finite number of open sets $U_{ij\ell}$ such that each $s_{ij}|_{U_{ij\ell}}$ comes from a section of a suitable $\mathcal{G}_{k(ij\ell)}$ over $U_{ij\ell}$. Doing this for all the s_{ij}, and taking $k = \max(k(ij\ell))$, we find that all the s_{ij} come from sections of $\mathcal{G}_k$. Hence $\mathcal{G}_k = \mathcal{G}$, and so our sequence is stationary. □

Corollary 7. *Let X be a locally noetherian prescheme. Then any direct sum of injective $\mathcal{O}_X$-modules is injective.*

Corollary 8. *Let X be a locally noetherian prescheme. Then any injective $\mathcal{O}_X$-module can be written uniquely as a direct sum of indecomposable injective $\mathcal{O}_X$-modules.*

Definition 24. *Let X be a locally noetherian prescheme, let $x \in X$, and let x' be a specialization of x, i.e., $x' \in \overline{\{x\}}$. Let I be an injective hull of $k(x)$ over the local* *ring $\mathcal{O}_{X,x}$. We define $J(x,x')$ to be the restriction of the sheaf $i_x(I)$ to the closed subset $\overline{\{x'\}}$ of $\overline{\{x\}}$, as in [G, II Thm. 2.9.1]. If $x = x'$, we write simply $J(x)$ for $J(x,x) = i_x(I)$. Note that $J(x,x')$ is an indecomposable injective $\mathcal{O}_X$-module.*

Theorem 6. *Let X be a locally noetherian prescheme. Then the indecomposable injective $\mathcal{O}_X$-modules are precisely the sheaves $J(x,x')$ defined above, for every pair of points x specializing to x' of X.*

Lemma 12. *Let X be a locally noetherian prescheme, and let I be an injective $\mathcal{O}_X$-module. Then the stalk I_x of I at each point $x \in X$ is an injective module over the local ring $\mathcal{O}_{X,x}$ of x.*

Proof. Since $\mathcal{O}_{X,x}$ is a noetherian ring, it is sufficient to show that whenever $M \subseteq N$ is an inclusion of $\mathcal{O}_{X,x}$-modules of finite type, and $\phi\colon M \to I_x$ is a map, then ϕ extends to N. We can find coherent sheaves $\overline{M} \subseteq \overline{N}$ on X, with stalks M and N at x [EGA I 9.4.8], and we can find a map of $\overline{M}$ to I in a suitable neighborhood $\overline{U}$ of x, extending ϕ, since $\overline{M}$ is a sheaf of finite presentation [EGA 0 5.2.6]. This gives a map of $\overline{M}_U$ to I. Now since I is injective, this extends to a map of $\overline{N}_U$ to I, whose stalk at x extends ϕ. □

Proof of Theorem 7.11. Let I be an indecomposable injective $\mathcal{O}_X$-module. We define $\operatorname{Supp}(I)$ to be the set of points $x \in X$ such that the stalk I_x is non-zero. Let $x' \in \operatorname{Supp}(I)$ be a maximal point (i.e., one which is not a specialization of any other point in $\operatorname{Supp}(I)$). Then by the lemma, the stalk $I_{x'}$ is an injective $\mathcal{O}_{X,x'}$-module. Let I_0 be an indecomposable injective direct summand of $I_{x'}$. Then by Proposition 7.4, I_0 is of the form $I(\mathfrak{p})$ for some prime ideal $\mathfrak{p}$ of $\mathcal{O}_{x'}$. Let $\mathfrak{p}$ correspond to the point $x \in X$. Then $i_x(I(\mathfrak{p})) = J(x)$, using the notation above. Furthermore, there is a natural inclusion of $\mathcal{O}_Z$ in $J(x)$, where $Z = \overline{\{x\}}$. Since the stalk $J(x)_{x'} = I_0$ is mapped into $I_{x'}$, we have a map $\mathcal{O}_{Z,x'} \to I_{x'}$, and hence can find a map $\phi\colon \mathcal{O}_Z \to I$ extending it in a suitable neighborhood U of x'. Let $Z' = \overline{\{x'\}}$. Then the image of ϕ has support in Z', since x' was chosen maximal in $\operatorname{Supp}(I)$, and by construction, ϕ is injective at x'. Now applying Lemma 7.7 to the kernel of ϕ, we see that ϕ factors through $\mathcal{O}_{Z,Z'}$, the restriction of $\mathcal{O}_Z$ to the closed subset Z', to give a map $\psi\colon \mathcal{O}_{Z,Z'} \to I$ (defined on U), and that by shrinking U a bit, we may assume that ψ is injective on U. In other words, we have an injection of sheaves on X,

$$\psi_U\colon \mathcal{O}_{Z,Z',U} \longrightarrow I$$

where $\mathcal{O}_{Z,Z',U}$ is the restriction of $\mathcal{O}_{Z,Z'}$ to the open set U.

Now it is easily seen that $J(x,x')$ is an injective hull of $\mathcal{O}_{Z,Z',U}$, so ψ_U extends to give a map $J(x,x') \to I$, which is necessarily an inclusion since ψ_U is. Now

$J(x, x')$ being injective, is a direct summand, so must be equal to I since I was indecomposable. $\square$

Corollary 9. *Let X be a locally noetherian prescheme. Then every injective $\mathcal{O}_X$-module is uniquely a direct sum of injectives $J(x, x')$ defined above.*

Corollary 10. *Let A be a noetherian ring, let I be an injective A-module, and let $X = \operatorname{Spec} A$. Then $\widetilde{I}$ is an injective $\mathcal{O}_X$-module.*

Proof. Follows from Theorem 7.3, Propositions 7.4 and 7.6, and Corollary 7.9. $\square$

Corollary 11. *Let A be a noetherian ring, let M be an A-module of finite type, and let N be any A-module. Then*

$$\underline{\operatorname{Ext}}^i_X(\widetilde{M}, \widetilde{N}) = \operatorname{Ext}^i_A(M, N)^\sim$$

for all i, where $X = \operatorname{Spec} A$.

Proof. Since M is of finite type,

$$\underline{\operatorname{Hom}}_X(\widetilde{M}, \widetilde{N}) = \operatorname{Hom}_A(M, N)^\sim$$

for any N. Take an injective resolution $I^\bullet$ of N. Then $\widetilde{I}^\bullet$ is an injective resolution of $\widetilde{N}$ by the previous Corollary. Since $^\sim$ is an exact functor, we have the result. $\square$

Lemma 13. *Let X be a prescheme, and let F be an $\mathcal{O}_X$-module. Then F is injective in $\operatorname{Mod}(X)$ if and only if there is an open cover $\{U_\alpha\}$ of X such that for each α, $F|_{U_\alpha}$ is injective in the category $\operatorname{Mod}(U_\alpha)$.*

Proof. If F is injective, then F restricted to any open set is injective. Indeed, given $G' \subseteq G$ on U and a map $G' \to F|_U$, we deduce a map $G'^X \to F$, where G'^X is the sheaf G' extended by zero outside U. This extends to a map $G^X \to F$, hence $G \to F|_U$.

On the other hand, to test whether F is injective, it is enough, by Zorn's lemma, to show that for every sheaf G in a family of generators of the category, and for every map $\phi\colon G' \to F$ of a subsheaf G' of G to F, ϕ extends to G. Since the sheaves $\mathcal{O}_U$ (which is $\mathcal{O}_X$ on U and 0 outside), for U arbitrarily small, form a family of generators of $\operatorname{Mod}(X)$, we see that the question of injectivity is local, as required. $\square$

Proposition 45. *Let X be a locally noetherian prescheme, and let F be a quasi-coherent $\mathcal{O}_X$-module. Then the following conditions are equivalent:*

(i) F is an injective $\mathcal{O}_X$-module.

(ii) *F is isomorphic to a direct sum of sheaves $J(x)$ for various $x \in X$.*

(iii) *For every $x \in X$, the stalk F_x of F at x is an injective $\mathcal{O}_{X,x}$-module.*

(iv) *For all coherent sheaves G on X, $\operatorname{Ext}^1(G, F) = 0$.*

(v) *There is an open cover $\{U_\alpha\}$ of X such that $F|_{U_\alpha}$ is an injective $\mathcal{O}_{U_\alpha}$-module for all α.*

Proof. (i) $\Longrightarrow$ (ii) We know by Corollary 7.13 that F is isomorphic to a direct sum of $J(x, x')$. Since F is quasi-coherent, each $J(x, x')$ must be also. But $J(x, x')$ is quasi-coherent if and only if $x = x'$, i.e., $J(x, x') = J(x)$.

(ii) $\Longrightarrow$ (iii) Clear.

(iii) $\Longrightarrow$ (iv) Since G is coherent, Ext commutes with passage to stalks, and all the stalks are zero.

(iv) $\Longrightarrow$ (v) On a noetherian affine $U = \operatorname{Spec} A$ of X, $\operatorname{Ext}^1(G, F)|_U = \operatorname{Ext}^1_A(M, N)^\sim$ where $G = \widetilde{M}$ and $F = \widetilde{N}$. The result then follows from [EGA I 9.4.8], Corollary 7.14, and the well-known fact that on a noetherian ring, N is injective if and only if $\operatorname{Ext}^1_A(M, N) = 0$ for all A-modules M of finite type.

(v) $\Longrightarrow$ (i) by the lemma. □

Theorem 7. *Let X be a locally noetherian prescheme. Then every quasi-coherent $\mathcal{O}_X$-module F can be embedded in a quasi-coherent, injective $\mathcal{O}_X$-module I.*

Proof. Indeed, we will show that the injective hull I of a quasi-coherent sheaf F is quasi-coherent. Let I be written as a direct sum of sheaves $J(x, x')$, by Corollary 7.13. For each such $J(x, x')$, since I is an injective hull of F, there is a neighborhood U of x', and a section $s \in J(x, x')(U)$ which is also a section of $F(U)$. By shrinking U if necessary, we may assume that the support of s is just $U \cap Z'$, where $Z' = \overline{\{x'\}}$. Since s is a section of a quasi-coherent sheaf F, it must be annihilated by some power of the ideal $\mathcal{O}_{Z'}$ of Z'. This implies $x = x'$, for if $x \neq x'$, no section of $J(x, x')$ is annihilated by any power of $\mathcal{O}_{Z'}$. Thus $J(x, x') = J(x)$ is quasi-coherent, and so I is quasi-coherent. □

Corollary 12. *Let X be a locally noetherian prescheme. Then the natural functor*

$$\mathrm{D}^+(\mathrm{Qco}(X)) \longrightarrow \mathrm{D}^+_{\mathrm{qc}}(X)$$

is an equivalence of categories.

Proof. (Cf. §1 for notations.) Follows from [I.4.8]. □

As an application, we give the following result on complexes of finite injective dimension.

Proposition 46. *Let X be a locally noetherian prescheme, let $A = \mathrm{Mod}(X)$, and let $F^\bullet \in \mathrm{Ob}\,\mathrm{K}^+_{\mathrm{qc}}(A)$. Then the equivalent conditions (i), (ii), and (iii) of [I.7.6] are also equivalent to the following:*

(i)$_{\mathrm{qc}}$ *$F^\bullet$ admits a quasi-isomorphism $F^\bullet \to I^\bullet$ into a bounded complex of quasi-coherent injective $\mathcal{O}_X$-modules.*

(ii)$_{\mathrm{qc}}$ *The functor $\underline{\underline{\mathrm{R}}}\underline{\mathrm{Hom}}^\bullet(\cdot, F^\bullet)$ from $\mathrm{D}(A)^\bullet$ to $\mathrm{D}(A)$ is way-out left.*

(iii)$_{\mathrm{qc}}$ *There is an integer n_0 such that $\underline{\mathrm{Ext}}^i(G, F^\bullet) = 0$ for all $G \in \mathrm{Mod}(X)$ and all $i > n_0$.*

(iii)$_{\mathrm{c}}$ *There is an integer n_0 such that $\underline{\mathrm{Ext}}^i(G, F^\bullet) = 0$ for all $G \in \mathrm{Coh}(X)$ and all $i > n_0$.*

Proof. (i)$_{\mathrm{qc}}$ $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iii)$_{\mathrm{c}}$ are all easy as before. It remains only to prove (iii)$_{\mathrm{c}}$ $\Rightarrow$ (i)$_{\mathrm{qc}}$. This is similar to the proof of (iii) $\Rightarrow$ (i) in loc. cit. using Proposition 7.17(iv) and Theorem 7.18 above. □

Example 1. *If X is a locally noetherian prescheme, the category $\mathrm{Qco}(X)$ of quasi-coherent sheaves on X may not be locally noetherian. Thus we do not know the structure of injectives in that category, and we do not know whether every injective object of $\mathrm{Qco}(X)$ is injective in $\mathrm{Mod}(X)$.*

Here is the example. Let X_1 be the projective plane, let E_1 be a line in X_1, and let x_1 be a closed point of E_1. Having defined X_n, E_n, x_n, define X_{n+1} to be X_n blown up at the point x_n, let E_{n+1} be the exceptional curve, and let x_{n+1} be a closed point of E_{n+1}. Define

$$X = \bigcup_{n=1}^{\infty} (X_n - x_n),$$

where we glue $X_n - x_n$ to the open subset $X_{n+1} - E_{n+1}$ of $X_{n+1} - x_{n+1}$. Then X is an integral, locally noetherian scheme.

However, $\mathrm{Qco}(X)$ is not a locally noetherian category. Indeed, let $\mathfrak{I}$ be a non-zero sheaf of ideals of $\mathcal{O}_X$, and let F be a noetherian generator of the category which admits a map into $\mathcal{O}_X$ not factoring through $\mathfrak{I}$. Then the image of F must be a noetherian non-zero ideal $\mathcal{G}$ of $\mathcal{O}_X$.

For each $k = 1, 2, \ldots,$ *let* Y_k *be a closed subset of* X,

$$Y_k = \bigcup_{n=k}^{\infty} (E_n - x_n).$$

Then $Y_1 \supset Y_2 \supset \cdots$. *Let* $\mathcal{J}_k$ *be the sheaf of ideals of* Y_k. *Then* $\mathcal{G}\mathcal{J}_1 \subset \mathcal{G}\mathcal{J}_2 \subset \cdots$ *which is a contradiction to the statement* $\mathcal{G}$ *is noetherian.*

We conclude that the category $\mathrm{Qco}(X)$ *does not have a family of noetherian generators.*

Remark. We do not know if the analogue of Corollary 7.19 is true for unbounded complexes, i.e., whether the natural functor

$$\mathrm{D}(\mathrm{Qco}(X)) \longrightarrow \mathrm{D}_{\mathrm{qc}}(X)$$

is an equivalence of categories. However, we conjecture it to be true when X is a *regular* noetherian scheme of finite Krull dimension, because in that case the category $\mathrm{Mod}(X)$ has finite injective dimension.

Chapter 4

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Index of Notations

Of course there is the usual collection of variable notations: A, B for abelian categories or rings, functors or sheaves, X, Y for preschemes, x, y for points of them, f, g for morphisms of preschemes, ϕ for morphisms of functors on sheaves, etc.

In general no underline denotes a group, as $\mathrm{Hom}(\mathcal{F}, \mathcal{G})$, one underline denotes a sheaf, as $\underline{\mathrm{Hom}}(\mathcal{F}, \mathcal{G})$ or $\underline{\mathrm{Tor}}(\mathcal{F}, \mathcal{G})$, two underlines denotes an object in a derived category, as $\underline{\underline{\mathrm{R}}}\,\mathrm{Hom}^\bullet(\mathcal{F}^\bullet, \mathcal{G}^\bullet)$, and a dot denotes a complex.

In the list of stable notations below, we have distinguished five categories: Latin alphabet, other alphabets, superscripts, subscripts, and arbitrary symbols.

Latin alphabet

Symbol	Meaning	Reference
Ab	the category of abelian groups	
acc	ascending chain condition	V.2
Ass	associated primes	
B	boundaries of a complex	
Coh	category of coherent sheaves	II.1
Coz	category of Cousin complexes	
D	dualizing functor	V.0
$\mathcal{D}(\mathcal{A})$	derived category	I.4
$\mathcal{D}(X)$	$= \mathcal{D}(\mathrm{Mod}(X))$	II.1
dcc	descending chain condition	V.2
dim	dimension	
Dual	category of dualizing complexes	VI.1
E	associated Cousin complex	IV.2, IV.3
Ext	Ext groups	I.6
$\underline{\mathrm{Ext}}$	sheaf Ext	II.3

Symbol	Meaning	Reference
fid	finite injective dimension	I.7.6, II.7.20
FR1–FR5	axioms of multiplicative systems	I.3
H	cohomology	I.1, IV.1
Hom	Hom groups	I.6
$\underline{\mathrm{Hom}}$	sheaf Hom	II.3
i_x	closed immersion	II.7
Icz	category of injective Cousin complexes	
id	identity map	
im	image	
$J(x), J(x, x')$	standard injectives	II.7
$\mathcal{K}(A)$		II.7
$k(x)$	residue field of a point x	
$K_\bullet(\underline{f}), K^\bullet(\underline{f}; F)$	Koszul complexes	III.7
ker	kernel	
L1–L3	axioms of direct systems	I.3
$DerL$	left derived functor	I.5
lffr	locally free of finite rank	II.5
Lno	category of locally noetherian preschemes	III.8
Mod(X)	category of $\mathcal{O}_X$-modules	II.1
Ob	objects of a category	
pfid	pointwise finite injective dimension	V.8
Ptwdual	category of pointwise dualizing complexes	VI.1
Q	functor into derived category	I.3
Qco	category of quasi-coherent sheaves	II.1
Qis	quasi-isomorphisms	I.4
$\underline{\underline{\mathrm{R}}}$	right derived functor	I.5
Res	category of residual complexes	VI.1
	residue symbol	
	residue of a differential	III.9
Spec	spectrum of a ring	VII.1
Supp	support of a sheaf or module	II.7
T	translation functor	I.1
$\underline{\mathrm{Tor}}$		
Tr	trace map	III.10, VI.4, VII.1
TR1–TR4	axioms of a triangulated category	I.1
TRA	axioms of trace	III.10.5, VI.4.2

Symbol	Meaning	Reference
tr. d	transcendence degree	
$\mathrm{Tr}\, f$	trace for a finite morphism	III.6
$\mathrm{Tr}\, p$	trace for a projective morphism	III.11
VAR	axioms for $f^!$ or f^Δ	III.8.7, VI.3.1
Z	cycles of a complex	

Greek and other alphabets

Symbol	Meaning	Reference
γ	trace for smooth morphism	III.3.4, III.11.2, VII.4.1
Γ	global sections of a sheaf	II.1.5
ζ	change of differentials	III.7.3
η	fundamental local isomorphism	III.7.3
θ	duality morphism	III.7.3
ξ	map of a functor to its derived functor	III.7.3
$\sigma_{\geq n}$	truncated complex	I.7
$\tau_{\geq n}$	truncated complex	I.7
ψ	residue isomorphism	III.8
ω	sheaf of differentials	III.1
Ω	sheaf of differentials	III.1
$\mathcal{O}_X$	structure sheaf of a prescheme	
$\mathfrak{m}$	maximal ideal of a ring	
$\mathfrak{p}$	prime ideal of a ring	
$\mathbb{A}$	affine space	
$\mathbb{P}$	projective space	
$\mathbb{Q}$	rational numbers	
$\mathbb{Z}$	integers	

Subscripts

Symbol	Meaning	Reference

$\mathcal{A}'$	as in $\mathcal{K}_{\mathcal{A}'}(\mathcal{A}), \mathcal{D}_{\mathcal{A}'}(\mathcal{A})$	I.4
c	as in $\mathcal{D}_c(X)$	II.1
fid	as in $\mathcal{D}(X)_{\mathrm{fid}}$	I.7
fTd	as in $\mathcal{D}(X)_{\mathrm{fTd}}$	II.4
Gor$(Z^\bullet)$	as in $\mathcal{D}(X)_{\mathrm{Gor}(Z^\bullet)}$	IV.3
$\mathfrak{p}$	as in $A_\mathfrak{p}$, $M_\mathfrak{p}$: localization at a prime ideal	
qc	as in $\mathcal{D}_{\mathrm{qc}}(X)$	II.1
x	as in $\mathcal{O}_x$: local ring at a point x	
	as in $\mathcal{F}_x$: stalk of a sheaf at x	
	as in H^i_x: local cohomology	
Z	as in Γ_Z, H^i_Z	IV.1
I, II	as in $\underline{\underline{\mathrm{R}}}_{\mathrm{I}}, \underline{\underline{\mathrm{R}}}_{\mathrm{II}}$	IV.1
$\geq n$	as in $\sigma_{\geq n}, \tau_{\geq n}$	IV.1
$*$	as in f_*: direct image	IV.1

Superscripts

Symbol	Meaning	Reference
b	as in $\mathcal{K}^b, \mathcal{D}^b$: bounded	I.2, I.4
i	as in X^i:i-th term of a complex	
	as in H^i:i cohomology group	
	as in R^iF:i-th derived functor	
y	as in f^y	VI.2
z	as in f^z	VI.2
$\bullet$	as in $\mathcal{F}^\bullet, X^\bullet$: denotes a complex	
$\bullet\bullet$	as in $\mathcal{C}^{\bullet\bullet}$: double complex	I.7
$\bullet$	as in $\mathcal{K}(A)^\bullet$: opposed category	
$*$	as in f^*: inverse image	I.7, II.4
$+$	as in $\mathcal{K}^+, \mathcal{D}^+, \underline{\underline{\mathrm{R}}}^+$: bounded below	I.2, I.4, I.5
$-$	as in $\mathcal{K}^-, \mathcal{D}^-, \underline{\underline{\mathrm{R}}}^-$: bounded above	I.2, I.4, I.5
$!$	as in $f^!$	Intr., III.8, VII.3
$\flat$	$f^\flat$	IX.6
$\sharp$	$f^\sharp$	VI.3
Δ	f^Δ	VI.3
$\vee$	as in $L^\vee$: dual sheaf	II.5.16, III.1
$-$	as in $\{x\}^-$: closure	

$\sim$	as in $\tilde{M}$: sheaf associated to a module
$\wedge$	as in $\hat{A}$: completion of local ring

Arbitrary Symbols

Symbol	Meaning	Reference
$\vert$	restriction	
$\gg$	enough larger than	
$\times$	product	
$\otimes$	tensor product	
$\oplus$	direct sum	
$\coprod$	disjoint union / sum	
[ ]	shift operator	I.1
	closed interval	I.1
	reference to Bibliography or other chapter	
$\bigcup$	union	
{ }	the set of	
$\to$	morphism	
$\mapsto$	effect of a map on elements	
$\cdot\to$	distinguished side of triangle	
$\dashrightarrow$	morphism being constructed	
$\circ$	composition of functors ($F \circ G$)	I.2

Cohomologie à Support Propre, et Construction du Foncteur $f^!$

by P. Deligne

Errata